Point-Set Topology

— ***EYNTKA*** **Series** —

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When topology was initially thought about, the goal was to be able to know how to classify different surfaces: What properties can we use to distinguish a sphere from a plane? The closed interval from the open interval? One manifold from the other? Similarly, we may ask when two shapes are actually the same. For example, is a sphere the same thing as a sphere flipped inside out? Can a disk and a plane be seen as the same shape? What about a coffee mug and a doughnut?

These questions led to further investigation into the concepts used to classify these shapes. How do we define dimension? Can we use the concept of distance (a metric) to classify shapes? Can every space have a concept of distance defined on it? As these questions went on, mathematicians also asked how we can build new spaces by taking what we already know in set theory (subsets, equivalence relations, products) and using them to construct topologies. All of these questions asking about the fundamental nature of geometric objects grew into the domain of ***Topology*** [1]!

A good introduction to topology means understanding how these set-theoretic concepts translate to geometry, gaining some notion of why we settled on the definitions that have solidified over the last century, and analyzing properties that are *intrinsic* to a topology.

But what does it mean for a property to be *intrinsic* to a topology? In linear algebra, you learned that two vector spaces are fundamentally the same if there is an isomorphism between them. In topology, we adopt a similar philosophy: we care about spaces and their properties *up to homeomorphism.* Two spaces are homeomorphic (which we will denote as $X \cong Y$) if we can continuously deform one into the other by stretching, twisting, crumpling, or bending—but never tearing or gluing.

If you already have some background in basic proofs and are impatient to know the answer, here is a summary of this categorical way of thinking:

> A property is "topological" if, whenever one space has it, any space homeomorphic to it must necessarily have it as well. If one doesn't have the property, the other necessarily lacks it too. For example, dimension is a topological property. So if $X \cong Y$, and X is one-dimensional, Y must *also* be one-dimensional. However, we will show that a space being bounded (ex. $(0, 1)$ is bounded, but $\mathbb{R}$ is not) is *not* a topological property. That is, if $X \cong Y$, and X is bounded, it needn't be the case that Y is bounded! [2]

The formalization of these topological properties is a huge part of this book.

Another central question in topology (and actually many fields of math) is: what is the right way to compare and relate different spaces? As mentioned, early topologists were concerned with properties preserved under continuous deformation. This geometric intuition will at first seem completely absent when we introduce the rigorous definition of a topology, which relies entirely on sets. However, we will cover examples to show how this set-theoretic machinery perfectly captures that geometric intuition. Topology is ultimately about mapping relations. Two topologies will be related (which we'll write as $X \to Y$ via a continuous function) if we can somehow "put" the topology of X into Y while preserving the local structure of X.

Most of this document is dedicated to understanding different topological properties (like connectedness and compactness) and how topologies relate. The goal is that if you can show one topological space possesses a property that another lacks, you know with absolute certainty that they are

[1] In mathematics, topology (from the Greek words $\tau o\pi o\varsigma$, 'place, location', and $\lambda o\gamma o\varsigma$, 'study').

[2] Boundedness is a *metric* property, not a topological one, and is only preserved under *isometric* functions (functions that preserve exact distance).

different shapes. In other words, you are learning to classify topological spaces—which is the natural foundation you need to ask bigger questions later!

For those who have taken calculus or real analysis, the following is another intuition that may be useful as we begin:

> In real analysis, we extensively work with *distance* to define limits and continuity. Essentially, real analysis feels like the study of constructing and manipulating objects through the ϵ-δ definition of closeness. Because of this, the sequential definition of continuity is often the most intuitive.
>
> However, there are spaces that are beautifully structured but entirely unsuited for measuring distance or taking limits of sequences (for example, uncountable product spaces like $\mathbb{R}^\omega$, or spaces of functions). Thus, the core ideas of analysis were abstracted, refined, and distilled into their absolute purest forms. We stripped away the concept of "distance" entirely, leaving behind only the concept of *open sets* (neighborhoods).
>
> By doing this, properties like compactness, connectedness, and continuity work in the fullest generality while remaining remarkably simple to work with (in a sense, the fact that these definitions can be so beautifully simple is a work of genius). Thus, studying topology is really studying what it means to be a "space" in the most general possible setting.
>
> We have essentially agreed that a space is defined by being *closed* under taking as many "unions" of open sets as we want. (I use "unions" in quotes because, in the right context, it really means "gluing," "combining," or "adding" local neighborhoods together). Conversely, when working with the "dual" of open sets (closed sets), we still need some flexibility, but in a much more restricted, finite way. If we allowed arbitrary intersections of open sets, or arbitrary unions of closed sets, spaces would gain too much flexibility and lose their underlying structure—the Euclidean topology would become completely degenerate. By carefully balancing infinite flexibility for open sets and finite restrictions for closed sets, we perfectly capture the essence of what makes a shape a shape.

1

Introduction to Topology

In this chapter, we will cover the definition of topologies, with some examples, how to compare topologies, and the concepts which will naturally arise given the definition of topology we will present (closure, interior, countability, etc.)

1.1 Basic definition and construction

Interestingly, with abstraction, the eventual definition of topology that is currently taken seems far away from any geometric notion. To my surmise, the direction in which the abstraction went was to think of topology as giving a notion of “localness” to an object. Be careful to not think of localness as distance: those two concepts will merge in nice topological spaces (as we’ll eventually show with metric spaces in section 2.5), but it does not merge nicely in general (like the finite-compliment topology). The idea of “localness” is linked to finding properties of a topological space by how a space behaves on different parts of the space. This is a common trick in math: using local information to find global information.

We will elaborate on all of this after the definition and examples:

Definition 1.1.1: Topological Space

Given a set X, a *topological space* is a pair $(X, \mathcal{T}_X)$ where $\mathcal{T}_X \subseteq \mathcal{P}(X)$ is a set of subsets of X such that:

1. $\emptyset$ and X are in $\mathcal{T}_X$
2. $\mathcal{T}_X$ is closed under arbitrary unions
3. $\mathcal{T}_X$ is closed under finite intersection's

An element inside $\mathcal{T}_X$ is called *open*. A set $S \subseteq X$ is *closed* if its compliment is in $\mathcal{T}_X$; $S^c \in \mathcal{T}_X$.

By convention, $Y \subseteq X$ is *open* if and only if $Y \in \mathcal{T}_X$. A better name for open and closed set could've been "local set" and "co-local set" which better reflects their intuition in more pathological topologies, however the euclidean-topology inspired stuck and there is no point in trying to change the vocabulary now.

Example 1.1.2: Topological Spaces

1. Given X being *any* set, then $\mathcal{T}_X = \{\emptyset, X\}$ is a topology. This is the *trivial topology*, or *indiscrete topology*. In terms of localness around points, these points are all considered to be in the one neighborhood. So in here, we can't really consider a *chunk* or *sub-part* of our geometric object, we only ever consider the whole: there are *no* local properties of points besides *whole thing*.

2. Given X to be any set, then $\mathcal{T}_X = P(X)$ is a topology. This is called the *discrete topology*. In terms of localness, we see every point as its own thing: the local property is so precise the smallest local level are points themselves. Since the union of points is also in the topology, every possible subset of X is also in the topology. It's like we're looking through every possible level at once, like we can see the quantum level, the atomic level, the chemical level, the biological level, and the empirical level all at once.

3. Given X to be any set, then $\mathcal{T}_X = \left\{Y \subseteq X \middle| Y = \emptyset \text{ or } |X \setminus Y| < \infty\right\}$. This set is called the *co-finite topology* or *finite-compliment topology*.

   1.1.3 Note Why not just say that U is open if it's finite? It's because the arbitrary union of U's would no longer be finite. In this compliment world, unions will *shrink* $(X \setminus (U_1 \cup U_2))$, making sure it's always finite, and intersection are *finite*, insuring our topology.

4. (If you have done some Analysis) Take $n \geq 1$, $X = \mathbb{R}^n$. Then

   $$\mathcal{T}_X = \{\text{Euclidean open sets in } \mathbb{R}^n\}$$

   Notationally written as $(\mathbb{R}^n)_{std}$, $(\mathbb{R}^n)_{euc}$, or just $\mathbb{R}^n$ if one abuses notation

> **1.1.4 Note** If you took analysis, when you first saw this topology, you probably saw it defined it with open balls. However, there is no metric here to produce the definition of an open ball. We *can* define a metric on $\mathbb{R}^n$, and it can define this very topology; that will be addressed once we reach metric spaces (section 2.5). For now, you only need to think about these as a collection of sets

5. The set $\mathbb{C}$ with all "open circles", similarly to $\mathbb{R}^2$. These two will actually be topologically equivalent, hence mathematically showing how these two are geometrically the same (In particular, any topological property $\mathbb{R}^2$ has, $\mathbb{C}$ has[a]). My current intuition on differentiating these two are *algebraic*: $\mathbb{R}^2$ and $\mathbb{C}$ are different because we put a different multiplication structure on them. We will make this intuition rigorous soon (definition 1.2.13).

6. Let X be any set, if $x_0 \in X$, then $\mathcal{T}_X = \{Y \subseteq X \big| Y = \emptyset \text{ or } x_i \in Y\}$. This topology is called the *particular point topology*

7. Let X be any set, with $<$ as be a [total relation] relation on X (so $\forall a, b \in X, a \neq b$, either $a < b$ or $b < a$ and if $a < b$, $b < c$, then $a < c$). With this, define
   (a) $(a, b) \coloneqq \{c \in x : a < c,\ c < b\}$
   (b) $(-\infty, a) \coloneqq \{c \in X : c < a\}$
   (c) $(a, \infty) \coloneqq \{c \in X : a < c\}$

   Then $\mathcal{T}_X$ consists of all *unions* of the above sets for all choices of $a, b \in X$, together with X itself. $\mathcal{T}_X$ is called the "*order topology*"

8. here is an image of all topologies on 3 points

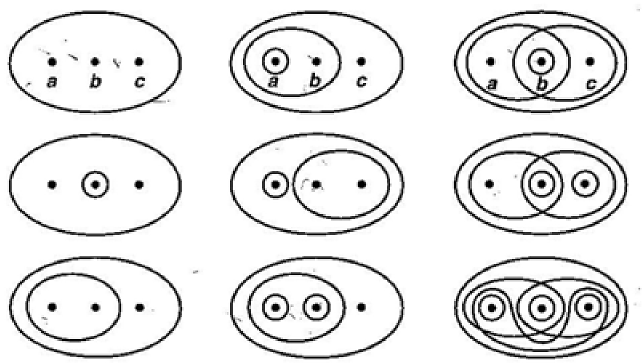

Figure 1.1: from Munkres p.76

[a]In a class on geometry, we shall see that $\mathbb{C}$ has some interesting differential structures that can be put on it that make it have a "different" geometry than $\mathbb{R}^2$, which would be explored in a class on Riemann surfaces or Simplectic Geometry

1.1.5 Note As an important technically, it *is* possible that an arbitrary of intersection of open sets are open, but it cannot be the case! For example $\bigcap_{i\in\mathbb{N}}(-i,1) = (-1,1)$, which is clearly open in the topology of example 6. This is a cheeky example, but there do exist more interesting examples when we get to *product spaces* (section 2.2).

To make this notion of local property more succinct, let's take $\mathbb{R}$ with the standard topology and $x \in \mathbb{R}$. Then we know that $(x-1, x+1)$ contains x, that is x is contained in the open set $(x-1, x+1)$. Topologists would usually say that $(x-1, x+1)$ *is the neighborhood of* x. However, there doesn't need to be a unique neighborhood for a point. The point x actually has uncountably many neighborhoods, since $(x-\epsilon, x+\epsilon)$ for all $\epsilon \in \mathbb{R}_{>0}$ is a valid neighborhood. So when thinking about localness, about *open neighborhoods* in topology, think of it as the fundamental starting point for what a topology does. This doesn't mean one should ignore points, for many times the underlying space will affect the topology too, but what distinguishes topologies are the open sets, the neighborhoods.

Example 1.1.6: Open neighborhoods on a Sphere

To understand this localness even more, let's take a sphere for a moment and consider some topologies on it. Taking the trivial topology on the sphere will make you *only* look at the entire sphere as a thing (you don't see points or subsets, you see the thing which results from all of them). However, taking the discrete topology is like looking at *all* possible subsets as local properties. So, you can think of this as the two extremes of "level of precision" with which we want to look at the local properties of the sphere. Any other topology on the sphere is looking at different *levels* of topologies, and different *types* of topologies will be different "types" of local ways of looking at the sphere.

For example, if we embed the sphere into $\mathbb{R}^3$ centered around 0, then we can define a topology on the sphere by taking all slices given by the xy-plane as the plane moves up the z-axis, and taking all possible unions (and finite intersections). Another topology we can put on the sphere is the topology given by intersecting all open sets of $\mathbb{R}^3$ with the sphere.

Also notice that open sets can be arbitrary union and remain open, while closed sets must at most be in finite unions to remain closed. This comes down to if you let "both" infinities exist (that is, both can be arbitrary, or even intersection of infinitely countable sets) then a lot of flexibility will pop up adding more sets than is useful. For example $(\frac{-1}{n}, \frac{1}{n})$ where $n \in \mathbb{N}$ is a collection of open sets in Euclidean $\mathbb{R}$. The intersection of all of them is $\{0\}$. We can do this around any point, and have every point be open sets. But then the arbitrary union of open sets is open, and since every one-point set is open, *every* possible combination of point is open. Thus, the topology of Euclidean $\mathbb{R}$ would just be $\mathcal{P}(R)$, that is the power set of $\mathbb{R}$. Because of this, a lot of topologies which would be distinguishable with different local properties and interesting would become identical. So, by restricting to "finiteness" for intersection, we essentially get rid of some wiggle room and force constraints. This is actually a very good example to show how bringing in infinities (of different types of infinities even, since we can have an *arbitrary* union, not just countable) will add flexibility to what you're dealing with – sometimes too much like what we just showed. This course will also make you gain an intuition of how infinities add a certain amount of flexibility in the right circumstances.

If you were curious, there is a domain in mathematics in which there are infinite union and intersection of sets. The domain is called *measure theory*, and the space with this property is called a σ-algebra. However, only *countable* unions and intersections are permitted. These spaces are the foundation for much of real analysis and defining important concepts like the [Lebesgue] integral.

With some notion of topologies, we can quickly go over the most basic relation between topologies: whether one contains the other:

Definition 1.1.7: Finer And Coarser

Let X be a set, and $(X, \mathcal{T}_1), (X, \mathcal{T}_2)$ be two topologies. Then

1. $(X, \mathcal{T}_1)$ is *finer* than $(X, \mathcal{T}_2)$ if $T_1 \supseteq T_2$
2. $(X, \mathcal{T}_1)$ is *coarser* than $(X, \mathcal{T}_2)$ if $T_1 \subseteq T_2$

Example 1.1.8: Finer and Coarser Topologies

1. The trivial topology is coarser than all other topologies on its respective X
2. The discrete topology is finer than all other topologies on its respective X
3. Take $\mathbb{R}$ generated by open balls and $\mathbb{R}$ generated by open squares. Then both of these topologies are finer *and* coarser than each other. This means that they contain each other, and so define the same "localness".

Note that two topologies need not be *comparable* with another, i.e., neither has to be finer nor coarser than the other.

Example 1.1.9: Incomparable Topologies

If we have $X = \{a, b, c\}$, and $\mathcal{T}_1 = \{\{a\}, \{b, c\}, X, \emptyset\}$, as well as $\mathcal{T}_2 = \{\{a, b\}, \{c\}, \emptyset, X\}$. Then these two are both topologies, However, $\mathcal{T}_1 \not\subseteq \mathcal{T}_2$ and $\mathcal{T}_1 \not\supseteq \mathcal{T}_2$

This next example is a bit of a hard, but worth taking the time to pour over. The topologies introduced in this example will come back often for counter-examples:

Example 1.1.10: Incomparable Topologies

Let $\mathbb{R}$ be the underlying set, and take the set generated by intervals $[a, b)$ to be the topology. What it means to be generated I'll come back to, but for now just think about taking all possible sets of the form $[a, b)$ and taking all possible unions of them. This topology is called the **lower-limit topology**, and is usually labeled $\mathbb{R}_l$. This topology is *finer* than the standard topology since every open interval can be written as a countably finite union of half-open intervals. Furthermore, the standard topology is *coarser* than $\mathbb{R}_l$, since it would require an arbitrary intersection of elements of the standard topology to create an element $[a, b)$

Now consider the following topology: Take all open intervals $A = \{(a, b) | a < b\}$, but additionally, take $K = \left\{\frac{1}{n} \middle| n \in \mathbb{N}\right\}$ and take all sets of the form $B = (a, b) \setminus K$. Then $A \cup B$ generates a topology. This topology is called the K**-topology**, and is usually denoted $\mathbb{R}_k$. By construction, this topology is *strictly finer* than he standard topology. Since A is the standard topology, A we have all of $\mathcal{T}_{\mathbb{R}}$. We also have sets of the form $(0, 1) \setminus K$, however, the fact that $0 \notin K$ will make the set $(-1, 1) \setminus K$ not constructible from open sets of $\mathcal{T}_{\mathbb{R}}$; around the point 0, there does not exist an open interval! So, $\mathbb{R}_K$ is strictly finer than $\mathbb{R}_{\text{euc}}$

Now, though the $\mathcal{T}_l$ and $\mathcal{T}_K$ are both finer than $\mathcal{T}_{euc}$, they are in fact *incomparable*. That is, one is not finer or coarser than the other! The set $[0, 1) \in \mathcal{T}_l$ is not in the K-topology: $[0, 1) \notin \mathcal{T}_K$, since again, it would require an infinite union of elements, and the sets of the form $(a, b) \setminus K$ have even less elements from the sets of the form (a, b) meaning they won't contribute in a meaningful way. Conversely, $(-1, 1) \setminus K \in \mathcal{T}_K$, but $(-1, 1) \setminus K \notin \mathcal{T}_l$. Since the closed part is on the *left*, it will follow the same reason as it did for the Euclidean topology! Note that if the closed part was on the *right*, the problem would have been avoided!

Note that this what was also meant earlier when I said there are topologies with different local properties! Not every topology defines the same idea of localness, of open neighborhoods, and you can even have topologies that define incomparable idea of openness.

1.2 Continuity

Continuous maps used to be that you can't lift your pen off your paper. Then it became the epsilon-delta definition, to deal with higher-dimensional functions $f : \mathbb{R}^n \to \mathbb{R}^m$. We now need a more abstract definition to account for the generalizations.

If you know algebra, you can think of continuous functions as the homomorphisms of topology. Homomorphisms tell you that the space you are mapping to will preserve some structure of the space you mapped from. This same criterion applies in topology, a continuous map between two set X, Y with different topologies $\mathcal{T}_X, \mathcal{T}_Y$ mean that $\mathcal{T}_Y$ has some similar "localness" properties as $\mathcal{T}_X$ that the continuous map brings over.

Definition 1.2.1: Continuous Functions

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. Then $f : X \to Y$ is said to be *continuous* if

$$\forall U \in \mathcal{T}_Y,\ f^{-1}(U) \in \mathcal{T}_X$$

1.2.2 Note Don't forget that functions map *points* from the underlying set, not open sets in the topology. The definition of continuous functions is dealing with sets, but don't let it confuse you when you're solving problems that there are points still involved.

Example 1.2.3: Continuous Functions

1. You can think of most functions in 1st year calculus: for example $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, or $f(x) = \sin(x)$, and so forth. Note that $f(x) = \log(x)$ is only continuous if the domain is positive [a]

2. Take the domain and co-domain to be $\mathbb{R} \times \mathbb{R}$. In the domain, the domain, the open sets are $(a, b) \times \mathbb{R}$, that is, the first position is the Euclidean domain, while the second is the entire line. The co-domain is the inverse: $\mathbb{R} \times (a, b)$. These two topologies are *incomparable*, however

$$f(x, y) = (0, x)$$

Is an open set. If we take any open set in the co-domain $\mathbb{R} \times (a, b)$, then

$$f^{-1}(\mathbb{R}, (a, b)) = (a, b) \times \mathbb{R}$$

which we know is open since that is a set in the topology of the domain. Since the open set in the co-domain was arbitrary, that means every open set has an open pre-image, meaning f is continuous.

3. A constant map $f(x) = y$ for some fixed y is *always* continuous function. If V is an open set in Y such that $y \in U$, then $f^{-1}(V) = X$. If $y \notin V$ then $f^{-1}(V) = \emptyset$. Both those sets are open sets, and thus, f is continuous.

[a] Or, if you already took complex analysis, $f : \mathbb{C} - \{0\} \to \mathbb{C}$

The question might be asked "why not the image of open sets is open"? This comes down to being okay with not preserve the entire topology of the *domain*, but miss some elements in the co-domain, and instead preserve the entire topology of the *co-domain* and miss some elements in the domain. This way, when working with the image of a continuous function, we know we can think of any open set there having an analogue in the domain (that is, I can think of $V \in \mathcal{T}_Y$ and know there is a $U \in \mathcal{T}_X$ such that $f^{-1}(V) = U$), instead of needing to be careful that there is an open set $U \subseteq X$ such that $f(U) = V$. The main reason prioritizing the co-domain is that we are usually trying to figure some information about the co-domain by mapping to it. Since if we have a function, $f : X \to Y$ it must be that $f^{-1}(Y) = X$, but it's not necessarily the case that $f(X) = Y$ (continuous functions need not be surjective), it's more natural to try to study topologies on spaces by putting them in the co-domain.

We might still want to give a name to maps whose image of open sets are open:

Definition 1.2.4: Open Maps

Let $f : X \to Y$ be a map such that if $U \subseteq X$ is an open set, then $f(U) \subseteq Y$ is an open set. We call f an *open map*.

At first glance, one might not think of an example on open map which is not continuous, so here is one to keep in mind

Example 1.2.5: Continuous but not open

Take

$$f : \mathbb{R} \to \mathbb{R} \qquad f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Since $(a, b) \in \mathbb{R}$, and if $a > 0$ then $f^{-1}((a, b)) = (a, b)$. If $a \leq 0$, then $f^{-1}((a, b)) = (-\infty, b)$. Since both pre-images are are open (the set $(-\infty, b) = \bigcup_{n \in \mathbb{N}} (-n, b)$) the map is continuous. However, The set $(-1, 1)$ maps to $[0, 1)$, which is *not* open in $\mathbb{R}$.

The other way is also not necessarily true:

Example 1.2.6: Open, but not Continuous

Let $f : \mathbb{R} \to \mathbb{R}$ $f(x) = x$, with the domain having the Euclidean metric, and co-domain having the discrete metric. Then it's clear that the image of every open set is open, but the pre-image of every set need not be open (like $\{0\}$).

Open maps are useful in terms of the continuity of f^{-1}, in particular, if f is an open map, then f^{-1} (when appropriately restricted/defined) is continuous.

Proposition 1.2.7: Swapping The Domain And Co-Domain

Let $f : X \to Y$ be a continuous function. Then X can have a finer topology, or Y can have a coarser topology

***Proof*:**

This is a simple carrying the definition question. The domain can be replaced with a *finer* topology (since the pre-images of open sets will remain there), while the co-domain can be replaced wit a *coarser* topology (since there are less open sets to take to the pre-image).

However, this does not imply that the co-domain or the domain are finer, coarser, or even comparable to each other:

Example 1.2.8: Continuous Function Between *Incomparable* Topologies

Take

$$T_1 = \{\emptyset, [a,b], [a,b]^c, \mathbb{R}\}, \qquad T_2 = \{\text{Euclidean topology}\}$$

And the function which takes $f([a,b]) = 0$, and $f([a,b]^c) = 1$. This function is in fact continuous. Take any open set $U \in T_2$. Then there are 4 cases:

1. if $0, 1$ are not in U, then $f^{-1}(U) = \emptyset$
2. if $0 \in U$ but $1 \notin U$, then $f^{-1}(U) = [a,b]$
3. if $1 \in U$ but $0 \notin U$, then $f^{-1}(U) = [a,b]^c$
4. if $1 \in U$ *and* $0 \in U$, then $f^{-1}(U) = \mathbb{R}$

This is true for any arbitrary open set $U \subseteq T_2$, and so f is continuous! Even though T_1 and T_2 are *not* comparable, this function is *still* continuous. This means that he "crux's" of what makes a function continuous is the function itself.

Thinking about preserving types of localness might feel strange here. This comes down to the map f being very particular. It maps entire parts of the domain to either 0 or 1, and 0 or 1 are about of uncountably many open sets. This shows how much the continuous map is important in defining the relation between the two spaces (in a sense similar to how algebraic homomorphisms are important in defining relation between seemingly "incomparable" algebraic structures)

Making a non-trivial continuous function between two bizarre space might make us want to know the limits of what can be done between topologies. This answer really comes down to the topologies in question, but here is an example:

Example 1.2.9: Limits of Continuous Functions Between Spaces
Take

$$\mathcal{T}_2 = \{\emptyset, (-\infty, 2) \cup (2, \infty), (-1, 1), (-\infty, 2) \cup (2, \infty) \cup (-1, 1), \mathbb{R}\}$$

Take $f : \mathbb{R} \to \mathbb{R}$ where the domain has the $\mathcal{T}_2$ topology, while the co-domain has the Euclidean topology, and f is *not* the constant function (for that will always define the trivial topology for the image). Since f is not the constant function, it must map to at least two values. Open sets containing those values must have an open pre-image. Let's say $f((-1, 1)) = 0$. So far, an open set in the co-domain containing 0 would be open in the pre-image. Since f is not a constant map, ew must map the other points to another place in the co-domain. Let's say $f(B) = 1$. We want to figure out what B works so that f is continuous.

A set V in the co-domain containing 1 would need to have an open pre-image. If it has 0 and 1, it's still open because the union of open sets is open. If we had some points $x \in (-1, 1)$ mapping to something else, say -1, then the pre-image of 0 wouldn't be open, but the pre-image of any open set must be open. Simiarly for b. However, now we must ask ourselves where do we map, say, 1.5? If we map it to 0 or 1, then it would be part of the pre-image of U of V, and that would make it not-open. Conversely, if we make it, say, -2 (i.e. some arbitrary un-picked number), then $(-3, 3)$ which is open in $\mathbb{R}$ would not be open in $\mathcal{T}_2$, since the pre-image would be $\{-2\} \cup (-1, 1)$, which is not open. Therefore, there is nowhere for 1.5 to map to wihtout breaking continuity. THerefore, in this instance, the "type" of topology we have actually limited us what type of topology we have!!

This mapping problem can be fixed if we have the open sets of our topology cover all of $\mathbb{R}$ (without considering $\mathbb{R}$ to make the covering work, because then every topology "trivially" covers the underlying set [a]). In our previous example, let's replace $(-1, 1)$ with $[-2, 2]$. Map that to 0, the other set to 1. This is now a continuous function.

[a] I'm not using the idea of open covering, or of a covering map here!

Though there is still a limitation to continuous functions. A continuous function in that scenario is still limited to having a finite image. Take any domain that has a finite topology. Since the pre-image has to be open, if one of those open sets in the domain "split" and some map to one point and some map to another, then taking an open set containing one of those points will have a non-open pre-image. Therefore, continuous maps rely heavily on what topology from it' underlying set it takes. It can always take the trivial topology and so be a constant function, but it might not be able to take other topologies depending on the domain and co-domain.

We concentrated a lot on the domain having a non-standard topology rather than the co-domain. However, both play their own role in a function being continuous. The co-domain plays the role of providing the sets whose pre-image have to be open, so it can determine the density of sets in the domain. However, that is holy dependent on the function itself, which as we've seen can have a finite topology on as the domain, and so the function will only map to finitely many elements, which in our case is still possible even if the topologies are incomparable.

Another example of the relation between domain and co-domain with continuous functions is if we take the same underlying set X, but have the discrete topology in the domain. Then for whatever topology we have in the co-domain, the pre-image of that open set is a set, which is open in the domain since the topology of the domain compromises the power-set. However, if the co-domain has the discrete topology, then there are *too many* open sets in the co-domain for most functions to be continuous. Since every arbitrary set in the co-domain is open, every pre-image also has to be open.

It is in fact interesting to take a quick look at topologies with discrete co-domains, cause in a sense they show continuous functions which work "universally", since if it's continuous in the discrete topology, then you can take coarser topologies in the co-domain, and it will still be continuous, and since every topology is coarser than the discrete topology , then what we're finding are the function that are "universally" continuous given any topology on the co-domain (given the same underlying set X for the domain and co-domain). For those interested, this is an example of a *universal property* in **Top**, the category of topological spaces.

With these ideas, we can ask if there are easy ways to manipulate continuous functions. We will use these time an time again to construct functions:

Proposition 1.2.10: Rules for constructing continuous functions

These 6 things:

1. **Constant function:** if $f : X \to Y$ maps all of X int othe single poitn $y_0 \in Y$, then f is continuous
2. **inclusion:** If a subsapce A is a subsapce of X, the inclusion $i : A \to X$ is continuous
3. **composition:** if $f : X \to Y$ and $g : Y \to Z$ are continuous, then $g \circ f : X \to Z$ is continuous
4. **Restricting the domain:** if $f : X \to Y$ if continuous, and if A is a subspace of X, then $f|_A : A \to Y$ is continuous with $U \subseteq V$ open if and only if there is a $U \subseteq X$ st $V = U \cap A$.
5. **Restricting or expanding the range:** Let $f : X \to Y$ be continuous. if Z is a subsapce of Y containing the image set $f(X)$, then the function $g : X \to Z$ obtained by restricting the range of f is continuous. If Z is a space having Y as a subsapce then the function $h : X \to Z$ obtaine by expanding the range of f is continuous
6. **local formulation of continuity**: The map $f : X \to Y$ is continuous if X can be written as the union of open sets U_α such that $f|_{U_\alpha}$ is continuous for each U_α.

***Proof*:**
These are all good exercises

1.2.11 Note taking a subspace is continuous, but that doesn't mean that continuous functions from the original space *to* another space will also be continuous from the subspace *to* that other space

example

The way I like to think about that is through this diagram:

$$\begin{tikzcd} A \arrow[r, "i"] \arrow[rr, bend right, "f"'] & X \arrow[r, "f"] & Y \end{tikzcd}$$

In the diagram, f is continuous between X and Y, and i is the inclusion map. With the example we showed earlier, we see how f need not be continuous on the subspace.

Here is one more rule that is so widely used that it's given it's own name:

Lemma 1.2.12: Pasting Lemma

Let $X = A \cup B$, where A and B are closed in X. Let $f : A \to B$ and $g : B \to Y$ be continuous. If $f(x) = g(x)$ for every $x \in A \cap B$, then f and g combine to give a continuous function $h :\to Y$, defined by setting $h(x) = f(x)$ if $x \in A$ and $h(x) = g(x)$ if $x \in B$

***Proof*:**
Good exercise, p.109 of Munkres

With this in mind, let's keep building on our notion of continuity by also adding that every open set maps to an open set. With this added condition, we almost have two identical topological spaces. Such a function is sometimes called *bi-continuous.* If a function is by-continuous, the topologies are basically comparable [1]. If the function is also bijective, then all open sets of the domain map to the co-domain, and vice-versa, meaning the two topologies are basically the same. When a functions is bi-continuous and bijective,we say it's is a *homeomorphism*

Definition 1.2.13: Homeomorphism

If $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ are topological spaces, a map $f : X \to Y$ is a *homeomorphism* if

1. f is bijective
2. $f^{-1}(\mathcal{T}_Y) = \mathcal{T}_X$ (i.e. continuous both ways)

$(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ are *homeomorphic* if there exists a homeomorphism between them

A homeomorphism is the strongest condition you can put between two topological spaces. It is the isomorphism of topologies [2]. You will encounter many other topological properties a space can have, and if a space is homeomorphic with another space, that space will have all the same properties, since we can exactly map one set of open sets onto another.

[1] If they are *actually* comparable I'll have to check later

[2] which is actually the vocabulary used in Category theory: homeomorphisms *are* the isomorphisms of the Top Category

Example 1.2.14: Homeomorphism
$\mathbb{R}^2$ and $\mathbb{C}$ are topologically the same. The map

$$\mathbb{R}^2 \to \mathbb{C} \qquad (x, y) \to x + iy$$

Will define a heomeomorphism.

If $f : X \to Y$ is a *homeomorphism*, then $f^{-1} : Y \to Y$ is also a homeomorphism It can intuitively be said that $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ are "homeomorphic" if there is a bijection between both (X, Y) and $\mathcal{T}_X, \mathcal{T}_Y)$, where this bijection must respect the property in the above definition. This is captured in the next corollary:

Corollary 1.2.15: Identical Topologies

$f : X \to Y$ are homeomorphic, then the two topologies are indetical ($\mathcal{T}_X = \mathcal{T}_Y$)

***Proof*:**
exercise

There is a slightly weaker condition than homeomorphism that is also used a lot. If a function is homeomorphic *onto it's image*, we call that an embedding

Definition 1.2.16: Embedding

Let X, Y be topological spaces. Then if $f : X \to f(X)$ is a homeomorphism, then f is called an *embedding*

Exercise 1.2.1

1. Is there a continuous bijection between $(0, 1)$ and $[0, 1]$?

1.3 Generating topologies (basis)

When you think about the topology on $\mathbb{R}$, you usually don't think about, say, the element $(1, 2) \cup (10, 11)$. This is because what you're is here is actually one for the *basis* of a topology. The idea of a basis is that you can take a smaller subset of open sets which will *generate* the rest of your topology.

Definition 1.3.1: Basis Of Topology

Let X be a set. A set $\beta \subseteq P(X)$ is a basis if

1. $\forall x \in X$, $\exists B_0 \in \beta$ such that $x \in B_0$
2. $\forall B_1, B_2 \in \beta$, $x \in B_1 \cap B_2$, then $\exists B_3 \in B$ such that $x \in B_3 \subset B_1 \cap B_2$

1.3.2 Note Don't confuse the idea of basis from linear algebra where you also need the basis elements to be the *unique* elements that some to any other element. Here, uniqueness is not a thing.

With the property a basis would need, we can ask what topology would be defined with a basis. This is *how* we will define topologies with a basis:

Lemma 1.3.3: Basis Generating A Topology

Let β be a basis. Then

$$\mathcal{T}_\beta \stackrel{\text{def}}{=} \left\{U \subseteq X \middle| \forall x \in U, \exists B_0 \in \beta \text{ such that } x \in B_0 \subseteq U\right\}$$

generates a topology

***Proof*:**
Take $\mathcal{T}_\beta$ as described in the lemma. Then $\emptyset \in \mathcal{T}_\beta$ vacuously. Next, $X \in \mathcal{T}_\beta$ since for every $x \in X$, there exists a basis element β_0 such that $x \in \beta_0 \subseteq X$. In fact, every basis element containing x will satify this condition.

Next, Let's take a collection of elements $U = \{U_\alpha\}_\alpha$. Is their union in $\mathcal{T}_\beta$? Take any $x \in U$. Then $x \in U_\alpha$ for some α. Since $U_\alpha \in \mathcal{T}_\beta$, then we know by definition that there exists a basis element β_x such that $x \in \beta_x \subseteq U_\alpha$. Since this was true for arbitray $x \in U$, then $U \in \mathcal{T}_\beta$, meaning $\mathcal{T}_\beta$ is closed under arbirary unions

Finally, Let's consider a finite number of open sets $\{U_1, ..., U_n\}$. Let's start with when there are two elemetns U_1, U_2, and take come $x \in U_1 \cap U_2$. Then by the definition of this topology, there exists a basis element such that $x \in \beta_1 \subseteq U_1$ and $x \in \beta_2 \subseteq U_2$. So then:

$$x \in \beta_1 \cap \beta_2$$

but then, by the 2nd condition of a basis, there exists a 3rd element such that $x \in \beta_3 \subseteq \beta_1 \cap \beta_2$. Since x was arbitrary, then $U_1 \cap U_2 \in \mathcal{T}_\beta$.

For the case with n elements, we can show this inductively. The case where $n = 1$ is trivial, so let's assume it works for $n - 1$ elements: $U_1 \cap \cdots \cap U_{n-1} \in \mathcal{T}_\beta$. Now add U_n. However, this is *exactly* the same case as we had with two basis elements (since $U_1 \cap \cdots \cap U_{n-1}$ is just another basis elements now!). Thus, we apply the same logic and get $U_1 \cap \cdots \cap U_n \in \mathcal{T}_\beta$, showing that $\mathcal{T}_\beta$ is indee a topology

Since this is *the* way of constructing a topology given a basis, we give this type of topology a name:

Definition 1.3.4: Topology Generated With Respect To β

Let β be a basis. Then we say that

$$\mathcal{T}_\beta \overset{\text{def}}{=} \{U \subseteq X \big| \forall x \in U, \exists B_0 \in \beta \text{ such that } x \in B_0 \subseteq U\}$$

is the topology generated by β

1.3.5 Note It is important to remember that *every* topology has a basis! If $\mathcal{T}_X$ is a topology, then Let $\beta = \mathcal{T}_X$. It is an exercise worth checking to see that β is a basis!

The way I like thinking about a basis is that you need to find a set in which all *unions* will be the topology you want.

Example 1.3.6: Topologies Generated by Basis

1. $\beta = \big\{\{x\} \big| x \in X\big\}$ generates the discrete topology
2. $\beta = \{X\}$ generates the trivial topology
3. $\beta = \{U \times V \mid U, V \subseteq \mathbb{R},\ \text{open}\}$ generates a topology homeomorphic to the standard topology on $\mathbb{R}^2$, and so it *is* the standard topology. More generally,

$$\beta_n = \{U_1 \times \cdots \times U_n \mid \text{open } U_i \subseteq \mathbb{R}, 1 \leq i \leq n\}$$

generates the standard topology on $\mathbb{R}^n$.

> ***Proof*:**
> Exercise! We'll show this after introducing lemma 1.3.9

Another way of generating this topology is by taking open balls (exercise):

$$\beta_n' = \{\|x - a\| < \epsilon \mid \epsilon > 0,\ x, a \in \mathbb{R}^n\}$$

This shows how a space (like $\mathbb{R}^n$) can have *different* bases. In general, there is *no* minimal or smallest basis generating a given topology.

You might ask yourself "why not have a set in which all unions and *intersection* create a topology"? We can define that too. There is no real advantage of using one over the other, it comes down to the amount of structure that is present in one versus the other. In the basis definition, since the intersection of elements is already inside it, then if we are using a basis, we have less to show when proving theorems with sub-basis. However, constructing topologies can be easier with sub-basis.

Definition 1.3.7: Sub-basis

Let X be a set. A set $\beta \subseteq P(X)$ is a basis if

1. $\forall x \in X$, $\exists B_0 \in \beta$ such that $x \in B_0$

Naturally, a basis is a sub-basis, and generating a topology from a sub-basis is essentially identical as generating a topology from a basis, with the added condition of intersecting the sets.

1.3.1 Simplification With Basis

Lemma 1.3.8: smallest possible topology containing β

Let β be a basis. Then $\mathcal{T}_\beta$ is the *smallest* topology on X for which elements of β are open

***Proof*:**
Simple proof; if smaller than contradiction.

If one finds the topology of a space, some proofs become easier to do:

Lemma 1.3.9: Simplification Due To Basis

Let X be a set ,β, β' be bases for X generating $\mathcal{T}, \mathcal{T}'$ respectively. then

$$\mathcal{T} \text{ is finer then } \mathcal{T}' \quad \Leftrightarrow \quad \forall B_0' \in \beta', x \in B_0', \exists B_o \in \beta \text{ st } x \in B_0 \subseteq B_0'$$

***Proof*:**
simple proof; exercise.

We can further use a basis to simplify continuity proofs:

Lemma 1.3.10: Continuity With Basis

Let X, Y be topological spates. Let β be a basis for the topology on Y. Then $f : X \to Y$ is continuous if and only if $\forall B \in \beta,\ f^{-1}(B)$ is open in X

***Proof*:**
expanding the pre-iamge of open sets is open using the basis generating deftinition of topology

If you have seen the Euclidean topology on $\mathbb{R}^n$, notice how this matches the epsilon-delta definition of continuity:

Definition 1.3.11: ϵ-δ Definition of Continuity

A function $f : \mathbb{R}^n \to \mathbb{R}^n$ is said to be continuous if

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ s.t. } \|x - a\| < \delta \Rightarrow \|f(x) - f(a)\| < \epsilon$$

Or equivalently, when generalized, the pre-image of open sets are open.

Parsing this definition, we see that the condition “for all epsilon” can be thought of as “for any basis element in in the image”. We want the pre-image of the basis element to be open. Recall that a set is open if for every point inside the set $x \in U$, there exists an open neighborhood $B_\delta(x)$ such that $B_\delta(x) \subseteq U$; this comes from the property of the basis. The reader should now be able to deduce how this definition matches the definition of continuity given by pre-images of open sets.

Also, we might ask ourselves if we have a topology, can we find a basis? The trivial is that yes, every topology is its own basis, but we want a better criterion for finding it coarser basis:

Lemma 1.3.12: Finding Basis When Given A Topology

Let X be a topological space. Supposer that $\mathcal{C}$ is a collection of open sets of X such that for each open set U of X, and each $x \in U$, there is an elemetn $C \in \mathcal{C}$ such that $x \in C \subseteq U$. Then $\mathcal{C}$ is a basis for the topology of X

***Proof*:**
p.80 of Munkres

Overall, you will use basis when thinking about topologies quite often to simplify the technicalities of proving things like continuous functions.

1.4 Closed sets

We have defined topology with *arbitrary* open sets, but having *finite* intersection. We can actually reverse the two, with *finite* union but *arbitrary* intersection. It turns out that what's key is that *one* of these two operations is finite (as we explained earlier), but it doesn't matter which!

What we'll explore in this section is how these two concepts are actually linked! That is, if you define it one way, you can get the other, and how can we use the properties of one to understand the other (like borders, connectedness, closure, interior, etc.). This will come down to the fact that

$$\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} (A_i)^c$$

This set-theoretic property will mean that we can be very interested in the compliments of open sets. We will give them a name:

Definition 1.4.1: Closed Sets

Let X be a set and $\mathcal{T}_X$ be a topology on X. A set $A \subseteq X$ is closed if it's compliment $A^c = X \setminus A$ is open

1.4.2 Note There is no good reason closed sets were picked over open sets, but they give a different flavour when examining topologies. So it just shows that we can look at topologies with two different sets and get the same thing.

Example 1.4.3: Closed Sets

1. $[a, b]$ is closed in $\mathbb{R}$ (it is the compliment of $(-\infty, a) \cup (b, \infty)$). All one point sets are closed in $\mathbb{R}$ (what is the complimentary open set?)
2. Let X be any topology. Then $\emptyset$ and X is always closed, since they are the compliments of each other and are by definition open. From this, we can conclude that in the trivial topology, every set is closed. Is every set in the discrete topology also closed?
3. Show that the arbitrary intersections of closed sets is closed, and the finite union of closed sets is closed.

The fact that finite unions and arbitrary intersections are closed should give an idea how closed is like being "co-open". You can basically define a topology as follows

Theorem 1.4.4: Thinking Of Topology In Terms Of Closed Sets

Let X be a topological space. Then the following conditions hold:

1. $\emptyset$ and X are closed
2. arbitrary intersections of closed set are closed
3. finite unions of closed sets are closed

There turns out to be no real advantage in using one set of axioms over the other. We could've had finite union and arbitrary intersection. However, one of those must be chosen (or else you can have $\bigcap_n (\frac{-1}{n}, \frac{1}{n}) = \{0\}$ which will start breaking how we think of basis, and a lot of other things) so by convention we choose arbitrary union and finite intersection.

As an example of how closed sets can be used basically interchangeably with open sets, here continuity with closed sets:

Theorem 1.4.5: Continuity And Closed Sets

The function $f : X \to Y$ is continuous if and only if for all $V \subseteq Y$ closed, $f^{-1}(V)$ is closed

***Proof*:**
Using some set-theory, this is simply the following line:

$$f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$$

prove this equality if it is unclear.

As an exercise, show that the notion of basis can be re-formulated in terms of closed sets. Since we have defined closed sets, we can define closed maps:

Definition 1.4.6: Closed Maps

Given topological spaces X, Y. Then $f : X \to Y$ is an closed map if the image of closed sets are closed

Note that the conditions of being continuous, open, or closed do not imply each other:

Example 1.4.7: Continuous, Open and Closed Maps

In example 1.2.5 we showed a map that is continuous but not open, and in example 1.2.6 we showed a map that is open but not continuous.

1. Show that example 1.2.5 is continuous, not open, and *closed*. Another good example of such a function is $f : \mathbb{R} \to \mathbb{R}$, $f(x) = c$ for some $c \in \mathbb{R}$.
2. Show that example 1.2.6 is not continuous, open, and *closed*.
3. Show that $f : \mathbb{R} \to \mathbb{R}$ where $f(x) = c$ for some $c \in \mathbb{R}$ is continuous, closed, but not open.
4. Give an example of a map that is not continuous, not open, and not closed. Give an example of a map that is continuous, open, and closed.

Here is an example of map that is continuous, open, but not closed: Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x, y) = x$, where $\mathbb{R}^2$ has the topology given by the basis:

$$\{S \times T \mid S \text{ open and } \subseteq \mathbb{R}, T \text{ open and } \subseteq \mathbb{R}\}$$

and $\mathbb{R}$ has the usual topology. Show this map is continuous and open, but not closed.

This is not all possible combinations of continuity, openness, and closedness. What was omitted, and can you find maps that match those conditions?

As we saw, $\emptyset$ and X are always closed and open for a topological space X. If a set $A \subseteq X$ has this property, we shall give it a name:

Definition 1.4.8: Clopen

In a topological space X, $A \subseteq X$ is *clopen* if A is open *and* closed

We will get back to these types of sets extensively when dealing with connectedness – it will become a defining feature.

1.4.1 Closure

With some basic idea of closed sets, we can take advantage of the fact that closed sets are closed under arbitrary intersections to formulate the next definition:

Definition 1.4.9: Closure

Let X be a topological space. If $A \subseteq X$, the *closure* $\overline{A}$ is the smallest closed subset of X containing X:

$$\overline{A} = \bigcap_{A \subseteq V,\ V \text{ is closed}} V$$

The set is non-empty if A is non-empty. Also note that for any closed set, trivially, $A = \overline{A}$

Note that

1. A is an arbitrary set, it need not be open
2. there is not necessarily a smallest open ball that contains that set. That's because openness is not closed under arbitrary intersection. However, because open sets have arbitrary union, we can define a dual notion for open sets

Similarly, since the arbitrary union of open sets is open, we have the following definition:

Definition 1.4.10: Interior

The *interior* of a open set A, labeled $\text{int}(A)$ is the largest open set contained in A:

$$\text{int}(A) = \bigcup_{A \supset V,\ V \text{ is open}}$$

Example 1.4.11: Closure

1. With this new way of looking at closure, try proving $\overline{(0,1)} = [0,1]$.
2. Similarly, try proving $\text{int}([0,1]) = (0,1)$.

With the idea of closure, we can bring up some logical consequences of the definition:

Theorem 1.4.12: Neighborhood Intersection Property

$x \in \overline{A} \Leftrightarrow$ every open set U containing x satisfies $U \cap A \neq \emptyset$

***Proof*:**
Very good exercise! For the inverse, contradiction is much easier

I say border-like property, because this idea gets us to the heart of what closure does. Notice that $\overline{A}$ is closed, and yet the theorem is centered around the *open* set A. So to say that for any *open* set U intersecting the *open* set A is non-empty gives information about the *closed* set $\overline{A}$ is actually quite interesting, especially because open sets are supposed to be some more granular way of looking at a geometric object than the entire object (like the trivial topology) but more coarse then looking at every point (like the discrete topology). This talk of open sets relating to this type of closed set will lead us to talking about the tiny difference between $\overline{A}$ and A in certain topologies with appropriate A. What I mean by this is that this condition is in a sense "trivial" for elements which are inside A since if $x \in A$ and $x \in U$ then there intersection is clearly non-empty. However, $\overline{A}$ is contains A, and so can be bigger. But it saying that if there is such a point, for any open set that contains this point, it *must* intersect with the potentially smaller A, meaning it must be, in a sense, arbitrarily close to A, but $\overline{A}$ is still possibly bigger than A. I keep saying things like "possibly" and "can be" because it is possible that A is open and $A = \overline{A}$ in certain topologies or certain sets in topologies. The interesting things happen when this is not the case, and in that case, $\overline{A}$ will be strictly bigger, *and* closed, so the compliment of an open set.

However, there are limits to thinking about that property as a border in the intuitive sense. For example, $\overline{\mathbb{R}} = \mathbb{R}$ (since $\mathbb{R}$ is closed) and so $\mathbb{R}$, which doesn't have a border as is intuitive, is still it's own closure. Another example would be the following topology

Example 1.4.13: Huge Closure

Let

$$T_{\text{large closure}} = \{\emptyset, (-1,1), (-\infty,2) \cup (2,\infty), (-\infty,2) \cup (-1,1) \cup (2,\infty), \mathbb{R}\}$$

In this topology, the closed sets are $[-2,2]$, $(-\infty,1] \cup [1,\infty)$, and $\mathbb{R}$. Therefore,

$$\overline{(0,1)} = [-2,2]$$

that is, the border is quite substantial! So a more general intuition is that we look at open sets near the *edge* of A. In our $T_{\text{large closure}}$ example, there isn't much locally going on locally around the borders, for the very construction of our set is particular in only caring for open sets that fit very particular properties so that the closure of $(-1,1)$ is purposely large.

Note that whatever "extra" we add to A when taking $\overline{A}$ is closed if A is open. Since $\overline{A}$ is closed then $X \setminus \overline{A}$ is open, and $(X \setminus \overline{A}) \cup A$ is open. Therefore $\left((X \setminus \overline{A}) \cup A)\right)^c$, which is equal to $\overline{A} \setminus A$, is closed.

To get more of an idea of what is meant by how open sets "look" like on the "border", notice that the "border" did not have any open set. This is in fact a result of the definition. Take for example:

$$T_{\text{weird example}} = \langle\{(0,1), (2,3), (-\infty,4) \cup (4,\infty), \mathbb{R}\}\rangle$$

which would be:

$$T_{\text{weird example}} = \{\emptyset, (0,1), (2,3), (-\infty,4) \cup (4,\infty), (0,1) \cup (2,3), (0,1) \cup (2,3) \cup (-\infty,4) \cup (4,\infty), \tag{1.1}$$

$$(0,1) \cup (-\infty,4) \cup (4,\infty), (2,3) \cup (-\infty,4) \cup (4,\infty), \mathbb{R}\} \tag{1.2}$$

In this example In this case, the two closed sets of interets are

$$[-4,4], [-4,2] \cup [3,4]$$

And so

$$\overline{(0,1)} = [-4,2]$$

When I was trying to construct these examples, you might've noticed that I had to make sure that *no* open sets can be inside the border. This is another way of thinking about the theroem from above:

Corollary 1.4.14: Closure's border and open sets

Borders cannot have open sets inside them

***Proof*:**
Notice that the open set is not inside that border. Let's say as a hypothetical example that is was in there. then:

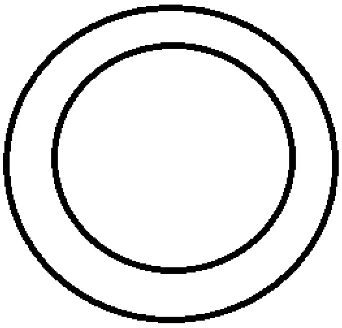

Figure 1.2

if there was an open inside it then visually we'd get

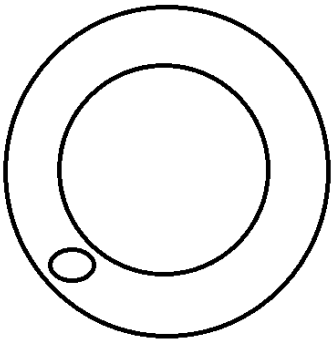

Figure 1.3

However, then we'd have all that's in black here:

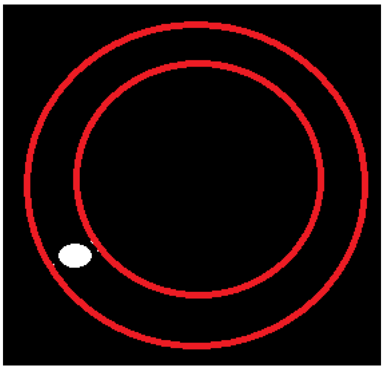

Figure 1.4

to be the closed, but then the new closure would be all that is black here:

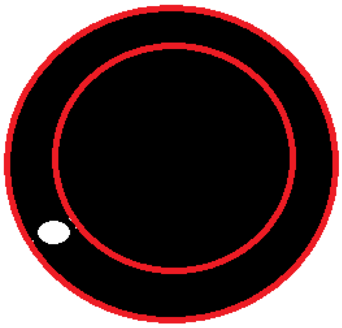

Figure 1.5

Therefore, another way of thinking about the relation between border and open sets is that the closure of a set is the smallest amount of room added so that an open set cannot fit in the border.

Finally, we leave off with the following useful property of continuous functions:

Theorem 1.4.15: Continuity and Closure

The function $f : X \to Y$ be continuous. Then $\forall A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$

***Proof*:**
good exercise

1.4.2 Limit Points

Limit points are a closely related concept. In this section, we shall go over them in generality, and in section 1.5.1 these concepts shall be related to sequences:

Definition 1.4.16: Limit Points

Let X be a topological space, and $A \subseteq X$. We say that $x \in X$ is a *limit point* of A if and only if $\forall U \subseteq X$ open such that $x \in U$, then $U \cap (A \setminus \{x\}) \neq \emptyset$. We'll denote $\lim(A)$ to be the set of limit points of A.

Note that it is $x \in X$, not $x \in A$. The limit point does not need to be in A, that is, there is the notion of a set being able to be *arbitrarly close* to it's edge, but not have it.

Limit points are similar to closure, in that they both capture some idea of "the edge", as we've discussed the intuition behind the definition of closure. The difference can be illustrated by looking at examples which contrast it to closure of sets:

Example 1.4.17: Limits Points

let $X = \mathbb{R}$

1. $A_1 = [0, 1]$, then the limit points are $[0, 1]$
2. $A_2 = (0, 1)$, then the limit points are $[0, 1]$
3. $A_3 = \{0\}$, then there are no limit points. This differs from closure since the closure of A_3 here would be A_3.
4. $A_4 = \left\{\frac{1}{n}, n \in \mathbb{Z}_+\right\}$, then the limit point is only 0. If this were the closure, then this would instead be $A_4 \cup \{0\}$
5. Let $A = \{x\}$ for $x \in \mathbb{R}$ where $\mathbb{R}$ has the trivial topology. Then (exercise)
$$\lim(A) = \mathbb{R}$$
In fact, for any non-empty $A \subseteq \mathbb{R}$,
$$\lim(A) = \mathbb{R}$$
If $\mathbb{R}$ had the trivial topology and $A \subseteq \mathbb{R}$ is an arbitrary subset, what is $\lim(A)$?

If you are already familiar with sequences, then limit points can be thought of as the limit (in the way it was defined in analysis) of sequences that *do not* include the limit point as part of the sequence; consequently constant sequences are eliminated.

That's why 1 in the last set does not converge in A_4. It should also be noted that we're more interested when a point is *not* part of A, because we're interested in seeing if there are points *around our point* x which will get "closer" to x, so in a sense limit points capture some nature of the neighborhood around our set A. This idea of capturing what's going on in the neighborhood of a point is key to topology, which is why this definition is worth intuitively understanding

We can see how closure relates to this. From the point of view of limit points, closure also lets in constant sequences. From the point of view of closure, limit points focus on edges which are in a sense "surrounded" by other points. The following captures the relation between the two concepts:

Theorem 1.4.18: Relating $\overline{A}$ With Lim(A)

$$\overline{A} = A \cup \lim(A)$$

The definition of limit points will often include the interior of many sets. The following focuses only on point that are on the edge:

Definition 1.4.19: Border of Set

$$\partial(A) = \overline{A} \setminus \operatorname{int}(A) = X \setminus (\operatorname{int}(A) \cup (\operatorname{int}(X \setminus A))) = \overline{A} \cap \overline{X \setminus A}$$

1.5 Hausdorff

We already mentioned Hausdorff spaces a couple of times. The idea of Hausdorff is that around any two arbitrary points, there exists a neighbourhood which contains each point respectively and doesn't intersect. This gives some forced "localness" properties to our space, especially if there are uncountably many points. In order to properly understand them, introducing them with sequences might make it intuitive on why we will defined it the way it is. How sequences act in Hausdroff spaces are in fact the motivation for the creator of the axiom which creates Hausdorff spaces.

We start with the definition of Hausdroff space:

Definition 1.5.1: Hausdorff Space

A topological space X is called *Hausdorff* if for all pairs of distinct points $x_1, x_2 \in X$, there exists open sets U_1, U_2 such that $x_1 \in U_1, x_2 \in U_2$ and $U_1 \cap U_2 = \emptyset$.

Figure 1.6: visualization

This would mean that we can think of spaces which are fine enough to have some "space" between points.

Example 1.5.2: Hausdorff Spaces
These are the basic examples

1. $\mathbb{R}$ is clearly a Hausdorff space
2. the discrete topology is Hausdorff
3. the trivial topology is *not* Hausdroff for $|X| \geq 2$
4. $\mathbb{R}^n$ is Hausdorff
5. $S^1 \subset \mathbb{R}^2$ is Hausdorff (the unit circle)
6. If $\mathcal{T}$ is a Hausdorff space, and $\mathcal{T}'$ is finer than $\mathcal{T}$, then $\mathcal{T}'$ is also a Hausdorff space.

Example 1.5.3: Not Hausdorff
An example of how a space is not Hausdorff, how not all points have "space" between them, is

$$\mathcal{T}_{\mathbb{N}} = \{(-n, n) \big| n \in \mathbb{N}\}$$

In this topology, the points -0.5 and 0.5 is not separated by an open set. Even if we do:

$$\mathcal{T}_{\mathbb{R}} = \{(-n, n) \big| n \in \mathbb{R} - \{0\}\}$$

It still doesn't work for, say $2, 4 \in \mathbb{R}$.

1.5.4 Note As an exercise to show how connected closed and open sets are, we can do the following:

Definition 1.5.5: Hausdorff In Terms Of Closed Sets

$\forall x_1 \neq x_2$, there exists closed $M_1, M_2 \subseteq X$ such that $(X \setminus M_1) \cap (X \setminus M_2) = \emptyset$ then

1. $(X \setminus M_1) \cap (X \setminus M - 2) = \emptyset$
2. $x_1 \in X \setminus M_1$
3. $x_2 \in X \setminus M_1$

Which is equivalently

1. $M_1 \cup M_2 = X$
2. $x_1 \notin M_1$
3. $x_2 \notin M_2$

Lemma 1.5.6: Finite Sets In Hausdroff Spaces

If X is Hausdorff, and $A \subset X$ is finite, then A is closed

***Proof*:**
It is enough to show singletons are closed, since closed sets are closed under finite unions. Let $x \in X$. Take all $g \in X$ such that $g \neq x$. then there exists $U_g \subset X$ which is open such that $g \in U_g$ and $x \notin U_g$ by Hausdorff property. Then

$$\bigcup_{g \neq x} U_g = X \setminus \{x\}$$

which is open, and hence is complement is closed, and hence $\{x\}$ is closed, and hence any finite set is closed

1.5.7 Note If a topology has finite sets which are closed, this does not mean that the topology is Hausdorff. The finite-compliment topology has closed sets, but it is not Hausdorff (take $\mathbb{R}$ with this topology, then any convergent sequence must converge to an open set which has all but finitely many elements). The axiom which says that finite sets are closed is called the T_1 axiom.

1.5.1 Sequences

Definition 1.5.8: Sequence

Let X be a set. Then a *sequence* is a function $f : \mathbb{N}_{>0} \to X$ that assigns to each positive natural number an element of X. The element $f(n)$ is often denoted x_n, where x is often replaced with some letter of the alphabet, and f is often denoted $(x_n)_{n=1}^{\infty}$ or, (x_n), or if it is clear from context x_n. If it is clear from context, the notation $x_n \subseteq X$ shall be represent picking elements of X to form a sequence.

Definition 1.5.9: Convergent sequence

Let X be a topological space, and $a_n \subseteq X$. We say that this sequence *converges* to $a \in X$ if and only if for all open sets U such that for $a \in U$, there exists N such that $\forall n \geq N, a_n \in U$.

If we are in a Hausdroff space, then we get the following known behavior of sequences:

Lemma 1.5.10: Hausdorff Spaces Means At Most 1 Element When Converges

If X is Hausdorff, every sequence converges to *at most* 1 element.

***Proof*:**
simple proof by contradiction.

Being hausdroff is a sufficient, but not necessary criterion for a sequence to converge to a single point. (perhaps explore this a bit more?)

Another nice property about sequences is that a continuous function preserves sequences:

Proposition 1.5.11: Continuity With Respect To Sequences

Let $f : X \to Y$ be a continuous function and $(x_n)_{n=1}^{\infty}$ a convergent sequence. Then $(f(x_n))_{n=1}^{\infty}$ is a convergent sequence, i.e. continuity preserve's convergent sequences

***Proof*:**
exercise

Notice that the converse is not true: if a function preserves convergence, it does *not* imply the function is continuous. To get the converse we must either:

1. Strengthen the space on which we are taking the sequence. For example, if the space is something called *secound-countable* or *compact*, then it will be the case. This shall be explored in the next chapter.
2. Generalize the notion of sequences to the notion of *nets* which shall add the necessary flexibility for the converse. This shall be explored in 2 chapters.

1.5.2 Relation With Previous Concepts

Proposition 1.5.12: closure of open set in Hausdroff space

Let $A \subseteq X$ be a *open* set. Then $\overline{A}$ add's a "trivial" amount to A, that is

$$A \subseteq B \Rightarrow \overline{A} \subseteq B$$

Solution If it were not the case, then the border would have an open set that doesn't intersect A, a contradiction. ▼

Corollary 1.5.13: closed Hausdorff set's interior

Let $A \subseteq X$ be a *closed* set. Then int$A$ removes a "trivial" amount to A, that is

$$B \subsetneq A \Rightarrow B \subseteq \text{int}(A)$$

Exercise 1.5.1

1. Does a local homeomorphism preserve or force Hausdorffness. That is, if $f : X \to Y$ is a local homeomorphism and either X or Y is Haudroff, should the other be Hausdorff? (Hint: Experiment with the spaces $\mathbb{R} \sqcup \mathbb{R}$, $\mathbb{R} \sqcup \mathbb{R}/\sim$ where $\sim$ identifies the positive numbers of both lines, and $\mathbb{R}$.)

1.6 Countability Axioms

It was emphasized a few times now that it is important that different sizes of infinities must be taken into account when working with arbitrary topologies. However, it is often favourable that our topology can be generated by the "simplest" type of infinity. In this section, we define such a notion:

Definition 1.6.1: Second-Countability

A topological space X is called ***secound-countable*** if it has a countable basis.

Example 1.6.2: Second Countability

$\mathbb{R}$ is second-countable, because all open intervals with rational end-points will form a basis

2

Constructing Topologies

In this chapter, we'll mainly be covering how to take the product of topologies and still have a topology, and how to define the notion of a sub-topology – in other words, we're integrating two more fundamental ideas when analyzing objects.

2.1 subspace (or Induced) topologies

We'll start with seeing how to create a topology if we have a subset $A \subseteq X$. The idea here is that sometimes we want to look at smaller parts of our space and see the topological properties it has:

Definition 2.1.1: Subspace Topology

Let X be a topological space, an Y be a subset of X. We definite the *subspace topology* on Y as:

$$\mathcal{T}_Y = \left\{ U \cap Y \middle| U \text{ is open in } X \right\}$$

2.1.2 Note Y need not be open. if it is open, then the topologies are comparable.

Example 2.1.3: Sub-Spaces

1. basic examples
2. Give a word of warning on subspace topologies of product topologies (refer to homework)

Theorem 2.1.4: Basis For Y Relative To X

1. if β is a basis for the topology on X, then $\{B \cap Y | B \in \beta\}$ is a basis for the subspace topology for Y
2. Let $X \supseteq Y \supseteq Z$. Let X have a topology $\mathcal{T}_X$, and let $\mathcal{T}_Y$ be the induced topology on Y. Then the topologies induced on Z form X and Y are the same

Combining with product topologies:

Proposition 2.1.5: Product Of Subspace Topologies

Let X, Y be topological spaces, and $A \subseteq X$, $B \subseteq Y$. The subspace topology on $A \times B$ is the same as the product topology of the subspaces topologies on A, B.

Solution ▼

using the bases, it should be intuitive.

2.1.6 Note CONVENTION: If you see some arbitrary set (ex. $\{\frac{1}{n} | n \in \mathbb{N}\}$), then this set will be a subspace topology of $\mathbb{R}$.

And like before, we can also define the idea of subspaces by using closed sets:

Lemma 2.1.7: Subspace Topology And Closed Sets

Let $Y \subseteq X$ [equipped with the subspace topology]. Then $A \subseteq Y$ is closed in Y if and only if $\exists V \subseteq X$ closed in X such that $A = V \cap Y$

Proof. good exercise

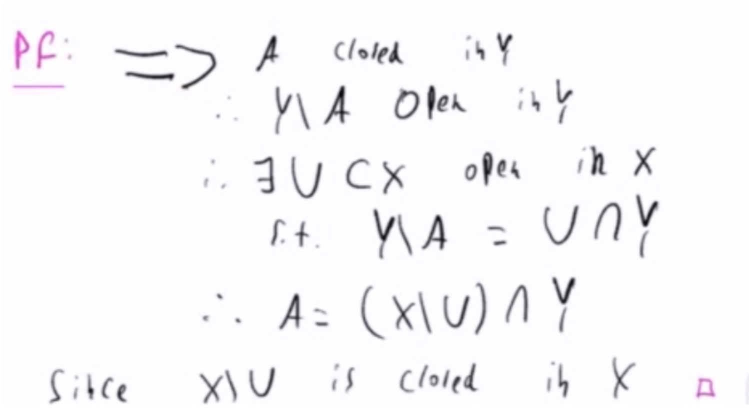

Figure 2.1: from lecture

□

Corollary 2.1.8: "transitivity" of closed sets between topology and subspace topology

Let X be a topological space. Suppose $Y \subseteq X$ is closed and $A \subseteq Y$ is closed in Y. The A is closed in X

Proof. Again, good exercice □

2.2 Product topology

This is a very common technique to generate new topologies, taking together smaller ones and producing a larger one

Definition 2.2.1: Product Topology basis

The *product topology* is generated by the basis

$$\beta = \{S \times T \big| S \subseteq X \text{ is open }, T \subseteq Y \text{ is open }\}$$

Corollary 2.2.2: Product Topology basis

The basis β define in 2.2.1 is indeed a topology

***Proof*:**
Recall $\cap$ and $\cup$ operation being "point-wise".

We can also generalise the top definition to larger products:

Definition 2.2.3: Product Topology on n Topologies

The *product topology* is generated by the basis

$$\beta = \{S_1 \times \cdots \times S_n | S_i \subseteq X_i \text{ is open }\}$$

2.2.4 Note Really make sure you have your intuition down on how to think about subsets of $X \times Y$. For example, the point $(1,1)$ is not covered by $(1,0) \cup (0,1)$ visually, we we get something like

here

Another thing to watch out for is that $A \subseteq X \times Y$, this doesn't mean that $A = U \times V$, $U \subseteq X$, $V \subseteq Y$. For example: $A \subseteq X \times X$, $A = \{x \times x | x \in X\}$. By no means is this set of the form $U \times V$. Be very wary of this! Especially when thinking about subspaces of $X \times Y$!

Example 2.2.5: Product Topologies

1. You are already familiar with $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. You might've seen this as open balls. However, this topology looks more like squares:

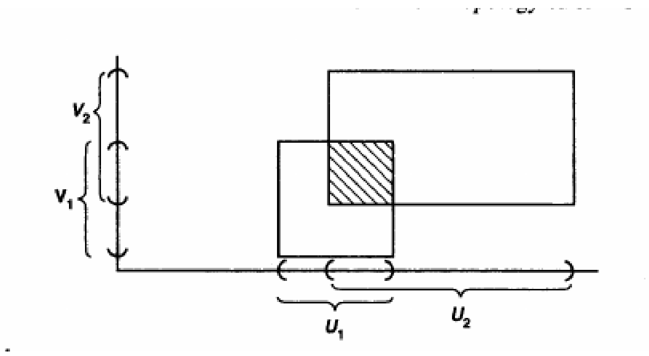

Figure 2.2: Notice that everything is a square

We will introduce open-balls once we define *metric-spaces*, and show how these two topologies are actually *homeomorphic*, and so can consider both interchangeably

2. Take the topology $\mathcal{T} = \{\emptyset, \{a, b\}, \{a\}\}$ on $X = \{a, b\}$. Then the product topology on $X \times X$ is $\{\emptyset, X \times X, \{(a, a)\}, \{(a, a), (a, b)\}, \{(a, a), (b, a)\}, \{(a, a), (a, b), (b, a)\}\}$. Notice that the last open set in the list is not in the basis.

With this construction, what we've done is start with two topologies, and combine them together to form a product. However, we often wonder if we can have an arbitrary topology Z and ask whether there exists two topologies X, Y such that $Z = X \times Y$. You can imagine that the analysis of a space can be greatly simplified if we can *project* down. This brings us to the following idea:

Theorem 2.2.6: Projection And Product Topologies

The product topology on $X \times Y$ is the coarsest topology for which π_X and π_Y are continuous

***Proof*:**
Let $\pi_X : X \times Y \to X$ such that $\pi_X(u, v) = u$. We'll first show that it's continuous. If we let $U \subseteq X$ be open, then $pi_X^{-1}(U) = (U, Y)$, and since U is open in X and Y is open in Y, then $pi_X^{-1}(U)$ is open. Since U was arbitrary, that means that π_X is continuous. Similarly for π_Y.

To shwo that $X \times Y$ is the corsest topology for which *both* π_X and π_Y are continuous (at the same time). Let's say we pick $\mathcal{T}_{X'} \subsetneq \mathcal{T}_X$. That means there is an open set $U \in \mathcal{T}_X$ such that $U \notin \mathcal{T}_{X'}$. THis will now become a problem, since if we pick $U \in X$, and consider $\pi_X^{-1}(U) = (U, Y)$, this will no longer be open, since $U \notin \mathcal{T}_{X'}$. Similarly for Y. Thus $X \times Y$ *must* be the coarsest topology for which *both* π_X and π_Y are continuous.

If you are more Categorically inclined, then what's going on here is that in the **Set**-Category, we can define product sets $X \times Y$ (as sets) by the following universal property

$$\begin{tikzcd} & X \arrow[d, "\pi_\alpha"] \\ Y \arrow[ur, "f"] \arrow[r, "f_\alpha"] & X_\alpha \end{tikzcd}$$

When working with product topologies, we want this same diagram to commute, except now we have continuous functions instead of regular function. And it turns out that the (categorical)*limit* which makes us find this universal property is the projection functions. The reason to formulate in this manner is to show that the following universal property is satisfied:

Lemma 2.2.7: π_I Is An Open Map

here

***Proof*:**
Simple proof

2.2.8 Note THis example requires Hausdorffness
It need not be that π_X or π_Y be closed maps! It might seem tempting, since $id : X \to X$ is open and closed, so $\pi_X : X \times Y \to X$ would be seem to be closed. However take positive portion of the graph $1/x$ (label this set Γ). It can be shown that in Hausdorff spaces, graphs are closed. However, this image of this set is $(0, \infty)$, which is open! This will hold if Y is compact.

We showed that if $f : Z \to X \times Y$ is continuous, there must exist two "coordinate" functions $f_1 : Z \to X$, $f_2 : Z \to Y$ which are continuous, but what if we have the converse: $f : X \times Y \to Z$; does the "coordinate" function here need to be continuous? The answer is no, we can find a counter-example

Example 2.2.9: not continuous function for $X \times Y \to Z$
p.112 exercice 12

Also, here are some easy mistakes to make the first time you work with the product topology:

1. $(1, 0) \cup (0, 1)$ does not cover $(1, 1)$. The product topology defines a whole new grid of points.
2. If $U \subseteq \mathbb{R}^2$ is open, this doesn't mean that $U = V \times W$, V, W open in $\mathbb{R}$! It's true that $V \times W$ is an elemetn of $\mathbb{R}^2$, but if U is open in $\mathbb{R}^2$, then the union of elemetns of the form $V \times W$ (i.e., the basis for $\mathbb{R}^2$) are elements in $\mathbb{R}^2$. That is, not all open sets of $\mathbb{R}^2$ can e written as $V \times W$!

Example 2.2.10: More Open Sets
ake

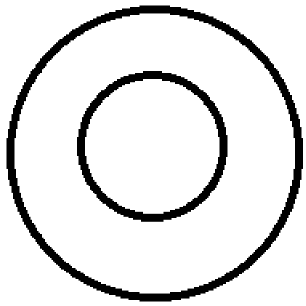

Figure 2.3: counter-example

To be some open set somewphere in $\mathbb{R}^2$. This set is the *arbitrary* union of open sets of the form $V \times W$! It can get even worse. In $\mathbb{R}^2$. take $V = \{(x, x) | x \in \mathbb{R}\}$. This set is closed (as it can be shown to be the compliment of an open set), but it is in no way represented by closed sets of the form $U \times V$. where U, V have more than 1 element. It is true that $x \times x$ is closed, and the *arbitrary* union of these closed sets are indeed closed, but as at this point know, the arbitrary union of closed sets need not be closed.

Proposition 2.2.11: Product Of Hausdroff Spaces

X, Y are Hausdorff if and only if $X \times Y$ is Hausdroff.

***Proof*:**
easy proof

Proposition 2.2.12: Subspace Of Hausdroff Space Is Hausdroff

here

***Proof*:**
here

2.2.1 The Infinite Case

When we defined the product topology for more than 2 topologies, we were sneaky in insuring the definition limited to *fintie* products (notice the n in the definition). This was purposeful! We'll Now explore what happens in the infinite case.

The goal is the show what the flexibility of letting there be an infinite number of topologies bring. We'll give this topology a new name, and also figure out a definition of the product topology on infinitely many topological spaces.

Here is the first generalization of the product topology:

Definition 2.2.13: Cross product of infinite topologies (Box Topology)

Let $\mathcal{I}$ be an indexing set for topologies X_i, $i \in \mathcal{I}$. Then let us take as basis for a topology:

$$\prod_{i \in \mathcal{I}} U_i$$

where U_i is open in X_i. This topology is called the ***box topology***.

And here is the second keeping the finiteness of product topologies:

Definition 2.2.14: Cross product of infinite topologies (Product Topology)

Let $\mathcal{I}$ be an indexing set for topologies X_i, $i \in \mathcal{I}$. Then $\prod_{i \in \mathcal{I}} X_i$ has as basis all sets of the form $\prod U_i$, where U_i is open in X_i, for each i, and U_i is *equal* to X_i for all but finitely many values of i.

2.2.15 Note Notice that definition 2.2.14 is named *product topology*. This definition is actually the *same* as product topology! Notice in both cases we have a finite amount of open sets in the cross product which (which are non trivial). Since in any element, most spaces are X_i (the entire space) while finitely can be U_i, these two spaces are *topologically the same*: **all** properties that apply to one apply to the other, they are indistinguishable topologically.
Note also that the box topology is thus generally *finer* than the product topology.

So why would one be preferable to the other? What does this restriction to only finitely many positions in any element being U_i? This comes down to the universal property of products *failing* on the box topology, that is, you can't split a continuous function $f : A \to \prod_{i \in \mathcal{I}} X_i$ to component functions $f_i : A \to X_i$! That is, The product topology is more useful than the box topology since continuous maps are continuous if and only if the component maps are continuous. We will generalize theorem 2.2.6 to the infinite case, then show how it fails for the box topology:

Theorem 2.2.16: Continuous Maps On Product Topologies

Endow X with the product topology. Let $f : Y \to X_I$, where $f(y) = (f_i(y))_{i \in I}$, f splits, that is:

$$\begin{tikzcd}
 & X \arrow[d, "\pi_i"] \\
Y \arrow[ur, "f"] \arrow[r, "f_i"] & X
\end{tikzcd}$$

where π_i is the projection map. Then f is continuous if and only if $\forall i \in I$ f_i is continuous

***Proof*:**
here

Example 2.2.17: Failing on the Box-Topology

Consider $\mathbb{R}^\omega$, the countably infinite product of $\mathbb{R}$ with itself. Let $f : \mathbb{R} \to \mathbb{R}^\omega$ be the function st

$$f(t) = (t, t, t..., t, ...)$$

This functions splits into coordinate functio $f_n(t) = t$. As we've shown in 2.2.16, if $\mathbb{R}^\omega$ has the product topology, every f_n is continuous. However, if $\mathbb{R}^\omega$ has the box topology, f is no longer continuous, even if every component function is continuous. Take the open set:

$$V = \prod_n \left(\frac{-1}{n}, \frac{1}{n}\right) = (-1, 1) \times \left(\frac{-1}{2}, \frac{1}{2}\right) \times \cdots \in \mathbb{R}^\omega$$

and consider $f^{-1}(V)$. If $f^{-1}(V)$ were open ,in $\mathbb{R}$, it would containg some interval $(-\delta, \delta)$ around the point 0. This would mean that $f((-\delta, \delta)) \subseteq B$, so that, applying π_n to both sides of the inclusion:

$$f_n((-\delta, \delta)) = (-\delta, \delta) \subseteq \left(\frac{-1}{n}, \frac{1}{n}\right)$$

for all n. However, for n larger enough, δ will be too big ($n > \frac{1}{\delta}$). Thus, the inverse image will be will be $\{0\}$, which is not open. Thus f is not open!

This here is an example of what we mentieond when discusing why you can only have either unions or intersections be arbitrary applied: Adding the 2nd infinity makes it too flexible! Here's adding that many open sets will sneakily introduce "infinite intersections", which as we mentioned before is not a property we want when dealing with topologies.

Though the box-topology fails in this important criterion, there are a couple of theorems which work on Box topologies, and so on product topologies:

subspace works well, Hausdorff works well.

Proposition 2.2.18: Closure Of A Product Is Product Of Closure

Let X be the product topology of the sets $\{X_j | j \in J\}$ is family of topological spaces and $A_j \subseteq X_j$. Then $\overline{\prod_{j\in J} A_j} = \prod_{j\in J} \overline{A_j}$

Solution Proof here ▼

2.3 Quotient Topology

As usual as in most categories we work in, we can also define the dual notion of sub-object with quotient-object. In our case, that would be the quotient topology. What's interesting about the **Top** category is that the quotient objects don't really behave well. You can't generally take the product of quotient topologies, you can't take the subspace of a quotient topology and still have a topology. However, unlike the **Grp** Category, you do not need a *congruence relation* to define a quotient object, only an equivalence relation. To refresh your memory, a *congruence relation* is an equivalence relation which *preserves* the algebraic structure (so $[x \cdot y] = [x] \cdot [y]$). in the **Top**

Category, we don't need an equivalent notion of *congruence relatoin*; defining quotients in term of equivalence relations *only* will suffice! In this way, quotient objects in the **Top** category are more flexible, but at the cost that many related objects we like to work with (ex. sub-objects, Cartesian product objects) will not be well-defined without extra conditions.

2.3.1 Basic Properties And Definition

The intuitive first example given with quotient topologies is by "glueing" surfaces together. so we can take an interval, and glue together the two end points, and get a circle. Or we can get a rectangle, and glue opposite edges together in the right way, and get a torus. Or we can glue all the points together and get a sphere.

To understand how to combine these points and treat them as *one* points, we start by defining a quotient map, which will let us define the quotient topology:

Definition 2.3.1: Quotient Map

Let X and Y be topological spaces, and let $p : X \to Y$ be a surjective map. The map p is say to be a ***quotient map*** if

$$\text{A subset } U \text{ is open in } Y \text{ if and only if } p^{-1}(U) \text{ is open in } X$$

This condition is also sometimes called *strong continuity.* For a continuous function, if $p^{-1}(U)$ is open, then it it need not be that U is open. In this way, the function must preserve *more* open sets from the domain.

It's kind of a weird extra condition at first. The intuition given by the prof is that if you have space to walk around once you've glued stuff toegether, then once you pull that back, you should still have space to walk around. Also can be thought of as the smallest set which the equivalence class of the quotient space is preserved.

Another intuition for quotient maps is that its *almost* a homeomorphism. If it is injective, then it is a homeomorphism

We can also formulate the equivalent definition with closed sets with the usual strategy ($f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$)

There are a couple of other ways of thinking about quotient maps that might be intuitive. Let $f : X \to Y$ be a function, and take $A \subseteq X$. we say that A is ***saturated*** (with respect to f) if A contains every set $p^{-1}(y)$ that it intersects – or as the professor said it $f^{-1}(f(A)) = A$. That is, if $y \in Y$ such that a point $x \in p^{-1}(y)$ is also in A, then A must contain all of $p^{-1}(y)$, for every y. With this formulation, we can equivalently say that p is a quotient map if it a continuous and p maps *saturated* open sets of X to sets of Y (or saturated closed sets of X to closed sets of Y)

There are also two special kind of quotient maps. If p is a surjective continuous function that is also *open* or *closed*, then it is also a quotient map. However, the condition of open/closed is too strong of a condition too: there are quotient maps which are neither open or closed. This is because the $p^{-1}(U)$ being open condition is weaker than the image of open sets are open condition, and so the converse of open map is not the converse of $p^{-1}(U)$ being open since if a map is not an open map, it might still satisfy the $p^{-1}(U)$ being open condition.

The idea of a quotient map is similar to quotient groups. Since the function is continuous, we already got the "homomorphic" property. The extra condition of $f^{-1}(V)$ open then V open ensures us that the function is more than just projecting down to a smaller space, but it's also making sure that if the fiber of a set $f^{-1}(U)$ is open, then it had to come from an open set U.

Example 2.3.2: Gluing

1. This examples brings in this "gluing" idea that will come up often when looking at quotient topology.Let Let $p : [0,1] \cup [2,3] \to [0,2]$, both subspaces of $\mathbb{R}$. We'll define p so that it "glues" these two intervals together:

$$p(x) = \begin{cases} x & \text{for } x \in [0,1] \\ x-1 & \text{for } x \in [2,3] \end{cases}$$

   Notice that p is surjective, showing the gluing property. It is also closed. if we take a closed basis element $[a,b]$ in the domain, then $p([a,b])$ will either the a closed interval. Finally, this map is continuous. If we take a closed interval $[a,b]$ in the co-domain [a], then the pre-image is clearly either a closed interval or the union of two closed intervals

   As an aside, this map is *not* an open map. $[0,1]$ is open in the subspace topology in the domain, but $[0,1]$ is not open the subspace topology of the co-domain. Furthermore, if we got rid of the point where we were about to glue (say $[0,1) \cup [2,3]$), then we no longer have a quotient map. It will still be surjective and continuous. It would no longer be closed ($p([0,1) = [0,1)$, where $[0,1)$ is closed in the co-domain), but that as we showed doesn't disprove enough. To show it's not a quotient map, notice that $[2,3]$ is open in the domain, and $p(p^{-1}([2,3])) = [2,3]$, meaning that it's saturated, but $p([2,3]) = [1,2]$ is *not* open in $[0,2]$.

2. Projection maps are quotient maps. Let $\pi_1 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$. This map is surjective, continuous, and open (as we've shown before).

   As an aside, note that π_1 is *not* a closed map! The subset $C = \{(x,y) | xy = 1\}$ is closed in $\mathbb{R} \times \mathbb{R}$, but $\pi_1(C) = \mathbb{R} - \{0\}$ is *not* closed in $\mathbb{R}$. Furthermore, if we were to take $p : C \cup \{0\} \to \mathbb{R}$, then this map is continuous and surjective, but the one point set $\{0\}$ is open in A and saturated with respect to p, but its image is not open in $\mathbb{R}$

3. Here is a more general example. Let X be a topological space and $A \subseteq X$. if $r : X \to A$ is a continuous map such that $r|_A(a) = a$ (r is the identity on A), then r is a quotient map

   Solution We'll first show that if $r : X \to Y$ and $f : Y \to X$ are continuous maps such that $f \circ r \cong id$ (that is, f is a continuous right inverse),then r is a quotient map.

   the map r is already continuous and surjective by the right inverse; it's only left to show that if $r^{-1}(U)$ is open, then os is U. since f is continuous, it's pre-iamge is also continuous, that is

$$f^{-1}(p^{-1}(U))$$

is open. But now it's just a matter of going through the definition:

$$f^{-1}(p^{1}(U)) = (p \circ f)^{-1}(U) = id(U) = U$$

thus U is open. To show that r is a quotient map if we find the appropraite f. and indeed,

$$f : A \to X \qquad f(a) = a$$

is a right-inverse, thus r is a quotient map! Retraction makes will come back big-time when we talk about the Fundametnal group of the circle (definition 7.3.1) ▼

4. Let $\pi_1 : R \times \mathbb{R} \to \mathbb{R}$ be the projection othe first coordiante. Let $A \subseteq R \times R$ such that $x \geq 0$ or $y = 0$ (or both); let $q : A \to \mathbb{R})$ be obtained by restrictin π_1. Then q is a quotient map, but it is neither open or closed.

   Solution FIrst show not open or closed, then take from hw. ▼

[a]I said closed interval instead of closed set because I'm using the *basis* definition of continuity, combined with the fact that we can alternate between open and closed sets

With the idea of a quotient map, we can define the topology which is formed from it:

Definition 2.3.3: Quotient Topology

If X is a sapce and A is a *set* and if $p : X \to A$ is a surjective map, then there exits exactly one topology $\mathcal{T}$ on A relative to which p is a quotient map; it is called the ***quotient topology*** induced by p.

It should be quite natural to see how a topology would come from p. You would let $\mathcal{T}_A$ be be defined as consisting of subsets $U \subseteq A$ such that $p^{-1}(U)$ is open in X. [1].

Example 2.3.4: Quotient Topology

Munkre gave this example. Take $p : \mathbb{R} \to \{a, b, c\}$ by

$$p(x) = \begin{cases} a & \text{if } x > 0 \\ b & \text{if } x < 0 \\ c & \text{if } x = 0 \end{cases}$$

Then you can check that p is a quotient map, and so induces the quotient topology on $\{a, b, c\}$. So, by pushing open sets through p, we can see that the inducd topology is:

$$\{\{a, b, c\}, \{a\}, \{b\}, \{a, b\}\}$$

]

[1]which kind of makes me think of Algebra, that you're kind of pushing the structure forward onto the quotient group

Remember the equivalence intuition presented earlier? We can generalize this with the idea of partitioning our space, and letting the partitions collapse to a single point in the quotient topology (like he borders being part of the same partition)

Definition 2.3.5: Quotient Space

Let X be a topological space, and let X^* be a partition of X int odisjoint subsets whose union is X. Let $p : X \to X^*$ be the surjective map that carries each point X to the elemetns o X^* containing it. In the quotient topology induced by p, the sapce X^* is called the ***quotient space*** of X.

Personally, I like thinking about this in the categorical way. If there is a partition, then there is a surjective map from $X \to X*$. If we now map X into a topological space and $\to$ into a continuous function, what topology can I put on $X*$ in order for function to remain continuous? [2]. The answer is *yes*: Take the indiscrete topology on $X*$. Then $p^{-1}(X*) = X$ and $p^{-1}(\emptyset) = \emptyset$. The natural question then becomes, can we go finer? Even better, is there a finnet topology for which p is continuous? If we go too fine, the discrete topology then it's possible that p is no longer continuous. Fortunately we can answer the question: U is open in $X*$ if and only if $p^{-1}(U)$ is open in X. If we went any finer, then we would have a set that's *not* open in X! Thus, this is the finnest set. Notice how this matches the definition of a quotient topology! In fact, this is the *reason* for the definition of a quotient map when looking at math from a categorical perspective! The advantage of this is that we can relate our concept of "quotient" from other fields. For example, Thinking about quotient spaces is like thinking about G/N, the quotient group. One can show that $f : X^* \to \{\text{quotient sapce}\}$ is homeomorphic.

The overall takaway is that if we have a surjective map between X and $X*$, we will define the topology on $X*$ based on p.

Example 2.3.6: More Quotient Topologies

1. Let X be the closed unit ball
$$\left\{x \times y \middle| x^2 + y^2 \leq 1\right\}$$
in $\mathbb{R}^2$, and let X^* be the partition of X consisting of all the one-point sets $\{x \times y\}$ for which $x^2 + y^2 < 1$, along with the set $S^1 = \left\{x \times y \middle| x^2 + y^2 = 1\right\}$. Then we can induce a topology on X^* which is homeomorphic with the subspace of $\mathbb{R}^3$ called the ***unit 2-sphere*** defined by
$$S^2 = \left\{(x, y, z) \middle| x^2 + y^2 + z^2 = 1\right\}$$
Let $\varphi : X* \to S^2$,
$$\varphi(x, y) = \{$$
it's quite a complicated map, I'll do it later.
2. Let X be the rectangle $[0, 1] \times [0, 1]$. Define the partition X^* to be the singletons if it's an interio point, the piar $[x \times 0, x \times 1]$ and $[0 \times y, 1 \times y]$ for the edges, and the 2 corner points form their own set
$$\{0 \times 0, 1 \times 0, 1 \times 1, 0 \times 1\}$$

[2] or, categorically, I'm trying to establish a functor between the **Set**-Category and the **Top**-Category

This quotient space is homeomoprhic to the torus.

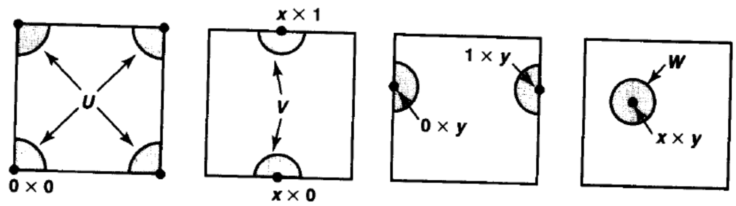

Figure 22.5

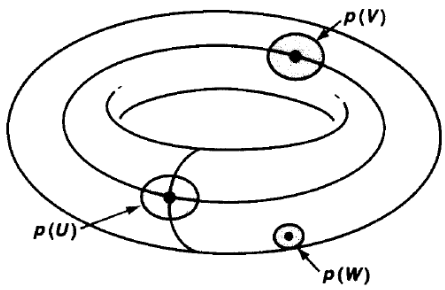

Figure 2.4: torusExampe

2.3.7 Note this partition construction relating to equivalent relations means we can define a quotient maps through equivalent relations!! Super useful and probably the way to think about it when construction quotient maps: Take f to be a surjective function and define the image through a relationhsip: $f(a) = f(b) \Leftrightarrow aRb$. Example:

Let S be a set, and $f : \mathbb{R} \to S$ be a surjective map of sets such that $f(a) = f(b) \Leftrightarrow a - b \in \mathbb{Q}$. Prove that the induce qutoeint topology on S is the trivial topology.

When we studied product spces, we had a criteron for determining whether a map $f : Z \to \prod X_i$ *into* a product spcae was continuous. Its counterpart in the theory of quotient spaces is a criterion for determining when a map $f : X^* \to Z$ *out of* e a quotient sapec is continous.

Theorem 2.3.8: Universal Property Of Quotient Maps

Let $p : X \to Y$ be a quotient map. Let Z be a space and let $g : X \to Z$ be a map that is contant for each set $p^{-1}(\{y\})$, for $y \in Y$. Then g induces a map $f : Y \to X$ such that $f \circ p = g$. The induced map f is continuous if and only if g is continuous; f is a quotient map if and only if g is quotient map

$$\begin{tikzcd} X \arrow[d, "p"'] \arrow[dr, "g"] & \\ Y \arrow[r, dotted, "f"'] & Z \end{tikzcd}$$

This theorem is useful to find continuous maps on the quotient space.Here's a really easy intution. Imagine you have the interval I which can be mapped to S^1 by the quotient map, and let's say you have another space Z:

$$\begin{tikzcd}
I \arrow[d, "p"'] \arrow[dr, "g"] & \\
S^1 \arrow[r, dotted, "f"'] & Z
\end{tikzcd}$$

Then in order to find a continuous map form S^1 to Z, it suffices to find a continuous map form I to Z, and we can find a continuous map f as well! We'd know that $g = f \circ p$

Proof. □

2.3.2 relationship between previous concepts

We can ask ourselves what properties we've explored previously do quotient maps hold. It turns out that not a lot. Though a quotient map is almost homeomorphic, a lot can break due to lack of injectivity.

Subspaces do not behave well. if $p : X \to Y$ is a quotient space, then taking A as a subpsace of X and considering $p : A \to p(A)$, then p need not be a quotient map. The following theorem fixes this as well as possible:

Theorem 2.3.9: Subspaces And Quotient Maps

Let $p : X \to Y$ be a quotient map; let A be a subsapce of X that is saturated with respect to p; let $q : A \to p(A)$ be the map obtained by restricting to p.

1. If A is either ope nor closed in X, then q i a quotien map.
2. if p is either an open map or closed map, then q is a quotient map.

Proof. Munkre p.140 □

Composition of maps still work well, which follow form the equation $p^{-1}(q^{-1}(U)) = (q \circ p)^{-1}(U)$

The product of two quotient maps need not be quotient. Example at p.143

The Hausdorff condition does not behave well. X can be Hausdorff, but X^* need not be Hausdorff.

One more fact that I will mention: Every topological space can be obtained by quotienting a Hausdorff space!

link here

The quotient of a Hausdorff space need not be Hausdorff:

Example 2.3.10: Counter-Example
here

Exercise 2.3.1

1. Show that $\mathbb{R}/\mathbb{Q}$ with the quotient topology is the trivial topology.

2.4 Weak Topology

We have given many constructions of topologies including the subspace, product, and quotient topology. All of these are special cases of one of two types of topologies: *initial topologies* and *final topologies*. These two types of topologies will generlize the results we've presented, and bring a new way of defining topologies that will give some minimal condition (for the initial topology) or maximal condition (for the final topology) for certain maps to be continuous:

Definition 2.4.1: Weak [initial] Topology

Let $F = \{f : X \to Y_\alpha\}_{\alpha \in A}$ be a collection of maps from a set A to topological spaces Y_α indexed by $\alpha \in A$. Then the coarsest topology on X which makes each f continuous is called the *weak topology*. In particular, a basis for the weak topology generated by F consists of $\{f_\alpha^{-1}(U)\}$ for all open sets $U \subseteq Y_\alpha$ for all $\alpha \in A$.

Note that it is not interesting to ask what is the finnest topology on X that makes all the maps continuous; that would simply be the discrete topology on X. Note too that since the weak topology $\mathcal{T}$ is the coarest topology on X that makes all the maps continuous, if $\mathcal{T}'$ is another topology on X that makes each map in F continuous, then $\mathcal{T}' \subseteq \mathcal{T}$. Note too that we showed in the definition how to actually create the weak topology, however for the skeptical reader it should be verified that the weak topology does indeed exist

Example 2.4.2: Weak Topology

1. Let $\{X_\alpha\}_{\alpha\in A}$ be some collection of topologies and $X = \prod_{\alpha\in A} X_\alpha$. Let $\Pi = \{\pi_\alpha : X \to X_\alpha\}$ be the collection of functions. Then the weak topology on X generated by F is *precisely* the product topology of X

2. Let X be any topological space and $A \subseteq X$ any subset. Let $\iota : A \to X$ be the inclusion map. Then the weak topology on A generated by ι is the *subspace* topology.

3. Let X be a normed space (over a topological field $\mathbb{F}$). Then X has a natural topology defined by the norm (the norm induces a metric, which induces the metric topology). let $X^* := \mathrm{Hom}(X, \mathbb{F})$ be the dual of X (the collection of all continuous linear maps from X to $\mathbb{F}$). Then we will define the weak topology on X to be the initial topology generated by X^*. In this topology, a net $x_\alpha \to x$ if and only if $f(x_\alpha) \to f(x)$ for all $f \in X^*$. We will sometimes write $x_\alpha \xrightarrow{w} x$ to emphaize weak convergence. If X is infinite dimensional, the weak topology of X is coarser than the norm topology on X.

(properties that are key to weak topologies can be found here)

One of the great advantages of of defining weak topologies is that we can make the collection of f_α and spaces X_α anything we want. This flexibility will the definition's of many interesting topologies on funcitonal spaces (ex. collection of smooth functions on compact spaces, collection of bounded continuous functions, or simply collection of bounded functions, and so forth). All such spaces are of key importance in funtional analysis. Weak topologies are also a special case of a much more general construction, which we'll explore at the end of this chatper.

Finally, there is also the dual of weak topologies where we take maps that map *into* X.

Definition 2.4.3: Final Topologies

Let Y be a set, $\{X_\alpha\}_{\alpha\in A}$ be a collection of topological spaces, and $F = \{f_\alpha : X_\alpha \to Y\}$ be a collection of maps into Y. Then the the finnest topology on Y that makes each f_α continuous is called the *final topology*. In particular, $U \subseteq X$ is open if and only if $f_\alpha(U)^{-1}$ is open for each $\alpha \in A$

Again, it should be verified that this is indeed a topology. While the weak (or initial) topology seeks for a domain with the coarsest topology, final topologies seek a codomain with the finnest topology. This is because the coarsest topology on the co-domain will always create continuous maps. Unlike the initial topology, the final topology is not given the name "strong" to contrast the name "weak" for initial topologies. There is in fact a topology known as the strong topolgoy on a space X: If $\|\cdot\|$ is a norm on X[3], the strong topology on X is the topology generated by all the balls $B_r(x)$ for all $x \in X$ and $r > 0$. The contrast between strong and weak topologies will make more sense when studying functional analysis, where there we would much rather a space have a strong topology, but it will sometimes be unfeasible (ex. the space of all smooth functions, or infinite dimensional vector spaces won't have a good norm defined on them), and so we will "weaken" the topology that will create topology that will not be induced by something as strong as a norm (usually, it the weak topology will be induced by a countable collection of seminorms, see Folland chapter 5 for more information).

Example 2.4.4: Final Topologies

Let X be an arbitrary topological space an $A \subseteq X$ a subspace. Let $\pi : X \to A$ be the projection map. Then the final topology on A is the *quotient topology*.

2.5 Metric Spaces

Metric spaces are different from topological spaces. While in topology, we have an underlying set X with a set of subsets of it $\mathcal{T}_X$, defining $(X, \mathcal{T}_X)$, a metric spaces a set X with a *metric* d_X defined on it (X, d_X) (which we'll define in a moment). This makes these two spaces categorically different [4]. However, we introduce these spaces since every metric space also creates a topological space! Hence, a very natural way of studying metric spaces is through topology. For example, we can show that every metric spaces (with the induced topology by the metric) is Hausdorff. However, continuity will not preserve all the properties of a metric space!! For example, a metric can define whether a sapce is bounded. But, we will show examples of a continuous function betweeen $f : X \to Y$, where X is

[3]See definition ref:HERE

[4]As in, metric spaces are objects in the **Met** Category, while topologies are objects in the **Top** Category

bounded, while Y is not! Another such example is completeness, where we can make a continuous map between $(-1, 1) \to \mathbb{R}$, but $(-1, 1)$ is clearly not complete, while $\mathbb{R}$ is.

If we take a further step back for a moment and see the bigger picture in Analysis, metric spaces are perhaps the most common topological space mathematicians use. There are some even more specific spaces, outlined in the bellow image:

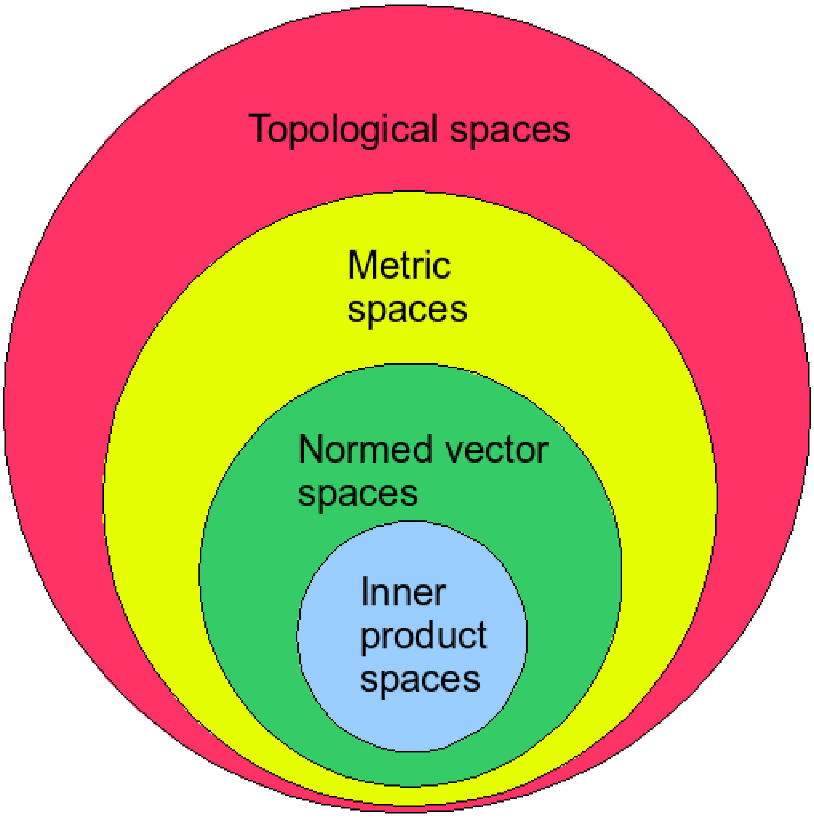

Figure 2.5: Hiarchy of spaces

and covering metric space's can also be seen as the first step into analysis.

This and more is covered in this chapter.

2.5.1 Basic Definition And Examples

We'll put the definition of metric here as a reminder

Definition 2.5.1: Metric

Let X be a set. A *metric* on X is the function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

such that

1. $d(x, y) = d(y, x)$
2. $d(x, y) + d(y, z) \geq d(x, z)$
3. $d(x, y) = 0$ if and only if $x = y$

2.5.2 Note the fact that it is an *if and only if* is in fact very important. Without that you can say

$$d(x, y) = 0 \qquad \forall x, y \in X$$

and that would be a metric, and you can go on to say it would define the indiscreet topology, but this was purposely left out because it would then not make all metric spaces Hausdorff (as we'll show soon)

From this, we can define the basis. Take $\epsilon > 0$. Then

$$B_d(x, \epsilon) = \left\{ y \middle| d(x, y) < \epsilon \right\}$$

Definition 2.5.3: Metric Topology

If d is a metric on X, we call the *metric topology* on X, to e the topology generated by the basis

$$\mathcal{B} = \left\{ B_d(x, \epsilon) \middle| x \in X, \epsilon > 0 \right\}$$

One might ask themselves why is this the definition of a metric space? I personally like to think of it Categorically

> If X has a metric d, then the coarsest topology which makes $d : X \times X \to \mathbb{R}$ continuous is the one we just described

To me, this is what it means to describe a metric topology, since we have reaosn to augment the notation of metric to a topological one in a way that ins't merely convention, but with purpose. If one were to continue categorically, there might even be a universal property hidden here.

Lemma 2.5.4: Metric topology is Hausdorff

The metric topology is Hausdorff

Proof. Let X be a metric topology. Take any two distinct points x_1, x_2. Since they're distinct, let $r = d(x_1, x_2)$ where r is non-zero. Consider $\epsilon = \frac{r}{10}$. Then $B_\epsilon(x_1) = \left\{ y \in X \middle| d(x_1, y) < \epsilon \right\}$ and

$B_\epsilon(x_2) = \{y \in X \mid d(x_2, y) < \epsilon\}$ Therefore:

$$B_\epsilon(x_1) \cap B_\epsilon(x_2) = \emptyset$$

by definition of open balls. Hence, metrizable spaces are metric topologies. □

Now, you can think that we might need to start carrying around the metric when defining a topological space which has a definable metric (ex. $(\mathcal{T}_X, X, d)$). However, many metrics will sometimes define the same space. And sometimes, we want to think of spaces which have a metric not as topological spaces but as their own space. So we want to make the distinction of a space with a particular metric, and the idea of a topology which *can* be defined by a particular metric. The following terms distinguish between these two concepts:

Definition 2.5.5: Metrizable Topologies

If X is a topological space, we say that X is metrizable if there exists a metric on X which induces the topology of X.

Definition 2.5.6: Metric Space

A *metric space* is a metrizable topological space together with a **specific** metric d inducing the topology of X. We usually write

"Let (X, d) be a metric space"

2.5.7 Note Sometimes, a metric space in other places of math is not defined in terms of metrizable space, but *just* in terms of a metric being definable on the space. It will always turn out that if you have a metric spaces, it is also a metrizable topology, but that can be considered a subtly.

2.5.8 Note Note also that d is a continuous function! So some topological property of $\mathbb{R}$ is important when thinking of metric spaces!

Example 2.5.9: Metric Spaces

The typical MAT257 metric spaces. Take $\mathbb{R}^2$. $\mathbb{R}^2$ is metrizable. For example

1. $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
2. $|x_1 - x_2| + |y_1 - y_2|$
3. $\max(|x_1 - x_2|, |y_1 - y_2|)$
4. $\min(1, \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2})$
5. 0 if $x = y$ and 1 if $x \neq y$ (the discrete metric, which creates the discrete topology)
6. Given a metric d, $\overline{d} = \min(d, 1)$. A cool analogy the prof used for this metric is as follows: Imagine you have a teleporter which can get you anywhere. It takes an hour to boot. If it takes less than an our to get somewhere, you'll walk. Otherwise you'll use the teleporter.

A quick observation on the metric on $\mathbb{R}$. say you have the Euclidean metric. Then take

$$\overline{d}_\epsilon = \begin{cases} d(x,y) + \epsilon & x \neq y \\ 0 & x = y \end{cases}$$

Then this metric is in fact equivalent to the discrete metric! If $B_{\overline{d}_\epsilon}(x, \frac{\epsilon}{2})$, then B only contains x, making it equivalent to the discrete metric.

In general, for finite domains or domains which are discrete subsets of $\mathbb{R}$ (relative to the Euclidean topology), something like this will hold.

We can also ask if some notion of the coarsest possible metric topology. However, The thing is that the moment you have the non-discrete metric, since we have that the co-domain is $\mathbb{R}$, and d is a constant, continuous

$$d(x,y) \leq d(x,z) + d(z,y)$$

meaning we just shrunk we we considered our open sets. So this does not seem to be a *natural* question to ask. Indeed, it might even be that there *isn't* a coarsest space

Lemma 2.5.10: Subspace Of Metric Space Is Metric Space

If (X,d) is a metric space, and $Y \subseteq X$, then $(Y, d|_Y)$ is also a metric space, and the subspace topology on Y coincides with the metric topology induced by $d|_Y$

***Proof*:**
the proof have 2 sets:

1. Can check that $d|_Y$ is a metric
2. $\forall g \in Y, \epsilon > 0, B_d(g,\epsilon) \cap Y = B_{d|_Y}(g,\epsilon)$

which would finish the proof

Definition 2.5.11: Continuity For Metric Spaces

Let (X, d_X), $Y, d_Y)$ be metric spaces. Then $f : X \to Y$ is continuous if and only if

$$\forall, x \in X, \exists \epsilon > 0, \exists \delta > 0 \text{ such that } d_X(x,x') < \delta \Rightarrow d_Y(f(x), f(x')) < \epsilon$$

which is the same definition in MAT257

We can also define some property which strongly rely on the metric:

Definition 2.5.12: Diamater

For $A \subseteq X$, define the diamater of A to be

$$\operatorname{diam}(A) = \sup(d(a_1,a_2) \big| a_1, a_2 \in A)$$

Definition 2.5.13: Bounded

We say that (X, d) is *bounded* if $\operatorname{diam}(X) < \infty$

Theorem 2.5.14: Bounding Any Metric Spaces

Given $\overline{d}$, then the metric topology for $\overline{d}$ is the same as for d.

Proof. □

We can also return to our earlier discussion about border of sets in metric spaces. Let say a set A is open in a metric topology. Consider $x \in \overline{A} \setminus A$. When x is not Hausdorff, we know that it is possible that the border of a set be weird. Hausdroff fixes that for us

Theorem 2.5.15: Border Of Set In Metric Space

Take A to be a non-empty open in the metric space X. Then $x \in \overline{A} \setminus A$ if and only if

$$\forall \epsilon > 0, B_\epsilon(x) \cap A \neq \emptyset \wedge B_\epsilon(x) \cap (X \setminus A) \neq \emptyset$$

intuition

Proof. recall that

$$x \in \overline{A} \setminus A \Leftrightarrow \text{every open set } U \text{ containing } x \text{ satisfies } U \cap A \neq \emptyset$$

We will be using that fact for this proof:

($\Rightarrow$) Let's start with $x \in \overline{A} \setminus A$. For the sake of contradiction, let's say

$$\exists \epsilon > 0 \qquad B_\epsilon(x) \cap A = \emptyset \vee B_\epsilon(x) \cap (X \setminus A) = \emptyset$$

That is, either the epsilon ball is completely seperate from A, completely seperate from the compliment of A. If it's completely seperate from A then it contradicts the equivalent definition of closure "every open set U containing x satisfies $U \cap A \neq \emptyset$", since we have a an open set $B_\epsilon(x)$ containing x but not intersecting A. Therefore, it would have to be that $B_\epsilon(x) \cap (X \setminus A) = \emptyset$, that is, it is completely seperate from the compliment of A. However, $x \in \overline{A} \setminus A$, meaning that x is in some subset of X which is not part of A, that is some comliment of A, and so $B_\epsilon(x) \cap (X \setminus A) \neq \emptyset$, a contradiciton again. Therefore, there does not exist an epsilon for which an epsilon ball around x is either completely inside A or outside of A, and therefore any epsilon ball containing A but intersect both A

($\Leftarrow$) Let's take any point x that satisfies that condition. If we consider any open set U that contains x and non-trivially intersects with A, then since X is a metric space generated by all epsilon balls, $\exists \epsilon_0$ such that $B_{\epsilon_0} \subseteq U$, and we know that by assumption, that any ball around x will intersect A, therefore, it must hold.

Therefore, it must always be the case in metric topologies that the border has this "infinitely thin" property □

What made me originally think of this is imaginning a thick border, thick enough so that $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq \overline{A} \setminus A$. That is, you can think an epsilon ball around the border. However, that would then contradict that for any point in the border, $U \cap A \neq \emptyset$, therefore, the border must be really thin.

2.5.2 Product Spaces and Metrics

Given two metric spaces, if we take the product of them, we can ask ourselves what would be the metric on $X \times Y$ which would define the natural product topology? It turns out to be the following metric:

Definition 2.5.16: product metric

Let (X, d_X), (Y, d_Y) be metric spaces. Then

$$d_{X\times Y}((x_1, y_1), (x_2, y_2)) = \max(d_x(x_1, x_2), d_y(y_1, y_2))$$

Lemma 2.5.17: Metric Product Space

1. $(X \times Y, d_{X\times Y})$ is a metric space
2. The metric space coincides with the product topology

Proof. Oct 1st lecture □

Theorem 2.5.18: Metrizablility Of Product Sapces

Let $(X_n)_{n\in\mathbb{Z}_+}$. Then every X_n is metrizable if and only if $X = \prod_{n\in\mathbb{Z}_+} X_n$ is metrizable (if X has the product topology)

***Proof*:**
Oct 1st lecture

3

Topological Invariances

We know begin the study of topological properties which are *invariant* between topologies. What invariance means is that if we have a homeomoprhism between two topologies, the properites will be present in both. We will make one exception in this chapter from this idea to deviate to metric spaces after we've studied Hausdorff spaces, since they present a very interesting example.

3.1 Connectedness

Connectedness is another property which is between points and the whole thing which talks about the nature of the geometry. This property gives us some idea of whether an open set can be "broken down further" to open sets. If a set is connected, then relative to that open set, you cannot break it down to the disjoint union of two (or more [1]) other open sets. An example of something that is clearly disconnected is two open balls $B_1(-10)$ and $B_1(10)$ in the Euclidean topology. Similarly, each respective balls are connected in the Euclidean topology.

It might be tempting to then say that it gives some idea of contiguity between points. However, that is not necessarily the case: In a topology which I'll outline later, we'll see how $B_r(x)$ will *always* be disconnected for any r and x with *any* underlying set with the exception of singletons. The topologies in which this holds is not even necessarily topologies which are intuitively disconnected. For example, in the Euclidean topology, we'll show how balls are connected, but if you also add the closure of balls to your open sets, then you will actually get that every set is disconnected.

[1] there's no real point in saying *or more* since then you can union all but one of the sets, and you're back to having 2 open sets

3.1.1 Basic Properties And Definitions

Recall clopen sets. We are bringing them back here to talk when sets *are* clopen, and what does it imply when they are:

Lemma 3.1.1: separation of sets regarding clopen sets

Suppose $\mathcal{T}_X$ be a topological space such that $X = A \cup B$ and $A \cap B = \emptyset$, where A and B are also arbitrary sets. Then the following are Equivalent:

1. A is clopen
2. B is clopen
3. A, B are closed
4. A, B are open

Proof. □

Recall earlier when we introduced clopen sets, with the example of $\emptyset$ and X as the "trivial" clopen sets, and then we introduced what I called the separation lemma. There is something intuition in saying that space X is completely disjoint (meaning $\overline{A} \cap B = A \cap \overline{B} = \emptyset$) if X is as described in that lemma. This will turn out to be the basis for our thinking about connectedness.

Definition 3.1.2: Connected

Let X be a topology, and take the set $A \subseteq X$. A is **connected** if and only if the only clopen subsets of A are $\emptyset$ and A.

I like to think of open sets of an idea of "mandatory" overlap between all open sets in A – that is, the intuition for connectedness should really come from the topology rather than the underlying set [2].

Example 3.1.3: Connected Sets

1. The interval $I = (0, 1)$ is connected in the Euclidean topology. Let's say, for the sake of contradiction, that $I = A \cup B$, $A \cap B = \emptyset$, A, B are open (equivalent to them being clopen by 3.1.1). Then A and B must be the union of open intervals. One of them must contain the points (0.., so let that be in A. Since A is the union of open intervals, there is a sub-interval in A such that $(0, \epsilon) \in A$, and some points $(\epsilon, ..$ are not in A (since B cannot be non-empty). Let's say $(\epsilon, \epsilon + \delta) \in B$. However, this would imply that $\epsilon \notin A$ and $\epsilon \notin B$. There are also no singleton intervals. Thus, $I \neq A \cup B$, $A \cap B \neq \emptyset$, A, B open, showing that I is connected in the Euclidean topology. A more formal proof of this fact is presented in theorem 3.1.22.
2. $X = (0, 1) \cup (2, 3) \subset \mathbb{R}$. Then A is clopen in X, and so disconnected.
3. The trivial topology is connected.

[2] naturally, sometimes the underlying set (like $\mathbb{R}$) has properties that are a great influence to a topology (like Euclidean topology), but then we chose a topology which has our properties, so again it comes down to topology

4. the discrete topology is totally disconnected (only singletons are connected)
5. Let X be any infinite set with the finite-compliment topology. Then if $A \subseteq X$ is infinite, it is connected (try taking $A \cup B = A, A \cap B = \emptyset$, A, B open)
6. A space is said to be *totally* disconnected if the only connected components are the open sets. Is the discrete topology totally disconnected? Are there others?
7. Is $\mathbb{R}_l$ connected? What sets are conected?

In this way, if we have a disconnected set X, so take $X = A \cup B, A \cap B = \emptyset$, A, B clopen. Then A and B are sort of independent from each other. Notice that $\forall c \subseteq X$, c is open if and only if $c \cap A$ is open and $c \cap B$ is open. So in a sense ,the openness of those sets don't have anything in common – they don't affect each other. Similarly for closed sets too.

Lemma 3.1.4: Connectedness and independence

See previosu paragraph

***Proof*:**
exercice

Connectedness might seem at first intuitive, but it is actually very much based on topology, and we *must* cover weird topological examples to get a better feel of connectivity:

Example 3.1.5: Super Clopen
Take

$$\mathcal{T}_{\text{super-clopen}} = \langle \{ \ldots, [-3, 2], (2, 1), [1, 0], (0, 1)[1, 2], (2, 3), \ldots \} \rangle$$

then the entire real line is covered in open sets. Now consider $[1, 2]$. This set is open. by definition. The compliment of it $(-\infty, 1) \cup (2, \infty)$ is the union of all other elements, so it is *also* open, and therefore $[1, 2]$ is also closed. Therefore it is clopen. In fact, every open set in this topology is clopen. Therefore, the union of any open sets will produce a set which has more than the trivial clopen subset in the subspace topology, and more generally, any set $A \subseteq X$ would have more than the trivial clopen subsets unless A is element of the basis of $\mathcal{T}_{\text{super-clopen}}$ topology. So this topology has a lot of connected components, but anything bigger than that is not connected.

There is an even more extreme example of this, where only single point sets are connected; such a topology is called *totally disconnected.* The discrete topolgoy is an obvious one, but much subtler ones exist! Will will show that a topology generated by intervals (a, b) are connected. BUt, $\mathbb{R}_l$ (the lower-limit topology, generated by $[0, 1)$ intervals) is totally disconnected! You can see that $(\infty, 0) \cup [0, \infty)$ is a disconnection, showing that $\mathbb{R}_l$ is disconneced, and you can then generalize this for arbitrary subset of $\mathbb{R}_l$! This shows that any geometric intuition about connectedness is depdent on the topology in which you aquired that intuition!

In the example above, you might think that this super disconnectedness occured because of a weirdly desinged topology. This need not be the case:

Example 3.1.6: Making the Euclidean Topology a Bit Finer
Take $\mathbb{R}$ as the underlying space. Take this as the basis for a topology:

$$\mathcal{T}_{\text{extremely disconnected}}\langle\{y \in X \big| d(x,y) < \epsilon, \forall\epsilon\}\rangle \cup \langle\{y \in X \big| d(x,y) \leq \epsilon, \forall\epsilon\}\rangle$$

That is, the set of all epsilon balls around every point, and the set of epsilon balls *with* their closure in the Euclidean topology. This set is strictly finer than the Euclidean topology, since every basis element of $\mathcal{T}_{eu}$, is in $\mathcal{T}_{\text{super disconnected}}$, but not the converse. Let's not take an arbitrary interval $I = (a,b)$, and let $r = b - a$. Intervals are connected in $\mathbb{R}$. However, consider

$$S = \left(a + \frac{r}{10}, b - \frac{r}{10}\right) \subseteq I$$

Take the compliment of it in the subspace topology

$$S^c = \left(a + \frac{r}{10}, b - \frac{r}{10}\right)^c = \left(a, a + \frac{r}{10}\right] \cup \left[b - \frac{r}{10}, b\right)$$

which is open in I since $I \cap \left[a - 1, a + \frac{r}{10}\right] \cup \left[b - \frac{r}{10}, b + 1\right] = S^c$, and therefore S is clopen, and so is a clopen subset of I, and so I is *not* connected!! You can do even weirder stuff too if you tried: you can split any ball in half. That could be a good exercice.

Fun fact! This space actually has a name!! spaces where the closure of open sets is open, it's called *extremely disconnected*!

We can even find examples of spaces with topologies we are really familiar with, but the underlying set forces the intersciton between the topology and the set to be disconnected:

Example 3.1.7: Eucdliean Topology on $\mathbb{Q}$
(from tutorial, put some place appropriate). Note that $(\mathbb{Q}, \mathcal{T}_{Eu})$ is totally disconnected (only singletons are connected) If we take any $A \subseteq \mathbb{Q}$, $|A| \geq 2$. Let's say for some two points $x < y$. Then we can find some $\epsilon \in \mathbb{R}$ such that $x < \epsilon < y$. Then $((-\infty, \epsilon) \cap \mathbb{Q}) \cup (\epsilon, \infty)$. Then for any A now, it will be the union of two open sets, meaning that it is there must be more than 1 trivial clopen set.

In all these cases, there was a very small gap between the disconnected spaces, which might lead you to believe all connected spaces have some idea os closeness in common. THIs nead not be the case:

Example 3.1.8: Connected space with large gaps
Let

$$I^2 = I \amalg I, A \subseteq I^2, \mathcal{T}_{I^2} = \langle\{\{(a,0),(0,a)\} \,| a \in \forall(c,d) \subseteq I\}\rangle$$

that is, I^2 is all elements of the form $(a,0)$ and $(0,b)$ for all $a, b \in I$, and the topology on it are all the intervals who "mirror" each other on both sides. Note that this topology strictly coarser than the subsapce topology on I^2 of the product topology. The point of this topology is to "glue" two open sets together so that what used to be two clopen sets in the relative subsapce will not be the only clopen set, making it connected.

In particular:

$$\{((a, b), 0), (0, (a, b))\} \qquad a \neq 0$$

is connected. I'll prove this later, got algebra homework.

Note also that some of the spaces mentioned before are Hausdorff, while others are not. So "oddities" in connectedness will not be fixed if you choose a hausdorff space.

Also, here is an exerice I thought to be cool:

Example 3.1.9: Not Homeomorphic

Show that $(0, 1)$, $(0, 1]$ and $[0, 1]$ are each not homeomorphic to each other

Solution The solution is ingeinous. Take two of them. Assume one is homeomorphic to the other. Then the iamge of a connected set is open. Rmove a single point from the co-domain (make the co-domain one of the closed ones). Then ▼

Suppose that there exists imbeddings $f : X \to Y$ and imbedding $g : Y \to X$. Give an example where that doesn't imply they are homeomorphic

Solution here ▼

Show $\mathbb{R}$ and $\mathbb{R}$ are not homeomorphic if $n > 1$

Solution Same principle as the first one! ▼

This last one shows how $\mathbb{R}^n$ and $\mathbb{R}$ are *not* topologically similar under the Euclidean topology!! Better intuition of this will come with the Fundamental group (chapter 7).

3.1.2 Properties of Connected Spaces

Lemma 3.1.10: Subspace Topology And Disconectedness

If $Y \subseteq X$, and $Y = A \cup B$, $A \cap B = \emptyset$, A, B are non-empty then if $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$, then Y is disconnected. The closure of $\overline{A}$ and $\overline{A}$ are taken in X.

The purpose of this lemma is that it never refers to the subspace topology on Y.

***Proof*:**
Consider $\overline{A} \cap Y$. We know that $A \subseteq \overline{A} \cap Y$, and so

$$B \cap (\overline{A} \cap Y) \subseteq B \cap \overline{A} = \emptyset$$

Where the second subset inclusion because getting rid of Y will naturally make that true, and the last equal is by assumption. Therefore, $A = \overline{A} \cap Y$ is closed. Likewise for B. And so we're done.

One last lemma before we learn how to construct connected spaces. This one should be intuitive

Lemma 3.1.11: Connected Sets Within Disconnected Spaces

Let $X = A \cup B$, $A \cap B = \emptyset$ A, B clopen. Let $Y \subseteq X$. Assume Y is connected. Then either

$$Y \subseteq A \qquad \text{or} \qquad Y \subseteq B$$

Proof. if A is clopen, then $Y \cap A$ is clopen in Y. But Y is also connected, so $Y \cap A = \emptyset$ or $Y \cap A = Y$. If $Y \cap A = \emptyset$ then $Y \subseteq B$. If $Y \cap A = Y$ then $Y \subseteq A$. □

Corollary 3.1.12: Singleton's Are Always Connected

Given any topology and any set, singletons are connected

Proposition 3.1.13: Subspace Of Connected Sets

Let X be connected. Then $A \subseteq X$ need not be connected

Solution Take $\mathbb{R}$ with the Euclidean topology and $[0,1] \cup [4,5] \subseteq \mathbb{R}$. Then $[0,1] \cup [4,5]$ are both clopen, meaning there there is more than the trivial clopen set. ▼

However, there is one condition in which a subsapce is guaranteed to be a connected space:

Proposition 3.1.14: Connected Subspace Of Connceted Space

LEt A be a connected subspace of X. If $A \subseteq Bsse\overline{A}$, then B is also connected. In other words, if B is A and some limit points, it's such that's still connected.

Solution p.150 ▼

Proposition 3.1.15: Finite Product Of Connected Sets

Finite product of connected sets are connected

***Proof*:**
Let X, Y be connected spaces, and take $X \times Y$. Let's say there exists two open subsets of $A, B \subseteq X \times Y$ such that $A \cup B = X \times Y$, and $X \cap B = \emptyset$. Then

$$(A_1, A_2) \cup (B_1, B_2) = (A_1 \cup B_1, A_2 \cup B_2) = (X, Y)$$

implying that $A_1 \cup B_1 = X$ and $A_1 \cap B_1 = \emptyset$ for open set $A_1, B_2 \subseteq X$. But then that would imply X has a seperation, implying its disconnected – a contradiction. Similarly for Y. Thus, $X \times Y$ is connected.

Proposition 3.1.16: Arbitrary product of Connected Spaces

Arbitrary products of connected spaces need not be connected

Solution p.151 example 6. ▼

Proposition 3.1.17: quotient maps and connectedness

LEt $p : X \to y$ be a quotient map. If eahc set $p^{-1}(\{y\})$ is connceted, and Y is connected, then X is connected

***Proof*:**
homework.

3.1.3 Constructing connected spaces

Proposition 3.1.18: points in common between connected sets

Let $\{A_i\}_{i\in I}$ be subsets of X which are connected. Assume $p \in \bigcap_{i\in I} A_i$. Then

$$Y = \bigcup_{i\in I} A_i$$

is connected

Solution ▼

For example $A = \left\{(x, y) \in \mathbb{R}^2 \middle| xy = 0\right\}$ is connected. You can use that previous Lemma

Proposition 3.1.19: Adding Limit Points Keeps You Connected

Suppose $A, B \subseteq X$, A is connected, and

$$A \subset B \subset \overline{A}$$

then B is connected

Solution Tuesday Oct 5th lecture ▼

This next proposition is essentially what the Intermediate Value Theroem is in general topology (theorem 3.1.24). We will get back to this when we get to connectedness of the real line.

Proposition 3.1.20: Connectedness And Image Of Continuous Function

The image under a continuous map of a connected space is connected, that is

$$f : X \to Y$$

continuous, and X connected, then $f(X)$ connected.

Solution This one is not too bad. Start with some $A \subseteq Y$ which is assumed to be clopen. You'll use properties of continuity to and connectedness to eventually reach that $A = Y$ or $A = \emptyset$. ▼

Theorem 3.1.21: Finite Product Of Connected Spaces

A finite product of connected spaces is connected

intuition

Proof. Thursday lecture, beginning □

3.1.4 Connectedness for The Real lines

Let $X = \mathbb{R}$ with the Euclidean topology. We'll now study some special properties of the connectedness on this space

Theorem 3.1.22: $[0, 1]$ Is Connected

The closed interval $I = [0, 1]$ is connected

Proof. This is Heine-Borel but in a different formulation.

As usual with connectedness proofs, we need to show there is only the trivial clopen sets. let $A \subseteq I$ be a clopen set by possibly replacing $A \leftrightarrow I \setminus A$. Without loss of generality, let $0 \in A$. Define

$$m := \sup(b \in I \big| [0, b] \subseteq A)$$

that is, m is the largest number such that $[0, m] \subseteq A$. We'll show that $m = 1$, showing that $A = I$, and thus I only have the trivial clopen sets, and so is connected

First, let's show that $m \in A$ (since the supremum doesn't *have* to be inside A). Assume then that $m \in I \setminus A$. Then $\exists \epsilon > 0$ such that $(m - \epsilon, m + \epsilon) \cap I \subseteq I \setminus A$. THe $\cap I$ is because of the subspace topology (recall that it's open in I if U is an open set, then $U \cap I$ is open in I). This would imply that $\forall b \in (m - \epsilon, m + \epsilon)$, $[0, b] \nsubseteq A$. However, that's a contradiction by the way we defined m, since that would imply that $b \leq m - \epsilon$.

Next, we need to show that $m = 1$. Let's assume $m < 1$. Since A is open, then there exists a $\epsilon > 0$ such that $[m, m + \epsilon) \subseteq A$, which means that $[0, m + \frac{\epsilon}{2}] = [0, m] \cup [m, m + \frac{\epsilon}{2}] \subseteq A$, again a conradiction since m must be the largest such number.

Thus, $A = [0, 1]$ □

Corollary 3.1.23: any interval is connected

$[r_1, r_2]$ is connected, and so is (r_1, r_2), and so $[r_1, r_2)$ and $(r_1, r_2]$ is connected

Note that $\mathbb{R}$ is connected since $\bigcup_{n\in\mathbb{N}}(-n, n) = \mathbb{R}$. We can also show it by

$$f(x) = \frac{1}{1-x} - \frac{1}{x}$$

since it is a homeomorphism between $(0, 1)$ and $\mathbb{R}$.

With that, we can also show that $\mathbb{R}^n$ is connected.

Finally, there's also the generalization of the famous 1st year analysis theorem:

Theorem 3.1.24: Intermediate Value Theorem (Ivt)

Suppose X is connected, and $f : X \to [0, 1]$ is continuous. then $f(X)$ is an interval. That is, for all $x \in [0, 1]$ there exists a $c \in X$ such that $f(c) = x$.

Proof. We've already shown that the image of connected sets are connected, and that what is connected in subspace topologies of $\mathbb{R}^n$ are intervals, or products of, and so we're done. □

Example 3.1.25: Intermedaite Value Theorem

Here are some examples that gave tricks I think are important to know:

1. Let $f : S^1 \to \mathbb{R}$ be continuous. Show that there exists a point $x, -x \in S^1$ such that $f(x) = f(-x)$

   ***Proof*:**

   Let's assume there does not exist a $x, -x \in S^1$ such that $f(x) = f(-x)$, that is, $f(x) \neq f(-x)$ for all $x \in \S^1$. Let A be the upper hemisphere $\cup(1, 0)$, and let B be the lower hemisphere $\cup(-1, 0)$. But, for every $x \in A$, $\exists y \in B$ such that $y = -x$. Then since S^1 is a subspace of a Hausdorff space, it is also Hausdorff, and so for every $x \in A$, we can find two open sets $U_{f(x)}$, $V_{f(-x)}$ such that $U_{f(x)} \cap V_{f(-x)} = \emptyset$. We can do this for every point, forming

   $$U = \bigcup_{a\in A} U_f(a) \qquad V = \bigcup_{b\in B} V_{f(b)}$$

   (Check that $U \cap V = \emptyset$, then take the pre-image, which would cover S^1, but still be disconnected)

2. Let $f : X \to X$ be continuous. Show that if $X = [0, 1]$, there is a point x such that $f(x) =$ x. The point is called a *fixed point* of f. What happens if $X = [0, 1)$ or $(0, 1)$?

***Proof*:**
Let's say $x \neq f(x)$ for all $x \in [0,1]$. The trick is to define this function: $g : [0,1] \to [-1,1]\,, g(x) = x - f(x)$. Then since $[0,1]$ is connected, $g([0,1])$ is connected. Since $0 \notin g([0,1])$, then it cannot be that $g([0,1]) = [-a,0) \cup (0,b)$, since that would be a discontinuity,

so it must either be in the lower or upper region. Without loss of generality, we can say that $g([0,1]) \subseteq (0,1]$. Here is then were we would use the fact that the limit point exists to pull an interesting conclusion!

3.1.5 Path Connectedness

In weird topologies, connectedness can produce results that we don't usually think as "connected" when we think in a Euclidean way (like example (# here once I get referencing working)). We still want to preserve our notion of connectedness from Euclidean space in some manner. If we stick to Euclidean intuitions, we can think something is connected if we never have to "leave" it to get to another point. This is actually how a general counter-example to this intuition for connectedness in general was constructed: by choosing a topology which in the Euclidean intuition is disconnected, but the sets are open.

This particular concept of openness is very useful and usually what most think when they say connected. Thus, we will make our intuition of connected in a Euclidean a name:

Definition 3.1.26: Path Connectedness

Let X be a topological space. A **path** in X is a continuous map $f : [0,1] \to X$. If every two points $a, b \in X$ can be connected by a path, we say that X is **Path-Connected**

3.1.27 Note Munkres lets f be from any closed interval to X, which is functionally the same, but makes book-keeping a lot harder

Example 3.1.28: Path Connected Sets

1. Given $\mathbb{R}^n$ with the Euclidean topology, then for every $x_0 \in \mathbb{R}^n$ $B(x_0)$ is a path-connected set.
2. The unit circle S^1 is path connected
3. Here's a more challenging one if you konw your Algebra. The space $GL_n(\mathbb{C})$ is path connected. (hint: link to the idnetity matrix. If you're still stuck: here)
   (a) If you have cracked the other one, then you can try to show that $GL_n(\mathbb{R})$ is *not* path connected (which will come down to the invertibility matricies in $\mathbb{R}$ are split between $+$ and $-$)

Lemma 3.1.29: Path-Connected $\Rightarrow$ Connected

If X is path-connected, then it's connected

Proof. We'll do the contra-positive: if it's not connected, it's not path connceted. Suppose $X = A \cup B$, $A \cap B = \emptyset$. Let $f : [0,1] \to X$. Then since $[0,1]$ is connected, $f([0,1])$ is connected, so it must be contained in either A or B. Thus, there is no path from $a \in A$ to $b \in B$, thus it's not path connected.

There is also a categorical proof the professor presented. $\square$

We have seen path-connectedness is stronger than connectedness becasue of this proof. The converse does not necessairly hold:

Example 3.1.30: Topologist Sin-curve

The topologists sin curve union $\{0\} \times (-1,1)$ is the typical example of connected, but not path-connected set.

Let $S = \{(0,0)\} \cup \{(x, \sin(1/x)) | 0 < x < 1\}$ and $f = (f_1, f_2) : [0,1] \to S$ is path-connected. We'll show that if $f(0) = (0,0)$, then $f(t) = (0,0)$ for all t.

For the sake of contradiction, assume that $f(t)$ is not always $(0,0)$. For convinence, remove any constant initial part of the interval, so that we get $0 = \sup\{t | f([0,t]) = \{(0,0)\}\}$ (so that we consider the equiavlent function which immediately leaves 0). By theorem 2.2.16, we know that if f is continuous, then so are it's component functions, meaning f_1 and f_2 is continuous. Since f_2 is continuous, then $\forall \epsilon, \exists \delta$, s.t. $|t - 0| < \delta \Rightarrow |f_2(t) - 0| = |f_2(t)| < \epsilon$. Since the supremum of f_2 is 1, let $\epsilon = 1$ so that $|f_2(t)| < 1$ for all $t < \delta$. Since $\delta > 0$, we can pick a t_0 such that $0 < t_0 < \delta$, and where $f_1(t) > 0$ (since this function is oscillating in a undestandable manner for $t > 0$). Since f_1 is continuous *and* it's domain is an interval, then by the IVT (3.1.24) the image of an interval is an interval. Thus, $f_1([0,t_0]) = [0, f_1(t_0)]$. Since $f_2(t) = \sin(1/f_1(t))$ for all t with $f_1(t) \neq 0$ and $\sin(1/x)$ aps $(0,\epsilon)$ onto $[-1,1]$ for all $\epsilon > 0$, it follows that $[-1,1]$ is in the image of f_2 restricted to $[0,t_0]$ – a contradiction to the assumption that $t_0 < \delta$. Thus, there cannot exist a a non-constant path from $(0,0)$.

Keep the topologist sin-curve when thinking about path-connectedness! For example, note that if we just take $(-1,1)$, then it is path connected. This implies that it's not true that if A is path-connected, then $\overline{A}$ is path-connected, sinc the closure of the interval here will produce a non path-connected space. This is interesting, for it shows that closure might not play nicely with path-connectedness since we might use a nasty infinity when approaching a point! [a]

At first, this might seem an unintuitive of a proof but the "finiteness" of this will come from the fact that $[-1,1]$ is something called *compact* in the Euclidean topology. This is important, for that "finiteness" of paths because of the compactness of closed interval in $\mathbb{R}$ will have great consequences.

[a] and since we're limited to $[0,1]$ in the domain, a compact set, messing with infinites like this can break things!

To make the converse hold, we need that a space is connected and *locally* path-connected. This idea that if it holds *locally* plus some extra condition gives us a global concept harkens back to what I said at the begining of section 1.1, where I said that local properties can play a role in prove larger things. We will get back to local-path connectedness shortly (section 3.1.6)

Proposition 3.1.31: Product Of Path-Connected Spaces Is Path-Connected

If $\{X_i\}_{i\in I}$, are path connected spaces, then so is $\prod_{i\in I} X_i$

***Proof*:**
Naturally, we give $\prod_{i\in I} X_i$ the product topology. This proof becomes very easy if we use theorem 2.2.16, where f is continuous to $\prod_{i\in I} X_i$ if and only if every component map is also continuous.

Proposition 3.1.32: image of path-connected spaces is path connected

If X is path connected, then $f(X)$ is path connected

***Proof*:**
This is quite easy if you recall that the composition of continuous function is continuous (proposition 1.2.10).

3.1.6 Local Connectedness

As we mentioned in 3.1.4, if a set is connected, open sets on it "don't interact with others". We make this concept more rigorous, by defining a way of looking at a space by seperating it into it's connected components

Definition 3.1.33: Connected Components

Given X, define an equivalence relation on X by settting $x \sim y$ if there is a connected subspace of X containing both x and y. The equivalence classes are called teh ***components*** (or the connected components) of X

Verifying it's an equivalence class is pretty simple

Theorem 3.1.34: Connected Components Of A Space

The components of X are connected disjoint subspace sof X whose union is X, such that each nonempty connected subspace of X interescts only one of them

***Proof*:**
simple proof – exercise

Definition 3.1.35: Path-Connected Components

Given X, define an equivalence relation on X by settting $x \sim y$ if there is a path-connected subspace of X containing both x and y. The equivalence classes are called teh ***path-components*** of X

Verifying this equivalence relation requries a bit more work, but is worth going over.

We have the equivalent theorem for path-components:

Theorem 3.1.36: Path-Connected Components Of A Space

The path-components of X are connected disjoint subspace sof X whose union is X, such that each nonempty path-connected subspace of X interescts only one of them

***Proof*:**
very similar to 3.1.34

Example 3.1.37: Connected, but not Path Connected

1. Take $\mathbb{Q}$ as a subspace of $\mathbb{R}$. Then since $\mathbb{Q}$ is totally disconnected, every component and path-component are just the singletons
2. Let $\overline{S}$ be the topologist sine-curve. Then $\overline{S}$ has 1 connected component, and 2 path-components: One is the curve S, while the other is eh vertical interval $V = 0 \times [-1, 1]$.

   Super interestingly, if you delete all poitns of V having rational second coordinate, then you get a space that has only one connected component, but *uncountably many* path-components!

Definition 3.1.38: Locally Connected

A space X is said to be ***locally connected at x*** if for every neighborhood U of x, there is a connected neigbhorhod V of x contained in U. If X is locally connected at each of its points, it is said simply to be *locally connected*. Similarly, a space X is said to be ***locally path-connected at x*** if for every neighborhood U of x, there is a path-connected neigborhood V of x contained in U. If X is locally path connected at each of its points, then it is siad to be *locally path connectd*

Example 3.1.39: Locally Connected Sets

very interaval in $\mathbb{R}$ is connected and path-connected, and hence locally connected/path-connacted. The subspace $[0, 1) \cup (0, 1]$ is not connected, but locally connected. The topologists's sine curve is connected but not locally connected. The rationals $\mathbb{Q}$ are neither connected or locally connected

Theorem 3.1.40: Locally Connected If And Only If Open Components

A space X is locally connected if and only if for eveyr open set U of X, each component of U is open in X

Proof:
simply carry out the definition (p.161 Munkre)

similarly

Theorem 3.1.41: Locally Path-Connected If And Only If Open Components

A space X is locally path-connected if and only if for eveyr open set U of X, each path-component of U is open in X

3.2 Compactness

(cool results from Tao's blog here)

Compactness really captures the idea that you only need a finite amount of open sets to represent a set. THis is a super-useful property, especially when you're trying to avoid oddities that come up with weirdness of uncountable spaces (ex. in a non-compact space, you can have a uncountable discrete subspace of X, which can be difficult to work with). My general way of thinking about compactness is that it gets rid of infinities, which can spoil some things.

3.2.1 Basic Properties And Definitions

Definition 3.2.1: Cover

A collection $\mathcal{A}$ of subsets of a space X is said to be a ***cover***, or to be a ***covering*** of X, if the union of the elements of $\mathcal{A}$ is equal to X. it is called an ***open coering*** of X if its elements are open subset of X.

Definition 3.2.2: Compactness

A space X is said to be ***comapct*** if every open covering $\mathcal{A}$ of X contains a finite subcolletion that also a cover

3.2.3 Note Notice that we're talking about the *space*, not a subset of X. We'll get to that soon.

Example 3.2.4: Compact Sets

1. The trivial topology is compact
2. finite spaces are compact
3. the real line is *not* compact ($\mathbb{R} = \bigcup_{n\in\mathbb{N}}(-n,n)$, so no finite subset of that union can cover $\mathbb{R}$)
4. $(0,1]$ is *not* compact.
5. When we worked with connectedness, we saw that adding open sets to a topology might break connectedness. Same in compactness. Let $X = \mathbb{R}$. If $\mathcal{T} = \{\emptyset, X\}$, and $\mathcal{T}' = \mathcal{P}(X)$, then X is compact in $\mathcal{T}$ (since $\mathcal{T}$ is finite), but $\mathbb{R}$ is *not* compact in $\mathcal{T}'$ (since the set $\{x\}_{x\in\mathbb{R}}$ doesn't have a finite-subvcover!) This shows how Compactness is a very topological property! The topology on a set can change whether a set is compact or not. Note that the converse is true: if $\mathcal{T}'$ is a finer topology than $\mathcal{T}$ and X is comapct in $\mathcal{T}'$, then X is compact in $\mathcal{T}$. Let A be a cover for X from $\mathcal{T}$. Then A is also a cover for A in $\mathcal{T}'$. Then A admits a finite sub-cover. Then X is also compact in $\mathcal{T}$.
6. here is an example of a space which has an "infinity" which we'll be able to take advantage to show it's not compact. Take $\mathbb{R}_K$, the K-topology. Recall that this topology is *finer* than the Euclidean topology since it has $\mathcal{T}_\mathbb{R}$ as well as all sets of the form $(a,b) - K$ where $K = \{1/n | n \in \mathbb{N}\}$. Take the set $[0,1]$. We'll show it has no finite subcover. Take:

   $$U = [0,1) - K \qquad n_1 = (\frac{1}{2}, 1], U_n = (\frac{1}{n+1}, \frac{n-1}{)}, n \geq 2$$

   the key here is that U is open!! Take $\mathcal{C} = U \cup \{U_n\}_{n\in\mathbb{N}}$. Then it should be clear that *no* proper subcover (including finite ones)! This should make more intuitive the idea that compactness is about there being no weird infinites, cause if there are we can produce this.

(bonus) You can show that $\mathbb{R}_K$ is connected (Munkre's hint was that $(-\infty, 0), (0\infty)$ inheret their usual properties from $\mathbb{R}$), and show it's *not* path connected

<++>

Definition 3.2.5: Sub-Cover

If $Y \subseteq X$, S is a collection of subsets of X, we say S ***covers*** Y if $\bigcup_{B\in S} B \supseteq Y$

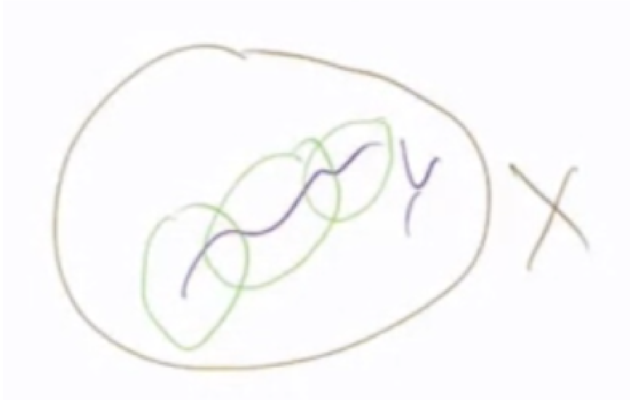

Figure 3.1: subcover

Lemma 3.2.6: Compactness With Cub-Cover

Let $Y \subseteq X$. THen Y is compact if and only if ***every*** S which open covers Y has a finite subset wihch also open covers Y.

Proof. proof both ways:

($\Rightarrow$) We know that $Y = \bigcup_{B \in S}(B \cap Y)$. Therefore, there exists $S' \subseteq S$ such that $Y = \bigcup_{B \in S'}(B \cap Y)$, which means that $Y \subseteq \bigcup_{B \in S'} B$

($\Leftarrow$) Let T be an open cover of Y, so $T = \{S_i \cap Y, i \in I\}$ (I indexing set) for S_i an open set in X. And so

$$Y = \bigcup_{i \in I} T_i \subset \bigcup_{i \in I} S_i$$

And so $S = \{S_i, i \in I\}$ covers Y, and so by assumption there exists a finite $J \subseteq I$ such that $S' = \left\{S_j \middle| j \in J\right\}$ covers Y. so there's a T' which also covers Y.

$\square$

3.2.2 Properties of Compact Spaces

Here are a couple of properties of compact spaces:

Theorem 3.2.7: closed subsets of compact spaces are compact

Let $Y \subseteq X$ be closed, and X be compact. Thne Y is compact

Proof. Let S be a collection of open sets in X which cover Y. So let $T = S \cup \{X \setminus Y\}$ be that cover union the compliment of Y. So T is also an open cover in X, so we can use the fact that X is compact. We know there exists a finite $T \subseteq T$ which is an open cover of X. Finally, now take $T' \cap S$, which is a finite subset of S which covers Y, and so is compact. $\square$

This is actually incredibly powerful! If you are closed and a subspace of a compact space, you are *comapct*! This gives you an idea of the nature of compactness and it's relation to closed sets. So in a sense, we found another way of thinking about closed sets when our space is compact.

The converse is not necessarily true:

Example 3.2.8: Compact But Not Closed

1. Let $X = \{1, 2, 3\}$ with the trivial topology. Then $Y = \{1\}$ is commpact, but *not* closed. You can also take $\mathcal{T} = \{\emptyset, \{1\}, X\}$, in which case, $\{1\}$ is compact, *open not closed*.
2. A little less trivial of an example. Let X be an infinite set with the finite-complement topology. Then every subset of X is compact (worth checking). [a]

[a] The cool thing about this example is that it's T_1 (recall what I said singletons are closed in the weaker T_1 condition?), so it's an example which is almost Hausdroff, which will be the proper condition for the converse.

We will talk about a condition for the converse when we add Hausdorffness.

Next, we will prove an incredibly powerfull theorem. This theorem will essentially be the proof of the Extreme Value Theorem (EVT) once we think of compactness in $\mathbb{R}^n$ (theorem 3.2.30)

Theorem 3.2.9: Continuity and Compactness

The image of a compact space under a continuous map is compact. That is, if $f : X \to Y$ is continuous, and $U \subseteq X$ is comapct, then $f(U)$ is compact.

Proof. Easy proof □

The converse is not necessarily ture:

Example 3.2.10: V compact, $f^{-1}(V)$ not Compact

Let

$$f : \mathbb{R} \to \mathbb{R} \qquad f(x) = 0$$

Then 0 is compact, however $f^{-1}(0) = \mathbb{R}$, which we shown earlier is not compact.

Theorem 3.2.11: Tychonoff's Theorem on finite spaces

If X, and Y are compact, then so is $X \times Y$.

Proof. This will also involve understanding the tube lemma. Worth going over for that lemma at least □

Example 3.2.12: Maps and Compactness

Earlier, we said that $\pi X \times Y \to X$ is *not* a closed map. However, if Y is compact, it *is* a closed map

Solution ▼

This final theorem is useful for characterising compactness using *closed* sets instead of open sets.

Definition 3.2.13: Finite Intersection Property

Let X be a set and $\mathcal{C}$ be a colleciton of subset of X. We say $\mathcal{C}$ has the ***finite intersection peroperty*** (FIP) if for every finite subcolleciton $\{C_1, ..., C_n\} \subseteq \mathcal{C}$,

$$\bigcap_i = 1^n C_i \neq \emptyset$$

Example 3.2.14: Finite Intersection Property

1. if we pick some $x_0 \in X$, then we can let $\mathcal{C}$ be the colleciton of *all* subsets containing x_0, and so any finite subcolelction would also contain x_0.
2. Another would be $\mathcal{C} = \left\{(n, \infty) \middle| n \in \mathbb{Z}\right\}$ since any finite subcollection of this set is non-empty because they're all rays.

Proposition 3.2.15: FIP And Compactness

Let X be a topological space. Then X is compact if and only if every collection $\mathcal{C}$ of *closed* subsets of X which has FIP satisfies $\bigcap_{c \in \mathcal{C}} c \neq \emptyset$

Solution p.169-70 ▼

3.2.3 Compact Hausdorff spaces

Compact Hausdorff spaces are used quite extensively, and so deserve some elaboration on their properties:

Theorem 3.2.16: compact subspaces of Hausdroff Spaces are closed

Suppose X is a Hausdorff space, and $Y \subseteq X$ is compact. Then Y is closed as a subset of X.

***Proof*:**
Let $x_0 \in X \setminus Y$. We will find an open set containing x_0 that's disjoint from Y. Since X is Hausdorff, then $\forall y \in Y \; \exists$ open U_y, V_y in X such that

1. $y \in Y, x_0 \in V_y$
2. $U_y \cap V_y = \emptyset$

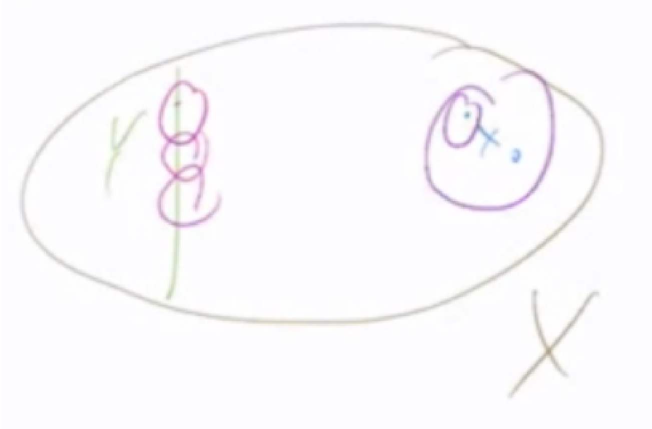

Figure 3.2: subspaceYcompactHausdorff

We can already see what's going on in Y: we're getting an open cover! That is $S = \{U_y | y \in Y\}$ covers Y. Therefore, $\exists S' \subseteq S$ such that S' is finite and S' still covers Y. Since $U_y \cap V_y = \emptyset$, and there is now only a *finite* number of U_y, then we can create a new open set which intersects every V_y:

$$\left(\bigcup_{y \in Y'} U_y\right) \cap \left(\bigcap_{y \in Y'} V_y\right) = \emptyset$$

which means that $V_{x_0} := \bigcap_{y \in Y'} V_y$ is disjoint from Y, so V is open and $x_0 \in V$.

From here, we simply repeated the process for every $x_0 \in X \setminus Y$. This will give us a collection set

$$\bigcup_{x \in X \setminus Y} V_x = X \setminus Y$$

meaning that $X \setminus Y$ is the union of open sets, and so is open. Thus, it's compliment Y is closed, as we sought to show.

This is actually a really powerful theorem if you think about it. If you have a compact Hausdorff space, *every* clsoed set is compact! So finiteness is a little bit everywhere.

Note that The converse is true by theorem 3.2.7. If we get rid of the Hausdorff condition, it is still possible to find spaces in which all clsoed sets are compact, and all compact sets are closed

Example 3.2.17: Compact Hausdorff Spaces
The trivial topology on any set is clearly always compact (and closed). Thus

3.2.18 ll closed sets are compact $\not\Rightarrow$ Compact Hausdorff Space

Lemma 3.2.19: Disjointness of points

If $Y \subseteq X$, Y is compact, X is Hausdorff, $x_0 \in X \setminus Y$. Then there exists open sets U, W such that $x_0 \in U, Y \subseteq W$. (i.e., there's a distance between the two by an pen set)

Corollary 3.2.20: making continuous function homeomorphisms

Let $f : X \to Y$ be bijective and continuous. If X is compact, and Y is Hausdorff, then f is a homeomoprhism

***Proof*:**

Let V be a closed set

So V is compact

So $f(V)$ is compact

So $f(V)$ is closed

so f is a closed map, meaning f^{-1} is continuous

So f is a homeomorphism.

We can use this fact to show the following result:

Corollary 3.2.21: Incomparability of Compact Hausdorff Spaces

Let $\mathcal{T}, \mathcal{T}' \subseteq P(X)$, where $\mathcal{T}$ is a Compact Hausdorff space

1. if $\mathcal{T}'$ is strictly finer than $\mathcal{T}$, then it is compact, but not Hausdorff
2. if $\mathcal{T}'$ is strictly coarser than $\mathcal{T}$, then it is Hausdorff, but not compact

Therefore, if we have the same underlying set X for two topologies, each of which is compact and Hausdroff, then they cannot be comparible.

***Proof*:**

($\mathcal{T}' \subset \mathcal{T}$) Let $\{U_i\} \subseteq T'$ such that $X = \cup_i U_i$. Since $\mathcal{T}'$ is finer than $\mathcal{T}$, then $\exists x_1, ..., x_n$ such that $X = \cup_{j=1}^n U_{x_j}$ still compact.

For the other way, let's take $id : (X, \mathcal{T}) \to (X, \mathcal{T}')$ Note that the identity is a continuous bijection, and so $\mathcal{T}'$ is *nto* Hausdorff, for it it were, then we'd have acontinuous bijjection from a compact space to a Hasudorff spaec, meaning the identity is a homeomorphism, meaning $\mathcal{T} = \mathcal{T}'$, a contradiction.

($\mathcal{T} \subset \mathcal{T}'$) means that $\mathcal{T}'$ is Hausdorff (for easy reasons), then use the identity map again.

We can also weaken compactness to local compactness, which will imply something called regularity. This will be covered once we introduce the concept of regular spaces.

Theorem 3.2.22: Compact In Metric $\Rightarrow$ Closed And Bounded

if (X, d) is a metric space and $Y \subseteq X$ is compact, then Y is clsoed and boudned

Proof. Clsoed because Hasudroff, bounded because compactness means take finite balls, then you can create a bigger ball with an epsilon bigger than all of of those. □

The converse is not necessarily true.

Example 3.2.23: Closed and Bounded, but not Compact
Let $X = \mathbb{R}$, and define

$$d(x, y) = \begin{cases} |x - y| & |x - y| < 1 \\ 1 & |x - y| \geq 1 \end{cases}$$

which makes $(\mathbb{R}, d)$ into a metric space (or the discrete metric too). $\mathbb{R}$ is not compact, but $\mathbb{R} \subseteq B_d(0, 2)$, so it is closed and bounded.

One last thing to note. Being a compact Hausdorff space is an insufficient condition to be a metric space. I couldn't find any example which I can put here without defining a new topology (the order topology), so maybe I'll do that in the introduction another time to explain this:

$\omega_1 + 1$ with the order topology is not metrizable

3.2.4 Compactness of real line

The fact that the any non-finite part of the real line is compact might at first very strange. Surely for an interval $[0, 1]$, there is *some* infinite covering which will not a finite sub-cover. As it turns out, the fact that the "end-points" are in $[0, 1]$, that is, there isn't any point in $[0, 1]$ which we can *perpetually approach without reaching* is the key to making it compact.

Theorem 3.2.24: $[0, 1]$ Is Compact (Heine-Borel)

$[0, 1]$ Is Compact

***Proof*:**
Let $X = [0, 1]$, and let S be an open cover for X. We want to show there exists a finite sub-cover for X. Let

$$T := \{t \in [0, 1] | \exists \text{ finite subset of } S \text{ which covers } [0, t]\}$$

We want to show that $1 \in T$. To do so, define $m := \sup(T)$. We'll show hat $m = 1 \in T$.

1. $m > 0$: Since S is a cover, then there exsits $U \in S$ such that $0 \in U$. Since U is open in $\mathbb{R} \cap [0, 1]$, then U is the union of open intervals intersected with $[0, 1]$, so there must exist a $a \in [0, 1]$ such that $[0, a) \subseteq U$, meaning $\left[0, \frac{a}{2}\right] \subseteq U$. Clearly, U covers $\left[0, \frac{a}{2}\right]$, so we must at least have $M \geq \frac{a}{2} > 0$. Thus $M > 0$.
2. $m = 1$. Let's say $m < 1$. Note that we don't know if $[0, m]$ is compact since m is the supremum, not the maximum. However, we know that $[0, m - \delta]$ *is* compact for appropriate

$\delta > 0$. We'll show that with the properties of the Euclidean topology, we can find a compact closed interval $[0, m + \delta]$ for appropriate $\delta > 0$. Since we're in the Euclidean topology, we know that there must exist some $U \in S$ such that $m \in U$. Thus, $\exists \epsilon > 0$ such that $(m - \epsilon, m + \epsilon) \subseteq U$. The fact that this exists is *key*: it is the core property of the Euclidean topology we're using to show compactness. The fact that there always exists this tinnier epsilon band, and that inside it, we'll have $\left[m - \frac{\epsilon}{2}, m + \frac{\epsilon}{2}\right]$, will carry the proof. Since m is the supremum of value such that all closed intervals $[0, m]$ are compact, take the finite sub-cover which covers $\left[0, m - \frac{\epsilon}{2}\right]$. Let $S' \subseteq S$ be the finite sub-cover which covers $\left[0, m - \frac{\epsilon}{2}\right]$. Then define

$$S'' := S' \cup U$$

Then

$$\left[0, m - \frac{\epsilon}{2}\right] \cup U = \left[0, m - \frac{\epsilon}{2}\right] \cup (m - \epsilon, m + \epsilon) \subseteq \bigcup_{V \in S''} V$$

$$\Rightarrow \left[0, m + \frac{\epsilon}{2}\right] \subseteq \bigcup_{V \in S''} V$$

meaning $m + \frac{\epsilon}{2} \in T$, but then m wouldn't be the supermom, a contradiction. Thus, $m = 1$.

3. $m \in T$. Finally, we must show that the supermom is inside T. As before, $\exists U \in S$ such that $m \in U$ $(1 \in U)$, so again by the property of the subspace topology of $[0, 1]$, exists some $\epsilon > 0$ such that $(m - \epsilon, m] \subseteq U$. Since m is the supremum, we know that there exists a finite sub-cover which covers $[0, m - \frac{\epsilon}{2}]$. Let this finite cover be $S' \subseteq S$. Then:

$$S'' = S' \cup \{U\}$$

Then S'' covers $[0, m] = [0, 1]$, completing the proof.

If you didn't understand this proof in MAT257, this is probably the time to take a moment and ponder on this proof here. It is in a sense a novel proof. Out of necessity, Now that we've proved this, you can re-visit the Topologist sin-curve and check how this "finiteness" is important in the proof of it

Corollary 3.2.25: $[a, b]$ Is Compact

$[a, b]$ Is Compact

Proof. $[a, b]$ is either empty, finite, or homeomorphic to $[0, 1]$ □

Corollary 3.2.26: higher dimensions

$[0, 1]n \subseteq \mathbb{R}^n$ is compact

Proof. We showed the product of two compact spaces are compact, so it's easy enough to show that the product of finitely many compact spaces are compact □

Theorem 3.2.27: Closed And Bounded In $\mathbb{R}^n$ Imply Compact

$A \subseteq \mathbb{R}^n$ is compact if and only if A is closed and bounded (for the Euclidean metric)

***Proof*:**

($\Rightarrow$) That it's compact means that impies it's closed and bounded comes from the fact that A is a subspace of a hausdorffspace $\mathbb{R}^n$, thus by 3.2.22 it is closed and bounded

($\Leftarrow$) Now we need to show that in $\mathbb{R}^n$, closed and bounded are also *sufficient* conditions for compactness in $\mathbb{R}^n$. This is actually an easy consequence of what we've already shown. If you are bounded, then there isn't any "rays" when taking the product of $[a, b]$. Since we are closed, A will be the union of:

$$[a_i, b_i]^n$$

And since every $[a_i, b_i]^n$ is compact, and the union of *finitely* many of these will remains compact (notice we're dealing with the finite union of closed sets, so we don't need to worry abou the infinite case), thus it will be compact.

3.2.28 Note You might've grown accustom to closed and bounded implies compact, and maybe you think in some nice spaces this is the case (like Hasudorff, or metric spaces). This really need not be the case! Take the Euclidean metric on the rational numbers. Closed and bounded is *not* compact. More generally, any non-complete space (that is, spaces were Cauchy sequences are not convergent sequences) will not have this implication hold in general. If you know some analysis already, then you might be interested in konwing that completeness is *not* a topological property, meaning this implication is not a completely topological one!
Even in cases where there is a complete metric, it's not necessarily the case, though examples of those are more complicated [a].

The condition for the converse being true is unfortunately beyond this course [b]

[a] no infinite-dimensional Banach Spaces as metric spaces have the property that closed and bounded imply compact

[b] A metric space has the "Heine-Borel" property if and only if it is complete, σ-compact, and locally compact. We don't define σ-compactness in this document

Example 3.2.29: Useful Example
The power of this theorem in $\mathbb{R}^n$ should no be understated. For example, S^n is compact. You can imbedd it in $\mathbb{R}^{n+1}$, and there is a closed and bounded set! This is thus a really powerfull tool in the right context.

These theorems make the following theroem a one-liner:

Theorem 3.2.30: Extreme Value Theorem (EVT)

If X is compact, and $f : X \to \mathbb{R}$ is continuous, then $f(X)$ achieves it's max and min

Proof. $f(X)$ is also compact, and so in $\mathbb{R}$ is closed and bounded. $\square$

3.2.5 Compactness and metric spaces

As with many things in topology, we like to find concepts which let's us detect properties of a space. Many times, this happens when two concepts merge given a nice enough space. This is exactly what happens in compact metric spaces, with the concepts we're about to introduce

Define uniform continuity. Why in metric space. Brief word on generalization.

next,

$$\text{uniform continuity} \Rightarrow \text{continuity}$$

example

when the two merge: compact metric spaces

$$\text{uniform continuity} \Leftrightarrow \text{continuity}$$

3.2.6 Locally Compact

This is not very important for the course, but it was introduced in tutorial and I personally find it intersting.

The coolest thing introduced in this section is the idea of taking a space, and making it a subspace of a *compact* space. One we get to Stone-Čech compactification.

Definition 3.2.31: Local Compactness

A space X is said to be *locally compact at x* if there is soe mcompact subsapce C of X that contains a neighborhood of x. If X is locally compact at each of its points, X is said simply to be ***locally compact***.

Local connectedness has that for *any* neigbhorhood around a point, there is a connected set that is contained in that neighborhood. This conditions is thus in a sense *weaker*. It will be satisfied once we add the condition that X is a Hausdorff space, but why weaken it in the first place? It will be to introduce the idea of *compactification*

Theorem 3.2.32: 1-Point Compactification

Let X be a space. Then X is locally compact Hausdroff if and only if there exists a space Y satisfying the following conditions

1. X is a subspace of Y
2. The set $Y - X$ consists of a singel point
3. Y is a compact Hasudroff space

If Y and Y' ar etwo spaces satisfying these conditions, then there is a homeomorphism of Y with Y' that equals the identity map on X

***Proof*:**
p. 183

Definition 3.2.33: One-Point Compactification

IF Y is a compact Hausdorff spaec and X is a proper subspace of Y whose closure equals Y, then Y is said to be the ***compactification*** of X. If $Y - X$ equals a single point, then Y is called the ***one-point compactification of*** X

Example 3.2.34: One Point Compactification

1. THe one point compactification of the real line is homeomorphic with the circle!
2. Similarly, the one-point compactification of $\mathbb{R}$ is homeomorphic to S^2
3. One might ask if the one-point compactification of $\mathbb{C}$ is the same as $\mathbb{R}^2$, since both are homeomorphic. And indeed they are, but since their algebraic structure differs, we usually don't think of it in this way, and say $\mathbb{C} \cup \{\infty\}$ is the compactification of $\mathbb{C}$, and often call it the *Riemann sphere* or *extended complex plane.*

The rest of these results are trying to match the usual notion of "locality" when a definition of "locally X" is given

Theorem 3.2.35: Locally Compact in Hausdroff Space

Let X be a Hausdorff space. Then X is locally compact if and only if given x in X, and given neigbhorhood U of x, there is a neigbhorhood V of x such that $\overline{V}$ is comapct and $\overline{V} \subseteq U$

***Proof*:**
p.185

Corollary 3.2.36: closed/open subsets of X also locally compact

Let X be a locally comapct Hausdrff space; let A be a subspace of X. If A is clsoed in X or open in X, then A is locally compact

***Proof*:**
p.185

Corollary 3.2.37: Compact Hausdorff space and locally compact

A space X is homeomrophic to an open subspace of a comapct Hausdorff space if and only if X is locally compact Hausdorff

***Proof*:**
This directly follows from theorem 3.2.32 and corollary 3.2.36

3.3 Nets

I'll eventually tranfer my notes on sequences from 257 here. Essentially,

1. If a space is Hausdorff then sequences converge to at most 1 point
2. a point is in $x \in \overline{A}$ if there exists a sequence $\{x_n\}$ in A such that $\{x_n\} \to x$
3. If $f : X \to Y$ is continuous, then f preserves sequences.

If we add the condition of first countability to each space, then these become if and only if's. However, we would want a construction that doesn't require us to add a condition to our space for these to be if and only if's. We Thus present this generalization of sequences

Definition 3.3.1: Net

A *net* is a function $\eta : S \to X$ whose domain is a directed set. A *directed set* is defined to be a pair $(S, \leq)$ where S is a set and $\leq$ is a relation on S satisfying

1. For all $s \in S$, $s \leq s$
2. for all $s, t, u \in S$, $s \leq t$, $t \leq u$ then $s \leq u$
3. For all $s, t \in S$ there exists $u \in S$ with $s \leq u$ and $t \leq u$

Definition 3.3.2: Filter

A *filter* on a set X is a collection $\mathcal{F} \subseteq 2^X$ that is

1. donward direct: $A, B \in \mathcal{F}$ implies there exists $C \in \mathcal{F}$ such that $C \subseteq A \cap B$
2. noempty: $\mathcal{F} \neq \emptyset$
3. upward closed ;$A \in \mathcal{F}$ and $A \subseteq B$ implies $B \in \mathcal{F}$

a *proper filter* has the additional property that

4) there exists $A \subseteq X$ such that $A \notin \mathcal{F}$

Definition 3.3.3: Net Converging

We say that η *converges* to x if and only if its eventually a filter

$$\mathcal{E}_\eta := \{A \subseteq X | \exists t \in S \text{ such that } s \leq t \Rightarrow \eta s \in A\}$$

Look at *Topology: A Categorical Approach* chapter 3 for more details. Should add here.

4

Expanding Hausdorff Property: Separations axioms

A lot of nice things started to happen when we introduce the Hausdorff condition on spaces. However, that still let's in some edge cases which do not confrom with the Euclidean topology intuitions we've built up in analysis courses. $\mathcal{T}_{\mathbb{R}}$ will have a stronger condition, called *normal* (or T_4) which will let us use our intuition we've built up before. We'll now build up from the Hausdorff condition to the normal condition to show what other nice results happen while strengthening the Hausdorff condition. Once we reach the normal condition, we'll show some really intersting consequences of normal the existence of normal spaces!

4.1 Regularity and Normality

Definition 4.1.1: Regular

Suppose X is a topological space in which one-point sets are closed. X is ***regular*** if given $x \in X$, and a closed set B in X such that $x \notin B$, then there exists disjoint open sets U, W such that $x \in U$ $B \subseteq W$.

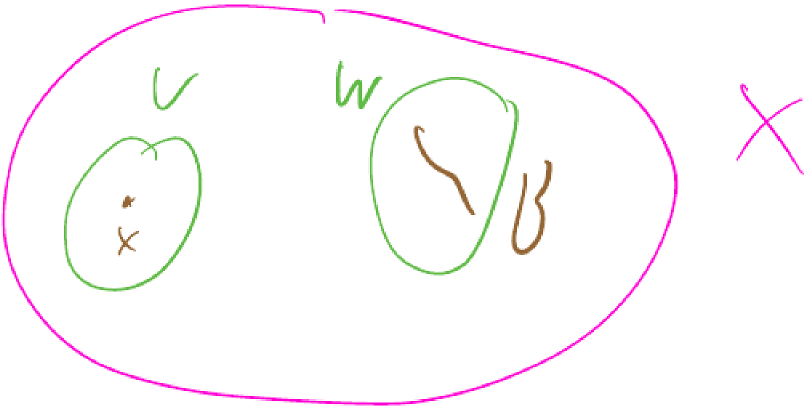

Figure 4.1: visual from lectures

4.1.2 Note Note that we are starting with the assumption that one point sets are closed. The converse of this, though it might seem intuitive enough, is not necessarily true (perhaps the trivial topology is a "pathological" example)

Example 4.1.3: Hausdorff but Not Regular Space

Here's a Hausdorff space that is not regular. One example is the space $\mathbb{R}_K$ constructed by Munkres. The underlying set is the reals, and the basis is chosen as the usual open intervals, along with all sets of the form $(a, b) \setminus K$ where $K = \{1/n \big| n \in \mathbb{N}\}$. This topology is clearly finer than the usual topology on the reals, and the reals form a Hausdorff space under the usual topology, so $\mathbb{R}_K$ is also Hausdorff (passing to a finer topology preserves Hausdorffness).

However, $\mathbb{R}_K$ is not a regular space. For instance, the closed subset K in this space (closed because its complement is open by construction in this topology) and the point 0 cannot be separated by disjoint open subsets.

Definition 4.1.4: Normal

X is ***normal*** if we give disjoint closed subspace $A, B \subseteq X$, then there exists disjoint open subsets u, W such that $A \subseteq U$ and $B \subseteq W$

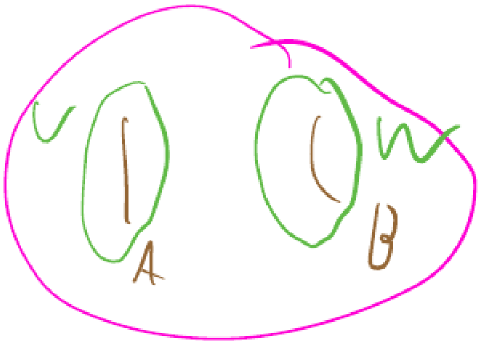

Figure 4.2: visual from lecture

Don't let image decieve you that we're going up "dimensions" of things (i.e. from point to line), but rather we're going from points to *closed set*. This might seem obvious if you just read it, but the visual is a bit miss-leading. There isn't something "higher" dimensionally than a open or closed set (perhaps the entire space, but you can't really define much there)

Normal is a stronger condition than regular. For example take

$$\mathcal{T} = \{\text{here}\}$$

Then $\mathcal{T}$ is regular, but not normal.

4.1.5 Chain of Implications iven that points closed, then:

$$\text{Normal} \Rightarrow \text{Regular} \Rightarrow \text{Hausdorff}$$

Lemma 4.1.6: equivalent formulations ("zoom-in" axiom)

Let X be a topological space such that one-point sets are closed then

1. X is regular if and only if given $x \in X$, and open U containing x, $\exists$ open set V such that $x \in V$, and $\overline{V} \subseteq U$
2. X is normal if and only if given a closed set $A \subseteq X$ and open set U containing A, then $\exists$ open set V such that $A \subseteq V$ and $\overline{V} \subseteq U$

Proof. **intuition**:

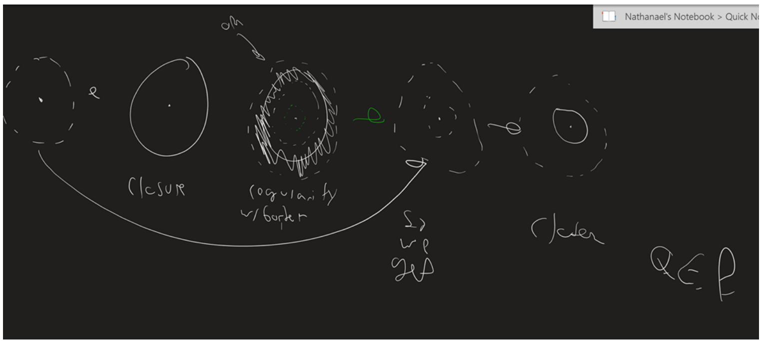

Figure 4.3: visual illustration of proof

Similarly for normality, except you can start with the closed set A instead of the center point □

Note that regular spaces behave nicely under subspaces and product spaces

Theorem 4.1.7: subspaces of Regular Spaces

A subspace of a regular space is regular.

Theorem 4.1.8: Product of Regular Spaces

A Product of regualr spaces is regular.

Proof. Munkre p.197 □

However, Though normal spaces have a stronger condition, they do not behave as nicely!

Example 4.1.9: Sorgenfrey Plane

For example, take the Sorgenfrey plane:

$$\beta_{\mathbb{R}_l^2} = \left\{[a, b) \times [c, d) \middle| a < b, c < d, a, b, c, d \in \mathbb{R}\right\}$$

This space is not normal. TO show this p.198 of Munrkes (long proof)

Note that Since $\mathbb{R}_l$ is regular, then $(\mathbb{R}_l)^2$ is also regular. Normal spaces are in a sense worse than regular spaces because they don't have this condition! I like to think of it as if one adds further restrictions on a space (or a concept) then there is less flexibility given because more conditions need to be satisfied in order for things to work.

However, one one must note that the converse is actually true, that is, if the product space is normal, then the spaces are normal

Proposition 4.1.10: $X \times Y$ Normal Implies X, Y Normal

title

Solution If A, A' are disjiont closed subsets of X, then $A \times Y$, $A' \times Y$ are disjiont closed of $X \times Y$. Therefore, we can use normality and find O, O' open disjoint neighbourhoods. Write $O = \bigcup_i U_i \times V_i$, and $O' = \bigcup_j U'_j \times V'_j$ where eac hU_i, U'_j is open in X and V_i, V'_j is open in Y. Pick $y \in Y$. Then

$$\bigcup \{U_i | y \in V_i\} \qquad \bigcup \{U'_j | y \in V'_j\}$$

are disjiont open neighbourhoods of A and A' ▼

There is one weird space I found that is said to be "vacuously normal" but not regular or Hausdorff, which confuses. me. Let $S = \{0, 1\}$ be the underlying set. Then Take the following is as the open sets:

$$\{\emptyset, \{1\}, \{0, 1\}\}$$

Note that the closed sets are:

$$\{\emptyset, \{0\}, \{0, 1\}\}$$

This space is not Hausdorff since $\{\}$ is not closed. However, normality looks like it applies.

4.2 What spaces are normal?

Though regular spaces act more nicely when being manipulated, normal spaces on themselves are strong enough to have lots of strong equivalent implications.

Lemma 4.2.1: Metrizable Spaces Are Normal

Metrizable spaces are normal

***Proof*:**
Let X be a metrizable space. Assign it the metric d. Let A, B be disjoint closed sets. We want to show that there exists two open sets (U, V), each containing their respective closed sets, each open. Take an $a \in A$. Then choose an $\epsilon_a > 0$ such that $B_{\epsilon_a}(a) \cap B = \emptyset$. This follows from $a \notin B$ and B being closed, so the limit poitns of A and B so they both have their border. If there was a point $a \in \partial A$ and an ball $B_\epsilon(a)$ that intersected B, then we can shrink the epsilon ball by the Hausdorff property of Metric spaces. So $d(a, B) > 0$. This is the key observation. From here the proof develops quite naturally. Take

$$U = \bigcup_{a \in A} B(a, \epsilon_a/2) \qquad \text{and} \qquad V = \bigcup_{b \in B} B(b, \epsilon_b, 2)$$

what's left to show is that they do not intersect. Take $x \in U \cap V$. Then

$$z \in B(a, \epsilon_a/2) \cap B(b, \epsilon_b/2)$$

for some $a \in A$, $b \in B$. Then by the triangle inequality, and some manipulating of inequalities, we get

$$d(a, b) < \frac{\epsilon_a + \epsilon_b}{2}$$

If $\epsilon_a \leq \epsilon_b$, then $d(a, b) < \epsilon_b$, so the ball $B(b, \epsilon_b)$ contains the point a, a contradiction. Similarly if $\epsilon_b \leq \epsilon_a$. Thus, the two open sets are disjiont, proving normality observation

There is a second proof that I found here.

Lemma 4.2.2: Compact Hausdorff Spaces Are Normal

Compact Hausdorff Spaces Are Normal

Proof. longer proof. Oct 22nd. □

If we weaken the condition of compactness a bit to locally compact, we'll get that this implies it's regular:

Lemma 4.2.3: Locally Compact Hausdorff Spaces Are Regular

Locally compact Hausdorff spaces are Regular

***Proof*:**
p.202

Theorem 4.2.4: Second-Countable Regular Implies Normal

Every second-countable regular space is normal

4.2.5 Note the converse is not true: normal spaces are *not* necessarily second-countable. Take the example from Munkres.

***Proof*:**
This is essentially the needed visual:

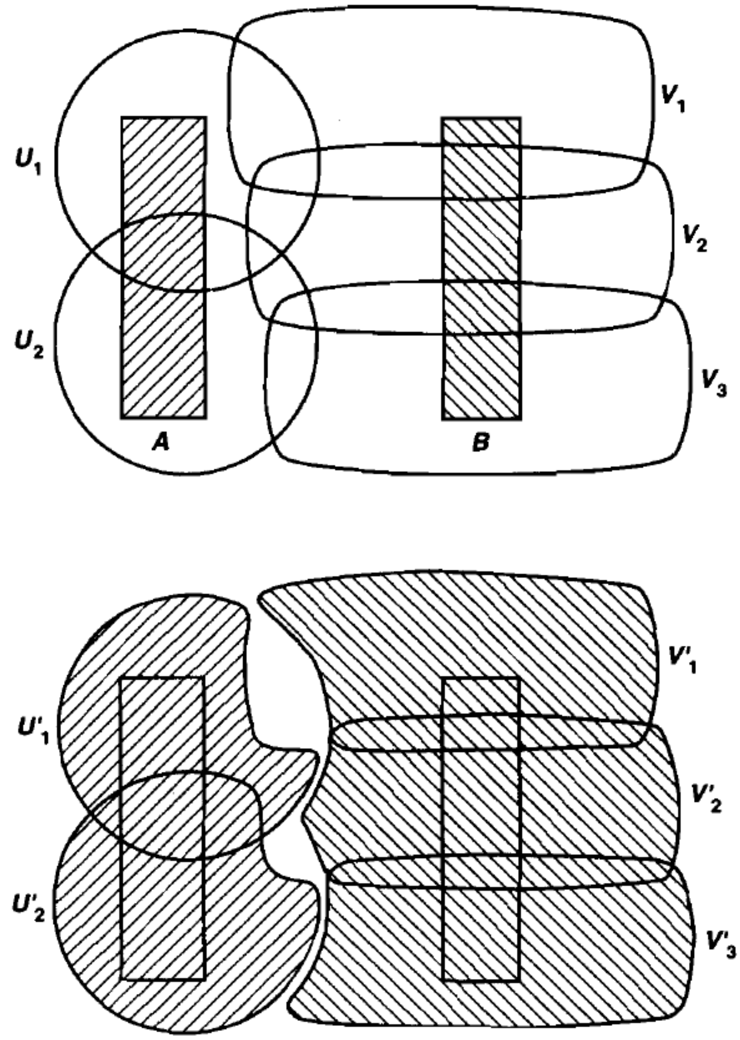

Figure 4.4: visual from Munkre

Proposition 4.2.6: Metrizable implies normal

Metrizable implies normal

***Proof*:**
here

However, normal does not imply metrizable:

Example 4.2.7: Normal, but not Metrizable, Spaces
https://math.stackexchange.com/questions/157906/normal-non-metrizable-spaces

4.3 Urysohn Lemma

Urysohn lemma is one of the larger topological insights, and has a great multitude of applicatoin in other very large and important thoerems (hence why it's called a lemma, it's like the key step for many important theorems).

Theorem 4.3.1: Urysohn

Let X be a normal space. Let A, B be disjoint closed sets. THere exists a continuous function $f : X \to [0, 1]$ st $A \subseteq f^{-1}(0)$, and $B \subseteq f^{-1}(1)$

Proof. p.207-210 □

Example 4.3.2: Urysohn's Lemma
Let X be a normal connected space. Then X must have uncountably many poitn

Solution Let X have more than 1 point. Then $x, y \in X$. Since X is Normal, then ▼

4.4 Metrization Theorem

As a consequence of Urysohn' Lemma, we have a couple of very neat results:

Theorem 4.4.1: Urysohn Metrization Theorem

If X is regualr and second-countable, then X is metrizable

Proof. here □

4.5 Tiezte Extension Theorem

Theorem 4.5.1: Tietze Extension Theorem

Let X be a normal space and let $A \subseteq X$ be a closed subset. let $f : A \to [0, 1]$ be a continuous funciton. THen there exists a continuous function $F : X \to [0, 1]$ such that $F|_A = f$.

Similarly if we replace $[0, 1]$ by $\mathbb{R}$

Proof. here □

5

Elements of Functional Analysis

See Munkres topology

6

Dimension Theory

See Munkres Topology

7

Fundamental Group

Recall how connectedness captured some idea of not being able to break down your set, and path-connectedness gave us the stronger notion that you must be able to "get" to any point within a "finite" amount of time (refer to topologist sin-curve for intuition). The key idea in this section is the generalization of the concept of path-connectedness from points to other topological spaces – in particular, we'll be interested in the loops we can form on spaces. This concept will turn out to be more useful because there will be many spaces which can be classified by seeing what loops can actually collapse down to a single points in a continuous fashion. For example, on a sphere, any loop can collapse to a single point, but on a torus, the loop going around the way you would grab it cannot. This turns out to be a naturally topological distinction. [And this generalization of path-connectedness is the right way to go because we are looking for some properties of a component, maybe? Will think about how this ties in later because I think the intuition tied to path-connectedness will be really cool].

Note that from now on, for notational simplicity, we'll take the convention that

$$I = [0, 1]$$

7.1 Long build-up

There is a lot of little pieces that need to fit together to see the use of the fundamental group. First, here's the basic ontology for this area:

Definition 7.1.1: Homotopy

Let X, Y be topological spaces, and suppose we have two maps f, f' which are continuous maps from X to Y. We say that f is ***homotopic*** to f' if there is a continuous map $F : X \times I \to Y$ such that

$$F(x, 0) = f(x), \qquad F(x, 1) = f'(x)$$

F is called a homotopy between f and f', and we write this as $f \simeq f'$

Basically, if we can continuously deform one shape into another, the two shapes are homotopic! Note that the topology from which we define continuity ($X \times I$) has *both* the topology of X and the Euclidean topology to define it. I suspect the "continuous deformation" is more reliant of the Euclidean portion, though I'll confirm this with further pondering. Also, it is worth pointing out that though I said its continuous deformation of the *shape*, technically, two *functions* are homotopic – homotopy is defined on the function space. Thinking about the graphs of functions is where the continuous deformation of shapes intuition comes in.

Example 7.1.2: Homotopy

1. Let $f, g : \mathbb{R} \to \mathbb{R}^2$, $f(x) = (x, x^3)$ and $g(x) = (x, e^x)$. Then the map $F : R \times I \to \mathbb{R}^2$,

$$F(x, t) = (x, (1-t)x^3 + te^x)$$

   Is a homotopy. $F(x, 0) = (x, x^3)$, and $F(x, 1) = (x, e^x)$, the way is easily checked to be continuous. Thus

$$f \simeq g$$

2. Take $f : I \to I$ $x \mapsto x^2$. Then

$$F(x, t)((1-t)x^2 + tx)$$

   is a homotopy. More generally, x^2 can be any polynomial, and this homotopy will work.

3. Take $f : \mathbb{R} \to \mathbb{R}$ be a continuous function, and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 0$. Then:

$$F : \{\mathbb{R}\} \times I \to \mathbb{R} \qquad F(x, t) = (1-t)f(x)$$

   is a continuous function, with $F(x, 0) = f(x)$ and $F(x, 1) = 0$. Thus, every continuous function on $\mathbb{R}$ is homotopic to the constant function.

   If we think of the graphs of the functions, the point $\{0\}$ with the trivial topology any shape of $\mathbb{R}$ with the Euclidean topology can be continuously deformed to on another.

When function's are homotopic to the constant function, we give function a special name:

Definition 7.1.3: nullhomotopic

If f is homotopic to a constant map, we say that f is ***nullhomotopic***

The idea of nullhomotopy is saying that if you deform your space in a continuous such that you can bring it to a constant point, then we can think of the two being, in some sense, topologically trivial [1].

[1] not necessarily trivial in the most general sense, because compactness, normality, connectedness, and all the other

You perhaps noticed I used the same idea for my homotopic examples of the $(1-t)a + tb$ amendment, that is, the usual function that creates *paths* between two things (recall in MAT257 how you used this function in the proof of MVT?). We give Homotopies that use this a name

Definition 7.1.4: Straight-Line Homotopy

If F is a homotopy such that it's construction is:

$$F(x,t) = ((1-t)f(x) + tg(x))$$

then it is a Straight-line homotopy

You might've also noticed that the notation was basically an equivalence notation ($\sim$). That is actually precisely how you should think about what a two homotopic paths are: *equivalent up to continuous deformation*

Lemma 7.1.5: Equivalence Relation With Homotopy

For fixed X, Y, $\simeq$ is an equivalent relation

***Proof*:**
We will prove the 3 properties for an equivalence relation:

(symetry) We need to show that $f \simeq f$. The identity map would do

(reflectivity) We need to show that $f \simeq f' \Rightarrow f' \simeq f$. the $1-t$ trick will do

(transitivity) We need to show that $f \simeq f', f' \simeq f'' \Rightarrow f \simeq f''$. Going twice as fast will do.

Since we have an equivalence relation, we an bring in our intuitions about them and form an equivalence class

Definition 7.1.6: Equivalence Class Of homotopies

We write $[f]$ for the equivalence class of f with respect to $\simeq$

A lot of study is on homotopies, which will eventually bring you to homotopy groups, homology, exact sequences of homotopic groups and Hopf fibration of the sphere. For now, we'll work with homotopies which are "1-dimensional" (which we can eventually generalize). Working with one-dimensional one's doesn't mean we won't be working with higher-dimensional objects, but the path in the space we use that will be homotopic will be one dimensional.

7.1.1 Path-Homotopy

The natural one-dimensional object to work with then is a path:

properties might still be present in this "trivial" case, but it's trivial when it comes to trying to form to bending it around

Definition 7.1.7: Path-Homotopic

Two paths $f, \tilde{f}$ in X are ***path-homotopic*** if

1. these have the same initial and final points: $f(0) = f'(0)$, $f(1) = f'(1)$
2. There exists a continuous function $F : I \times I \to X$ such that
$$F(s,0) = f(s), F(s,1) = f'(s), F(0,t) = x_0, f(1,t) = x_1$$
where $f(0) = f'(0) = x_0$ and $x_1 = f(1) = f'(1)$. We write
$$f \simeq_p f'$$
F is called a ***path-homotopy***

The professor also said something I thought interesting: That in a sense "we got one minute of time to change it, and we must do it in that time". I think there's some truth to that because you somehow can't go on "forever" perhaps.

Example 7.1.8: More Homotopies

1. If we take all function $I \to I$ such that $x \mapsto x^n$, then all of these are path-homotopic. In fact, any path such that $f(0) = g(0) = 0$, $f(1) = g(1) = 1$ will be path-homotopic by the straight-line homotopy
2. More generally, if C is a convex set, then any two paths with the same initial and ending points are path-homotopic by the straight-line homotopy
3. "There must be more than the straight-line homotopy" you might be thinking, and you're right.
4. "Are there paths that are not homotopic?". That can be the case too. Take $X = \mathbb{R}^2 - \{0\}$ and
$$f(s) = (\cos(\pi s), \sin(\pi s)) \qquad g(s) = (\cos(\pi s), 2\sin(\pi s))$$
then these two functions are path homotopic using the straight-line homotopy, as you see visually:

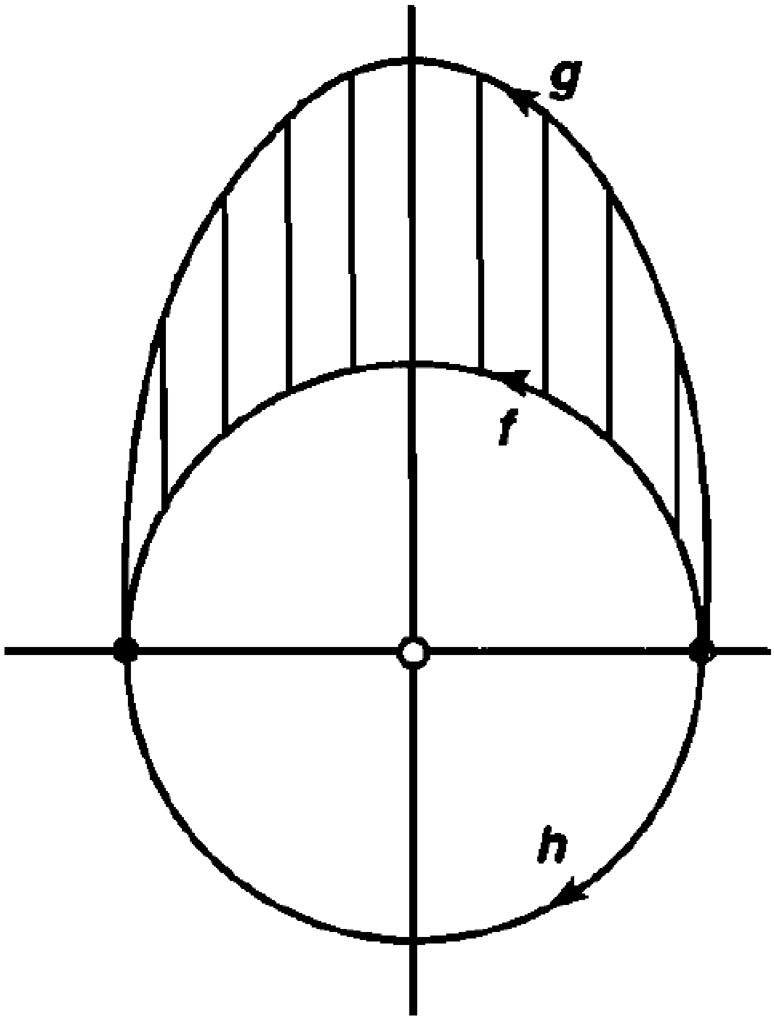

Figure 7.1: Same initial and end points, but one goes under point

In general, if the point 0 was not missing, we can have:

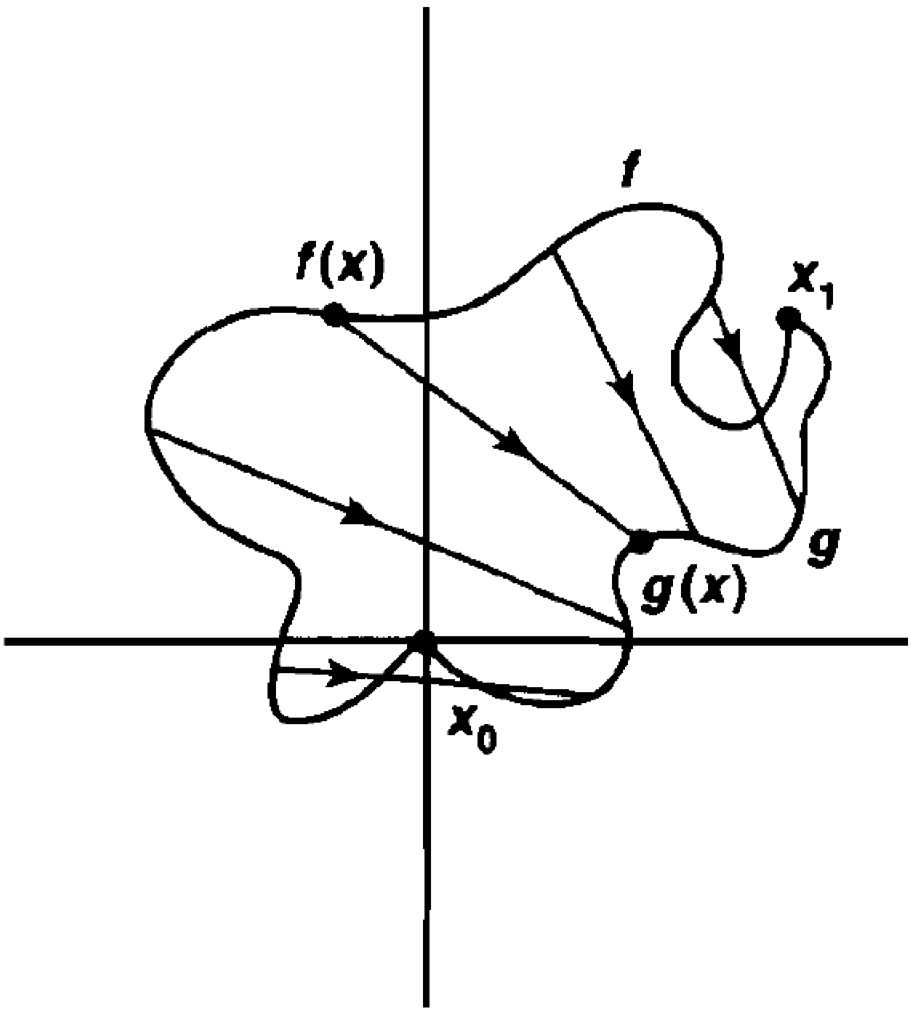

Figure 7.2: Convex space

However, the path $h(s) = (\cos(\pi s), -\sin(\pi s))$ is *not* path-homotopic. In the visual, this is the h path. It should perhaps seem intuitive that we we try to pass a continuous path *through* the point 0, we will introduce a discontinuity. The proof for this is a bit laborious at this point, so being satisfied with this "hole" intuition is useful.

Note that the equivalence relation constructions works just as well on path-homotopies, just more book-keeping. Because of the lower-dimension, the two will *not* define the same equivalence classes, so we define the equivalence

Definition 7.1.9: Equivalence Class Of Path

If f is a path in X, we write $[f]$ for the equivalence class of f with respect to $\simeq_p$

Example 7.1.10: Homotopy Equivalence

Recall all polynomials $x \mapsto x^n$? from earlier? These are in the same equivalence class

$$[f] = [g]$$

And since $F'(x,t) = (x, (t-1)x + t0)$ is also a path-homotopy,

$$[id_0] = [f] = [g]$$

meaning it is also *nullhomotopic*

7.1.11 Note This is an equivalence class, but it will not necessarily form a quotient spce: We are not forming an equivalence class on the points, but on the functions on the space. You'd neeed to define a topology on the paths in order to define a quotient topology on that function space.

7.1.2 Algebraic build-up

As mentioned in the note, this new collection of equivalences classes will not necessarily be a topology. However, we *can* form Algebraic structures one them! To start, we'll need to define the basic's that are present in algebraic structures: identities and a binary operation:

Definition 7.1.12: identity path

If $x \in X$, denote e_x to be the constant path $e_x : I \to X$ such that $e_x(t) = x$ for $t \in I$

Therefore, the *id* in our previous example would be relabeled as e_0.

Next, we need to figure out a way defining a relation between paths. The natural candidate for this would be to extend one path to the next. However, we can extend *arbitrary* paths from one to the next, since that will make discontinuous paths (because the end point of one path might not be the initial point of another). Thus, we we try and define paths by through "combination", we need to impose this limit:

Definition 7.1.13: combining paths Operators: $*$

If f, g are paths in X and $f(1) = g(0)$, define $f * g$ by

$$(f * g)(t) = \begin{cases} f(2t) & t \in [0, \frac{1}{2}] \\ g(2t-1) & t \in [\frac{1}{2}, 1] \end{cases}$$

Example 7.1.14: Paths
Try thinking of any two paths in $\mathbb{R}^2$ and combining them into 1.

Lemma 7.1.15: Well-defined: $*$ Preserves Equivalence Relation

if $f \simeq_p f'$, $g \simeq_p g'$, then
$$(f * g) \simeq_p (f' * g')$$

Proof. Let F, G be homotopies between f and f', g and g' respectively. Let $H : I \times I \to Y$. Basically, we'll just be doing just like in the equivalence relation, doubling the speed appropriately and using the pasting lemma to combine the two continuous functions:

$$H(x,t) = \begin{cases} F(2x,t) & t \in [0, \frac{1}{2}] \\ G(2x-1,t) & t \in [\frac{1}{2}, 1] \end{cases}$$

then note that $H(x,0) = f * g$ and $H(x,1) = f' * g'$, and H is continuous by the pasting lemma, and so we're done $\square$

Because of this, we can say
$$[f] * [g] = [f * g]$$

With the operation defined, we can look at the properties of it to see if it would follow all the axioms of a group!! Remember that when we defined $*$, it only works on appropriate paths (with end and initial points matching). This will need to be considered in our up-coming examination of its properties:

Theorem 7.1.16: Properties Of $*$

1. (Associativity) Assume f, g, h are paths in X, $f(1) = g(0)$, $g(1) = h(0)$. Then
$$([f] * [g]) * [h] = [f] * ([g] * [h])$$
2. (identity) If f is a path in x, $f(0) = x_0$, $f(1) = x_1$, then
$$[f] * e_{x_1} = [f] = e_{x_0} * [f]$$
3. (invertible) Given a path f in X, define $\tilde{f}(t) = f(1-t)$ Then
$$[f] * [\tilde{f}] = [e_{f(0)}]$$
$$[\tilde{f}] * [f] = [e_{f(1)}]$$

***Proof*:**
Some useful facts that will be used continuously

1. Let $k : X \to Y$ be a continuous map and F be a path homotopy between f and f'. Then

$k \circ F$ is a path-homotopy between $k \circ f$ and $k \circ f'$ where

$$(k \circ F)(s, t) = k(F(s, t))$$

2. composition and $*$ are basically distributive: $k \circ (f * g) = (k \circ f) * (k \circ g)$.

As you probably noticed in the construction, the identity wasn't unique! This means that $*$ just falls short of actually being a binary operation. In fact, it is not actually a function, but a *partial function*. That is you can't take two arbitrary paths, but you need to select paths with the appropriate mid-points (as we noticed when defining $*$ originally!). If we restrict to such paths, we will indeed have a binary function, but we usually want to think of $*$ being defined on our entire space.

We actually give a name to a "group" which uses a partial function (and hence, has multiple identities depending on how you must restrict the partial function to get a function):

Definition 7.1.17: Groupoid

A ***groupoid*** is a set (ex. G) with a ***partial function*** $* : G \times G \to G$ *and* a unary function $^{-1}\colon G \to G$ which then satisfy the group axioms (associatvitiy, inverse, identity). The slight difference between groups is that we now have:

1. (inverse) $a^{-1} * a$ and $a * a^{-1}$ is always defined
2. if $a * b$, is defined then $a * b * b^{-1} = a$ and $a^{-1} * a * b = b$ is also defined

How exactly the partial function $*$ is "partial" depends on the context – there is no one way of defining this partial function. Furthermore, we choose a unary function for the inverse because the "identity" because the identity is now no longer so "unique", as was shown in theorem 7.1.16.

Example 7.1.18: Groupoid

If we take $(\mathbb{R}, *, ^{-1})$ to be our groupoid with $*$ as we defined and $^{-1}(f) = \tilde{f}$ where $\tilde{f} = f(1 - x)$ (i.e. reverse f), then we defined a groupoid!

With the algebraic structure defined and labeled, we can define the elements of our groupoid:

Definition 7.1.19: Path Space

Let $P(X) = \{$equivalence classes of paths $[f]\}$. We will call this the ***path space***

there is a natural map $P(X) \to X^2$, $[f] \mapsto (f(0), f(1))$. This map is surjective if and only if X is path connected. So You can alos think of the groupoid from earlier as

$$(P(X), *, ^{-1})$$

7.1.3 Making it a group

As mentioned, we still don't have a "unique" identity nor a total function, so we are yet to totally define a group. The limiting factor is that we can just combine any function with any function. However, if the initial- an end-points where the same, this would solve our problem! For this reason, we will once again restrict our attention, this time to paths that *loop*:

Definition 7.1.20: Loop

Given $x_0 \in X$ a ***loop based at*** x_0 is a pth which begins and ends at x_0. that is

$$f : I \to X, \qquad f(0) = f(1) = x_0$$

There is actually an advantage to adding this resctriction and thinking about loops: Imagine you have a plane with a point missing. Then having a loop around that point would mean you can't contract that loop to the constant map (as we've seen before, where we sneakily introduced a loop). This idea of not being able to contract loops will be our key concept!

With this definition, we can consider the algebra on a subset of $P(X)$ such that the initial and end points match!!

Definition 7.1.21: Fundamental group at x_0

Let

$$\pi_1(X, x_0) = \{[f] | f(0) = f(1) = x_0\}$$

be the Path space with loops at x_0. Then $\pi_1(X, x_0)$ is a group under $*$! That is, if $[f], [g] \in \pi_1(X, x_0)$, then $[f] * [g] \in \pi_1(X, x_0)$, and $(\pi_1(X, x_0), *)$ is a group!!

Note that $\pi_1(X, x_0) \subseteq P(X)$. We call $\pi_1(X, x_0)$ the ***fundamental group*** of X *based at* x_0, or the *first homotopy group*

7.1.22 Note there *can* be more than one equivalence relation in $\pi_1(X, x_0)$. We will soon explore spaces where this is the case, and also explore cases where there is only one element.

Sometimes $\pi(X)$ is used to represent the groupoid of the space X.

One will naturally as if there is a 1st homotopy group, is there a 2nd? an nth? There is, and it's denoed $\pi_n(X, x_0)$. This higher value of n is where you can think of "dimensions" as I mentioned earlier. These, though, will not be studied here, and are the subject of homotopy theory.

Example 7.1.23: Fundamental Group of Euclidean Space

Let's consider $\mathbb{R}^n$ with the Euclidean topology and some $x_0 \in \mathbb{R}^n$ to define $\pi_1(\mathbb{R}^n, x_0)$. Then by the straight-line Homotopy, we can collapse every loop to the constant map, so there is only one element:

$$\pi_1(\mathbb{R}^n, x_0) = \{[e_{x_0}]\}$$

More generally, if you have a convex space, then it's fundamental group will be trivial (since you can use the straight-line homotopy).

Note that this group is *not* necessarily abelian! This example

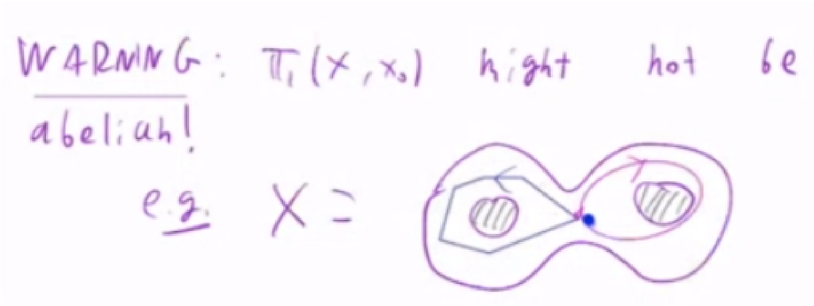

Figure 7.3: *not* abelian (i.e. homotopic) because of holes

Will eventually be used to illustrate why in section (HERE).

So how many fundamental groups are there on a space? Does every point have a distinct type of group or are some of them isomorphic, or better yet, all of them isomorphic on the same space. The different answers to this will actually give us the topological property we're looking for in groups! That is, when thinking about a space, we usually think about the fundamental groups *up to isomorphism* and then the topologically invariant property are the *types* of fundamental groups. That means that the fundamental group does not rely on the point x_0 on which it was originally defined to get it's structure. To that matter, it will also not rely on the *entire* space X, just the component spaces, so the interesting thing to study are the path-connected components.

To get to that intuition, we first, specify how to define the isomorphism on the group

Proposition 7.1.24: Group Isomorphism Between Fundemental Groups

Let α be a path X $\alpha(0) = x_1$, $\alpha(1) = x_1$, then define

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$$

where

$$\hat{\alpha}([f]) = [\tilde{\alpha}] * [f] * [\alpha]$$

where $\tilde{\alpha}$ is the reverse of the funtion.
In words, this means that you go back, to the loop, then go back forward.
This function is a group isomorphism.

The professor said this is a semi-direct product, and that it is an outer-automorphism. However, it looks like an inner-automorphism to me, so I'll have to check later.

***Proof*:**
This is basically an Algebra proof. To show it's homomorphic:

$$\begin{aligned}\hat{\alpha}([f]*[g]) = \hat{\alpha}([f*g]) &= [\alpha]*[f*g]*[\alpha] \\ &= [\tilde{\alpha}]*[f*e_{x_1}*g][\alpha] \\ &= [\tilde{\alpha}]*[f*\alpha*\tilde{\alpha}*g][\alpha] \\ &= ([\tilde{\alpha}]*[f]*[\alpha])*([\tilde{\alpha}]*[g]*[\alpha]) \\ &= \hat{\alpha}([f])*h(g)\end{aligned}$$

For bijectivity, we can define the two-sided inverse function $\hat{\alpha}^{-1}$ by reverseing $\tilde{\alpha}$, where it can be checked that

$$\hat{\alpha}(\hat{\alpha}^{-1}([f])) = \hat{\alpha}^{-1}(\hat{\alpha}([f])) = [f]$$

Making the homomorphism bijective, making it isomorphic.

This will imply that for every path-connected component of X, it will have the *same* fundamental group! Thus, when thinking of Fundamental groups, it is somewhat convention to thinkg about X as being a path-connected component (or at least, it will come down to breaking X up to it's path connected components)

For the more Categorically inclined, there is also no *natural* isomorphism between points! This will come down to the fact that not every space is commutative. I'll amend this once I actually know why it's true.

With this isomorphism at hand, we can now say what is meant when we say "*The* Fundemenatal Group" of a space X.

Corollary 7.1.25: Path-Connectedness And Isomorphism

If X is path-connected, $\pi_1(X, x_0)$ is isomorphic to $\pi_1(X, x_1)$ for any $x_0, x_1 \in X$

This means that if you have two path-connected spaces with different fundemental groups, they are topologically different. However, in general, different fundemental groups doesn't mean different spaces (ex. $\mathbb{R}$ and $\mathbb{R}^2$ have the same Fundemental group)

Definition 7.1.26: Simply-Connected

X is ***simply-connected*** if X is path-connected, and $\pi_1(X, x_0)$ is the trivial group for some (and hence all) $x_0 \in X$

Lemma 7.1.27: simply connected simplification

If X is simply connected, $P(X) \simeq X^2$ (i.e a bijection), that is,

***Proof*:**
First, we'll show all paths are homotopic. Let α, β be paths from x_0 to x_1. Then

$$\begin{aligned}
[\alpha] * [\tilde{\beta}] &\in \pi_1(X, x_0) \\
[\alpha] * [\tilde{\beta}] &= [e_{x_0}] \\
[\alpha] * [\tilde{\beta}] * [\beta] &= [e_{x_0}] * [\beta] \\
[\alpha] &= [\beta]
\end{aligned}$$

With this, we can show injectivity. If $\pi([\alpha]) = \pi([\beta])$, then

$$\alpha(0) = \beta(0), \qquad \alpha(1) = \beta(1)$$

then $[\alpha] = [\beta]$ by proof above

And or surjectivity, given $(x_0, x_1) \in X^2$, there exists a path $\alpha(0) = x_0$, $\alpha(1) = x_1$ such that

$$\pi([\alpha]) = (x_0, x_1)$$

completeing the proof
I will have to look over that surjectivity claim again because it looks like it's breaking well-definedness

7.1.4 Is it an invariance?

We've shown that we can define a fundamental group on a space, but what if we have another topological space Y, and a continuous function $f : X \to Y$, what of the group-structure we just build up? Well, we wouldn't of introduced fundamental groups if they weren't topological in nature, which means that continuous maps *preserve* them! This is also how the comment much earlier that in a sense topology is fundamentally the study of objects that don't change under continuation functions comes in.

So if we have a continuous function, can we define an appropriate *homomorphism* (the equivalent of continuous functions). This is what we'll do in this section.

Definition 7.1.28: induced homomorphic function

Suppose $h : X \to Y$ is continuous, with $h(x_0) = y_0$ (i.e. $h(X, x_0) \to (Y, y_0)$) . Then define

$$h_* : \pi_1(x, x_0) \to \pi_1(Y, y_0) \qquad h_*([\alpha]) = [h \circ \alpha]$$

7.1.29 Trivia otice that it's a topology isomorphism which also specifies a point is a point. This makes our function a little more than continuous, it's continuous *and* preserves that point. Categorically, this means we're working in a slightly different category than Topology (Top): It's technically the Base-topology (bTop) category.

Proposition 7.1.30: Properties Of h_*

1. h_* is a a group-homomorphism
2. If $i : (X, x_0) \to (X, x_0)$ is the identity, then i_* is the idnetity
3. THis map is functorial, that is, if $f : (X, x_0) \to (Y, y_0)$ and $(Y, y_0) \to (Z, z_0)$ then

$$(g \circ f)_* = g_* \circ f_*$$

what this means is that it's a functor between the **Top** Category and the **Grp** Category

Solution We'll prove each separately

1. $$\begin{aligned} h_*([\alpha][\beta]) &= h_*([\alpha * \beta]) \\ &= [h \circ (\alpha * \beta)] \\ &= [(h \circ \alpha) * (h \circ \beta)] \qquad = [(h \circ \alpha)] * [(h \circ \beta)] = h_*([\alpha]) * h_*([\beta]) \end{aligned}$$

2. $i_*([\alpha]) = [i \circ \alpha] = [\alpha]$, hence it's the identity

3. $$\begin{aligned} (g \circ f)_*[\alpha] &= [(g \circ f) \circ \alpha] \\ &= [g \circ (f \circ \alpha)] \\ &= g_*([f \circ \alpha]) \\ &= g_*(f_*([\alpha])) \end{aligned}$$

▼

Thus, we've shown that the fundamental group is invariant to continuous functions! This is excellent, for now we can properly study them as a topological property of our space.

7.2 Use of Fundamental group

So far, we only defined a fundamental group that was trivial: the fundamental group of a convex space. What we'll do now is classify ways of finding different fundamental groups on spaces.

7.2.1 Covering spaces

This first construction will allow us to find the fundamental group of a circle. As further motivation to understand covering spaces, they will come back when we study Riemann surfaces and Complex Manifolds!! So having a good grasp now is solidly useful

Definition 7.2.1: Evenly Covered and Slices

Let $p : E \to B$ be a continuous surjective map. The open set U of B is said to be ***evenly covered*** by p if the inverse image $p^{-1}(U)$ can be written as the union of disjoint open sets V_α in E such that for each α, the restriction of p to V_α is a homeomorphism of V_α onto U. The collection $\{V_\alpha\}$ will be called the partition pf $p^{-1}(U)$ into *slices*

if p does this to every neighbourhood in B, we call such a map p a covering map:

Definition 7.2.2: Covering Map

Let $p : E \to B$ be continuous and surjective. If every point b of B has a neighborhood U that is evenly covered by p, then p is called a ***covering map***, and E is said to be a *covering space* of B.

Example 7.2.3: Covering Maps

1. The identity map $i : X \to X$ is "trivially" a covering map
2. A generalization of the previous example, if $X \times \{1, ..., n\}$ is n disjoint copies of X, then $p(x, i) = x$ for all i is again a covering map, since all the spaces are disjoint and when you limit to any space, it's homeomorphic to X. This function gives a nice visual for how you can think of covering spaces in general:

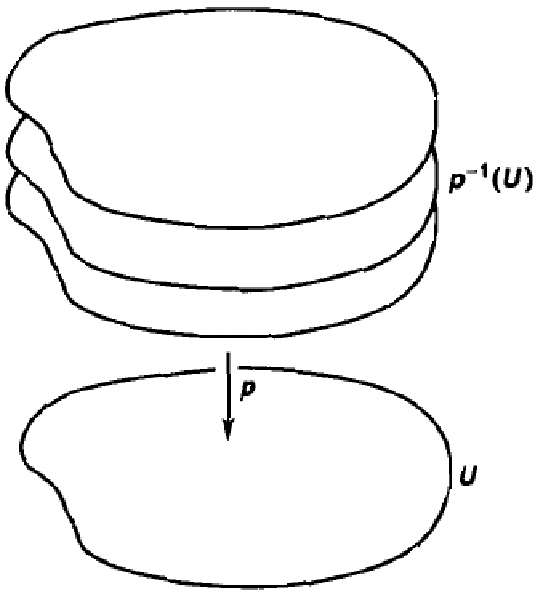

Figure 7.4: Covering space example

3. The function $\pi_x X \times Y \to X$ with Y given the discrete topology is a covering map. The

point of Y having the discrete topology is that that when taking the pre-image, and thinking about the homeomorphism, you need it to be *bijective*, and in particualr *injective*. This will be where using the discretenss of Y would be important.

More non-trivial examples can be found here

https://en.wikipedia.org/wiki/Covering_space#Examples

Corollary 7.2.4: properties of Covering Maps

1. Every $y \in B$, the subspace $p^{-1}(y)$, has the discrete topology
2. By the restrictions of a covering map, we will automatically get that it's an open map

***Proof*:**

1. This should be a straight forward test of understanding coverings. Take a neighborhood of y, $y \in U$. Since p is a covering map, then it's pre-image is partitioned into $\{V_\alpha\}$. Restricting our attention to any one slice V_α will lead to a homeomorphism (so for us, bijection) betweeen V_α and U, meaning one point of it maps to U. Since we can do this for every V_α, we get that $p^{-1}(U)$ is a collection of points intersected by open sets, meaning the subspace is discrete.
2. Let $A \subseteq E$. We'll show that $p(A)$ is open. If around every point $x \in p(A)$, there exists an open set V such that $V \subseteq p(A)$, then $p(A)$ is open (basis definition). Let's take a point $x \in p(A)$. Choose an open neighborhood V, $x \in V$. Then since p is an open cover, choose a neighbourhood that is evenly covered by p. Let $\{V_\alpha\}$ be the neighborhood. Since $x \in p(A)$, $p^{-1}(x) \in A$, so choose $y \in p^{-1}(x)$, and the appropraite V_β such that $y \in V_\beta$. Since A is open, $A \cap V_\beta$ is also open. Since p restricted to a V_β is a homeomorphism, $p(A \cap V_\beta)$ is *also* open. Note that $p(A \cap V_\beta) \subseteq p(A)$. Since we found an open neighborhood around x which is contained in $p(A)$, then $p(A)$ is open (by 1.3.4).

So with these in mind, define an interesting covering space? The examples were on disconnected spaces since that's kind of easy to construct: In general, we try to create covering spaces on path-connected spaces (as it's also what Fundamental Group is basically defined on). Here is our first non-trivial example:

Example 7.2.5: Another Example

The map $p : \mathbb{R} \to S^1$ given by the equation

$$p(x) = (\cos(2\pi x), \sin(2\pi, x))$$

is a covering map

Solution This comes down to geometric intuition about these trigonometric functions and circles. As an intuitive check that you get this function, if we take any interval $[n, n+1] \subseteq \mathbb{R}$, then that interval will wrap itself around the circle completely.

Let's start by showing that around any point for which the x values are positiev ($x > 0$) has a neighbourhood which is evenly covered. Let $U \subseteq S^1$ be that part of the circle. If you think about our function, that would mean that

$$p^{-1}(U) = \bigcup i \in \mathbb{Z}(n - \frac{1}{4}, n + \frac{1}{4})$$

or, visually:

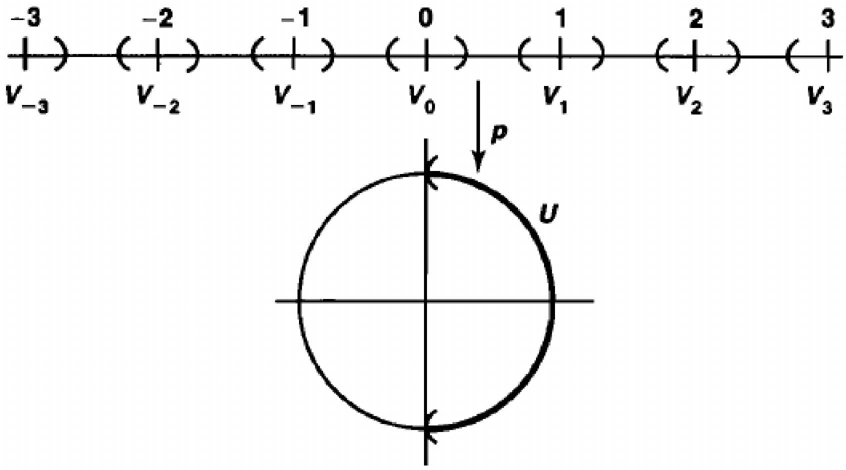

Figure 7.5: $x > 0$ U preimage

What we'll do now is take advantage of this functions properties and of compactness. Take some V_n from the preimage of $p^{-1}(U)$. Take it's closure $\overline{V_n}$. Then by 3.2.24 it is compact.Since our function is continuous and dealing with intervals, by the IVT (3.1.24), V_n maps surjectively onto U. It is already injective by construction. Then by 3.2.9 p_{V_n} is an open map from $\overline{V_n}$ to $\overline{U}$. Since it's Hausdorff, it means that we can get rid of the border in the domain and co-domain and it's still a homeomorphism (hw exercice). Since the function is already continuous, then it is a homeomorphism.

We restrice to $x > 0$, however we can do this for every half of the sphere $x < 0$, $y > 0$, $y < 0$, which will cover every point with an evenly covered neighbourhood, completing the proof. ▼

7.2.2 Properties of Covering Maps

As usual, we might wonder if we can restrict the domain and have a covering map, or take the product space and still have a covering map, or equivalent conditions for covering maps that are equivalent.

One of the two unique properties of covering maps is how p is a *local homeomorphism*: Around every point $e \in E$, there exists a neighbourhood $U \subseteq E$ such that when we restrict to that neighbourhood, our map becomes a homeomorphism to an open subset $V \subseteq B$. Is this condition itself enough to define a covering map? If you think about it, when doing the circle example, what was really key in the proof was proving that restricting will give us homeomorphisms: if two sheets overlap, we can consider them part of the same sheet, making it feel like we can still say it's evenly covered! However, this turns out not to be the case, for a subtle reason:

Example 7.2.6: Local Homeomorphism That's Not a Covering Map
Take p before but restrict the domain to $\mathbb{R}_+$:

$$p : \mathbb{R}_+ \to S^1 \qquad p(x) = (\cos(2\pi x), \sin(2\pi x))$$

This function is still surjective, and a local homeomorphism, and even is a covering map for *most* points. However, consider the point $b_0 = (1, 0)$ and some open neighborhood U around that point (not necessarily the same U as our example for the covering map of S^1). Like before, we'd have the same idea for the pre-images, except for V_0 which would be $(0, \epsilon)$ for some ϵ depending on the open neighbourhood U. If we restrict p to U, then we do not get a homeomorphism, but an *embedding*, that is, a function for which if we restrict the co-domain to it's image, it becomes a homeomorphism. If we restrict to $(0, \epsilon)$ we'd get the open neighborhood starting at 0 and ending at ϵ (relative to where sin and cos sends ϵ.)

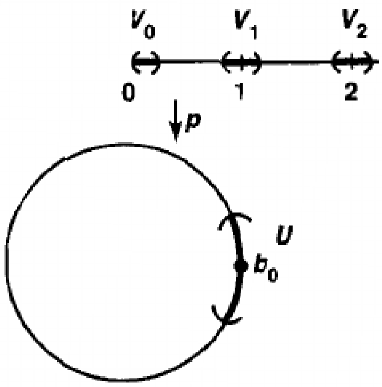

Figure 7.6: visual in Munrkes. Doesn't show embedding, but you can imagine the open set of the embedding starts at b_9

The professor also gave an easier example. If you take the projection map

$$E = [0, 1] \times \{0\} \cup (0, 1) \times \{1\}, B = [0, 1]$$

which is basically forcing surjectivity on the function $(0, 1) \to [0, 1]$. Then there isn't a neighborhood around $0 \in [0, 1]$, which is evenly covered. Again, this comes down to it being an embedding based on the domain rather than the co-domain.

This example shows how it's important to be weary when taking a subspace of the domain of the covering map. So how should we be careful when restricting the domain so that our function remains to be a covering map?

Proposition 7.2.7: Subspace Of Covering Space

Let $p : E \to B$ be a covering map. If B_0 is a subspace of B, and if $E_0 = p^{-1}(B_0)$, then the map $p_0 : E_0 \to B_0$ obtained by restricting p is a covering map

***Proof*:**
here

When taking the product of covering maps, we run into no problems

Proposition 7.2.8: Product Of Covering Maps

If $p : E \to B$ and $p' : E' \to B'$, are covering maps, then

$$p \times p' : E \times E' \to B \times B'$$

is a covering map

***Proof*:**
Let $b \in B$, $b' \in B$. Then we can take neighbourhoods U, U', $b \in U$, $b' \in U'$ which are evenly covered, so we have $\{V_\alpha\}$ and $\{V'_\alpha\}$. We can take the cross-product of the respective slices. They will still be disjiont, and restricting to slices still a homeomorphism, thus $p \times p'$ is a covering map

Example 7.2.9: Product Spaces Homotopy

1. Take
$$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$$
where p is the same covering map of the circle we introduced earlier. Then this function is a covering map for the torus! I'll put an image here soon.

2. Using lemma 7.2.7, and we restrict our attentiont to just the two circles we see in the image before, then we can think of covering the figure 8:

   figure here

   There are multiple ways of covering this surce. You can keep one circle and unwrap the other, or you can unwarp both, or you can have the universal covering space for it.

   figure here, really cool

Munkre present two more examples, and elaborates on the nature of the torus being represented in $\mathbb{R}^3$, but I'll get to that later

7.2.3 Fundamental group of a circle

We know have the tools to compute the fundamental group of a circle! The construction and the answer might at first seem unintuitive, but the prof gave this intuition for how to think about it:

> If you think about a circle, and a loop on the circle, you can think of the loop going around once to have had a 2π angle rotation. Similarly, the trivial loop had 0. You can't have something in between cause the you didn't have a loop. Similarly, you can have 4π, 6π, ..., $2k\pi$ angles of rotation. The fundamental group of a circle will capture the fact that each of these will be unique path homotopies!
>
> So in a sense, what the intuition is building up to for the fundamental group is asking the quesiton "how many ways can I wrap a piece of string around it, such that the start and end point are at some fixed point (like you put a tack to hold it down), and then you can move your string in the usual way you can move your string with a homotopy"

If you look at the examples at the end of last section, we see that the circle, the figure eight, the torus, where all covered by a "larger" space above it that got projected down. However, we didn't formally say how this space is found or achieved. We will start by that.

Definition 7.2.10: Lifting

Let $p : E \to B$ be a continuous function. Let $f : X \to B$ be a continuous map. A ***lifting*** of f is a continuous map $\tilde{f} : X \to E$ such that $f = p \circ \tilde{f}$. Similarly, we can say that this commutative diagram commutes

$$\begin{tikzcd} & E \arrow[d, "p"] \\ X \arrow[ur, "\tilde{f}"] \arrow[r, "f"] & B \end{tikzcd}$$

We can think of it as *lifting* in the following way: If we have a point $f(x) \in B$, then what the lift does is take our point x and puts it above one of the points in B which projects down to B – you can think of the corkscrew from the covering of S^1!

The reason we care about lifts and covering maps is because lifts with covering maps preserves paths!

Lemma 7.2.11: paths can be lifted

Let $p : E \to B$ be a covering map such that $p(e_0) = b_0$. Any path $f : [0,1] \to B$ with inital point $f(0) = b_0$. THen it has a unique lifting to a path $\tilde{f}$ in E with initial point $\tilde{f}(0) = e_0$

Proof:
(not the proof in Munkre since Munrke uses Lebegue number lemma which we didn't cover. Gonna use Profs proof)

(**Existence**) It is similar to how we showed connectedness. Let M be the largest elemetn of $[0,1]$ such that $f|_{[0,M]}$ has a lift $\tilde{f}$. Let $\tilde{f}(M) = e$, $f(M) = b$, then $p(e) = b$. Let U be an open set in B containing b such that U is evenly coevered $p^{-1}(U) = \bigcup_{\alpha \in I} V_\alpha$.
Let $\alpha_0 \in I$ be such that $e \in V_{\alpha_0}$. U is open, so $\exists \epsilon > 0$ such that $(M - \epsilon, m + \epsilon) \subseteq f^{-1}(U)$.

Thus, we define $\tilde{f}_{(M-\ \epsilon, M+\epsilon)}$ to be $i \circ f|_{(M-\ \epsilon, M+\epsilon)}$ is the where $i : U \to V_{\alpha_0}$ is the canonical isomorphism, so $M = 1$.

(**uniqueness**) in the slides, it's also intuitive (The Nov19th file)

We will now expand this to lift produce spaces:

Lemma 7.2.12: F can be lifted

Let $p : E \to B$ be a covering map, and let $p(e_0) = b_0$. Let $F : I \times I \to B$ be a continuous function such that $F(0, 0) = b_0$. Then there exists a unique lift $\tilde{F} : I \times I \to E$ such that $\tilde{F}(0, 0) = e_0$. If F is a path homotopy, then $\tilde{F}$ is a path homotopy.

***Proof*:**
TUsing lemma 7.2.11, we can lift every path of the homotopy. We need to then show its continuous. To do this, the prof did an interesting proof that is similar to the proof of compactness/connectedness with saying M is the largest value s such that $F(t, s)$ is continuous, and showed that it must be $M + \epsilon$, which lets us conclude that $M = 1$

Nov24th video

And now we use this to be able to lift path homotopies!

Lemma 7.2.13: Path Homotopies can be lifted

If F is a path homotopy, then so is $\tilde{F}$

***Proof*:**
IT follows since all lifts of constant paths are constant paths

Finally, we'll show that the endpoints of two path-homotopic functions also agree:

Lemma 7.2.14: Paths can be lifted

Let f, g be paths starting and ending at b_0. Assume $f \simeq_p g$. Then $\tilde{f}(1) = \tilde{g}(1)$

***Proof*:**
The path homotopy F between f, g lifts to a path homotopy between $\tilde{f}, \tilde{g}$. Therefore, $\tilde{f}(1) = \tilde{g}(1)$

7.2.15 Note Note that these are *paths* not loops. It will happen that when projected using the covering space, they will become loops!

With path-homotopies being conserved with liftings, we can now talk about fundamental groups when their lifted:

Definition 7.2.16: Lifting Correspondence

Let $p : E \to B$ be a covering map. Let $p(e_0) = b_0$. Given $[f] \in \pi_1(B, b_0)$, let $\tilde{f}$ be a lifting of f to E such that $\tilde{f}(0) = e_0$.
Then we can define:

$$\varphi : \pi_1(B, b_0) \to p^{-1}(b_0) \qquad \varphi([f]) = \tilde{f}(1)$$

and φ is called the ***lifting correspondence*** derived from the corvering map p, e_0.

7.2.17 Note it is important to choose and e_0 since usually the covering space might have many locally homeomorphic neigbhourhoods, so you have to make a choice. THe choice, however, doesn't really matter so much since there is a local neigbhourhood around e_0 will be homeomorphic to it's image.

Since we'll be basically working with path-connectd space:

Proposition 7.2.18: Lifting Correspondence and Path-Connected Spaces

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$. If E is path-connected, then the lifting correspondence is surjective. If E is also simply-connected, then the lifting correspondence is bijective

Solution Assume E is path-connected. Let $e_1 \in p^{-1}(b_0)$. Let f_E be a path in E such that $f_E(0) = e_0$, $f_E(1) = e_1$. Let $f = p \circ f_E$. Then f_E lifsts f, and

$$f(0) = p(f_E(0)) = p(e_0) = b_0 \tag{7.1}$$
$$f(1) = p(f_E(1)) = p(e_1) = b_0 \tag{7.2}$$

Thus, $\varphi(p([f])) = f_E(1) = e_1$, and so φ is surjective, as we sougth to show.

Now assume E is simply connected. Nov26th. Proving injectivity. ▼

What we've essentially proven is that we can deduce information of the fundamental group of a space given the covering map!

Example 7.2.19: Covering Spaces Properties

1. Let $B = E = S^1$, and take $p : E \to B$ $z \mapsto z^2$, and take $b_0 = e_0 = (1, 0)$. Visually:

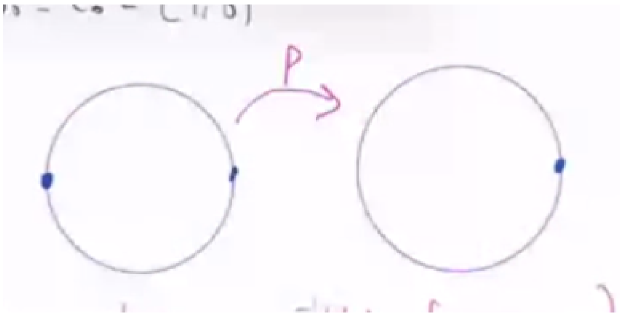

Figure 7.7: visual example of pre-iamge of b_0

Then $\varphi : \pi_1(S^1, b_0) \to p^{-1}(b_0) = \{(1,0), (-1,0)\}$.
(Show that it's a homomorphism?)
We'll take two function. Take: $e_{b_0} : [0,1] \to S^1$, $e_{b_0}(t) = b_0 = (1,0)$. Then the lifting correspondence of e_{b_0} is clearly $\tilde{e_{b_0}} = e_{e_0}$. Then:

$$\varphi([e_{b_0}]) = e_{e_0}(1) = e_0 = (1,0)$$

now, let:

$$f : I \to S^1, f(t) = (\cos(2\pi t)\sin(2\pi t))$$

which is the funciton which will go around the circle once. Then the lifting correspondence is:

$$\tilde{f} : I \to S^1, \tilde{f}(t) = (\cos(\pi t), \sin(\pi t))$$

So $\tilde{f}(1) = (-1,0)$ so $\varphi([f]) = (-1,0)$ (which also verifies surjectivity). So for the first time, we showed that we have a *non-contractibel loop* since the two homotopic classes are different!!! That is $[f] \neq [e_{b_0}]$

2. This time, we'll be a little less formal. take for granted for a moment that we're dealing with a homomorphism. Let X be the figure 8 as we described in a pervious example (from the torus). Let the intersection of the circles be b_0. Take one loop going the left circle counter-clockwise, and oen for the right circle, clockwise, so that $[f], [g] \in \pi_1(X, b_0)$. Let E represent the following lifting space:

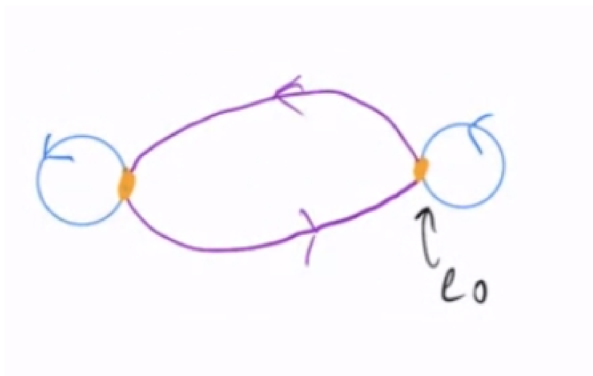

Figure 7.8: Lifting Space

and take e_0 as in the image. Then

$$\varphi([f]) = e_1 \tag{7.3}$$
$$\varphi([g]) = e_0 \tag{7.4}$$
$$\varphi([f] * [g]) = e_1 \tag{7.5}$$
$$\varphi([g] * [f]) = e_1 \tag{7.6}$$

okay, this told us htat $[f]$ and $[g]$ are different paths, but it seems that their product is the same. However, remember that φ is surjective, *not* injective, so φ might've only given us *partial* information on how the elements of this fundamental group interact. We can in fact work wtih a better covering space which will reveal more information:

3. Same X, same b_0, but this time, our lifting space E is different!

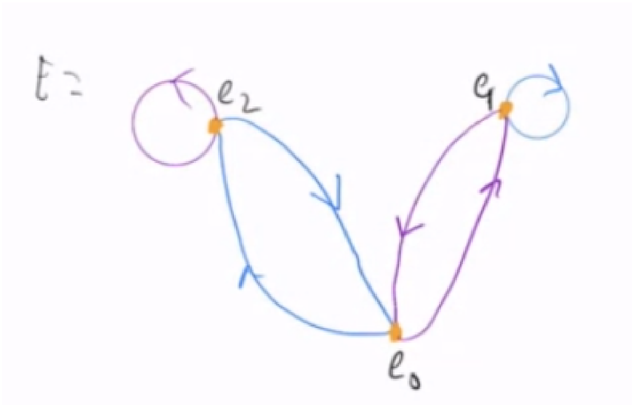

Figure 7.9: Different Lifting Space

Now,

$$\varphi([f] * [g]) = e_1 \tag{7.7}$$
$$\varphi([g] * [f]) = e_2 \tag{7.8}$$

Therefore, $[f] * [g] \neq [g] * [f]$, i.e. $\pi_1(X, b_0)$ is *not* abelian, as we claimed when $\pi_1(X, x_0)$ was introduced!!!

We'll finally present the fundamental group of the circle. For the long build-up of it, we'll make it a theorem

Theorem 7.2.20: Fundamental Groups Of The Circle

The fundamental group of S^1 is isomorphic to $(\mathbb{Z}, +)$

***Proof*:**
Let $p : \mathbb{R} \to S^1$, and $p(t) = (\cos(2\pi t), \sin(2\pi t))$. Then p is a covering map such that $p(r) = p(s) \Leftrightarrow r - s \in \mathbb{Z}$. Let $b_0 = (1, 0) \in S^1$. Then $p^{-1}(b_0) = \mathbb{Z}$. Let our starting point in E be $e_0 = 0$. Then the lifting corresponding gives a map $\varphi : \pi_1(S^1, b_0) \to \mathbb{Z}$. Moreover, since $\mathbb{R}$ is simply connected, it follows that φ is a bijetion! So, it remains to show that φ is a homomrophism:

$$\varphi([f] * [g]) = \varphi([f]) + \varphi([g])$$

Let $\tilde{f}, \tilde{g}$ be lifts of f, g starting at 0, so $\varphi([f]) = \tilde{f}(1) \in \mathbb{Z} = p^{-1}(b_0)$. Define $\tilde{g}_2(t) = \tilde{g}(t) + \tilde{f}(1)$. Then

$$p(\tilde{g}_2(t)) = p(\tilde{g}(t) + \tilde{f}(1)) = p(\tilde{g}(t) + k) = p(\tilde{g}(t)) = g(t)$$

since $k \in \mathbb{Z}$, so the covering map projects this path onto g, meaning it is also a lift of g, just shifted over by $f(t)$ to where the path starts. We can see this by $\tilde{g}_2(0) = \tilde{f}(1)$. Thus, we can combine these two paths, $\tilde{f} * \tilde{g}_2$, and by what we've just shown, this is a lifts of $f * g$. Thus

$$\begin{aligned}\varphi([f] * [g]) &= (\tilde{f} * \tilde{g}_2))(1) \\ &= \tilde{g}_2(1) \\ &= \tilde{g}(1) = \tilde{f}(1) = \varphi([f]) + \varphi([g])\end{aligned}$$

completeing the proof! Visually, this is what we just did with g_2:

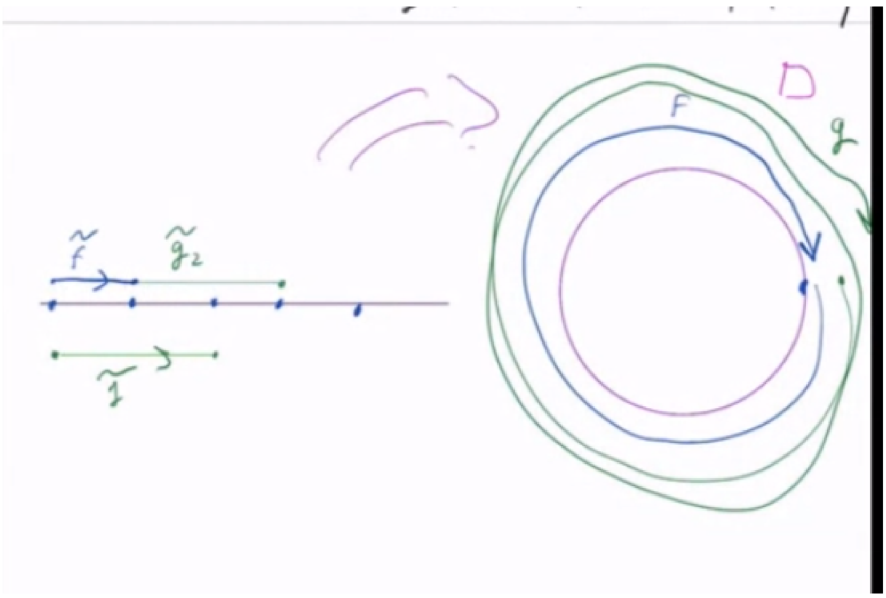

Figure 7.10: showing the logic for the lifting

This also means that the fundamental group of the circle is abelian, which should makes sense if you try and draw loops on a circle. You will get back and forth and will be able to reverse the order no problem. Trying this on the figure 8, you might see how you can get stuck. We just formalized such intuition!

There's one more theorem worth covering with cosets p.346.

7.2.4 Fundamental Group of finite space

I also found this interesting enough to create a section for it. Will fill in the details later:

https://en.wikipedia.org/wiki/Pseudocircle

7.2.5 Further explorations

http://mathonline.wikidot.com/list-of-fundamental-groups-of-common-spaces

7.3 Retractions and Contractions

We will not elaborate on important results that arrive as a consequence of the fundametnal group of a cricle. The eventuall conclusion of this is a theorem called *Browers Fixed point Theorem* which we will use to cool effect.

Retractions were introduced when we talked about quotient spaces as a good exampe of quotient maps. The relation between retraction maps and fundametnal groups of spaces their on is extremely intuitive, as we'll see, making them an excellent tool to analyze the fundamental group of spaces:

Definition 7.3.1: Retraction

If $A \subseteq X$, a ***retraction*** of X onto A is a continuous map $f : X \to A$ such that $r|_A$ is the identity. The supspace is the *retract* of X

Example 7.3.2: Retraction

1. A is a point
2. If $X = Y \times Z$, and $A = Y \times \{z_0\}$, then $f(y, z) = (y, z_0)$ is a retraction.
3. if A is an annulus, and we have a circle in the annulus, then pinching everything to A is a retraction.
4. $X = \mathbb{R}$, $A = \mathbb{R}_{\geq 0}$, then $\sigma(t) = |t|$ is a retraction

Lemma 7.3.3: induced maps and retractions

If A is a retract of X. for $a_0 \in A$, the induced map $j_* : \pi_1(A, a_0) \to \pi_1(X, a_0)$ is injective, where $j : A \to X$ is the inclusion

This means that X should have *at least* as many elements in it's fundamental group as A

***Proof*:**
Let $r : X \to A$ be a retraction. Then $f \circ j = id_A$. Thus, $(r \circ j)_* = (id_A)_* = id|_{\pi_1(A,a_0)}$. Thus

$$[f] = (r_* \circ j_*)([f]) = r_*(j_*([f]))$$

showing r_* is a left inverse, thus j_* is injective. Visually,

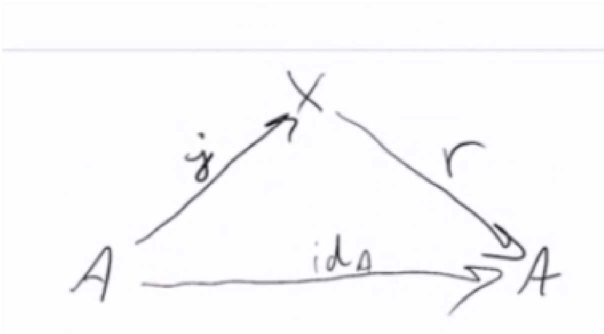

Figure 7.11: visual

We can finally prove a result you maybe heard jokes about all over the place:

Theorem 7.3.4: Can't make a whole

Let B^2 be the unit disc in $\mathbb{R}^2$. There is no retraction form B^2 to S^1. You can think of this as you cannot make a whole.

***Proof*:**
The fundamental group of the circle is infinite, but the fundamental group of B^2 has 1 element (because its star-convex). Since the fundamental group of S_1 would need to injectively map to B^2, this implies there is *no* retraction!

7.4 Browser's fixed point theorem

Theorem 7.4.1: Brouwer Fixed Point Theorem

if $f : B^2 \to B^2$ is continuous, then there exists $x \in B^2$ such that $f(x) = x$

***Proof*:**
Recall that $B^2 := \{(x, y)|x^2 + y^2 \leq 1\}$ and thus is $\approx I^2$.

Assume $f : B^2 \to B^2$ is continuous. For the sake of contradiction, let's say that for every $x \in B^2$, $f(x) \neq x$. We will use f to construct a retraction of B^2 to its boundary S^1.

The key idea is that since $f(x) \neq x$, and both are in B^2, we can draw a straiht line between x

and $f(x)$. We can continue this line to the boundary. If we give a direction (let's say from $f(x)$ to x), we can then make a function $g(x)$ which will take our point x, and given this line, it will bring it to the boundary.

The key idea here is that if we had that $f(x) = x$, then we woouldn't know what ray to draw. for example, if $f(x) = \frac{x}{\|x\|}$ then every point will have a clear ray exept $x = 0$. This unclearness of the ray is translated as f being discontinuous (in the case of the concrete f, we divided by 0).

More formally, we'll define $G : B^2 \times \mathbb{R}_{\geq 0} \to \mathbb{R}^2$.

$$G(x, r) = x + r(x - f(x))$$

since Everything is in $\mathbb{R}^2$, mutiplying, adding, subtracting is allowed. Note that

$$\|G(x, r)\|^2 = \|x\|^2 + r^2 \|x - f(x)\|^2 + 2rx \cdot (x - f(x))$$

so for $\|G(x, r)\| = 1$, we use the quadratic formula to solve r:

$$r = \frac{-2x \cdot (x - f(x)) + \sqrt{4(x \cdot (x - f(x)))^2 + 4(1 - \|x\|^2) + \|x - f(x)\|^2}}{2 \|x - f(x)\|^2}$$

This formula is complicated, but it's okay: everything is positive, so everyhing is okay (since f has no fixed points). We can make r into a formula which will make sure that $G(x, r) = 1$ by calling the right hand side $r(x)$.

THus, we can define

$$g(x) = G(x, r(x)) \in S^1$$

and by 1.2.10, $g(x)$ is continuous. Note that one the circle, $g(x) = x$, showing that it's a retraction. But we showed there can't be a retraction between the ball and the circle – a contradiction!

8

Topological Groups

A group and a topological space are, on their own, unrelated structures; a *topological group* is what results when the two are required to be compatible, so that the group operations respect the topology. The requirement is light (multiplication and inversion are asked only to be continuous), yet its consequences run through the whole chapter. Translation by a fixed element is a homeomorphism, so any two points of the group are topologically indistinguishable: the space is *homogeneous*, and its local structure is already determined by the neighbourhoods of the identity. Most of what follows extracts consequences from this single observation.

Homogeneity makes the separation axioms of chapter 4 degenerate. For a group, the weakest of them, T_0, already forces the Hausdorff property, so a topological group is Hausdorff as soon as its points are topologically distinct. Homogeneity also tames the quotient construction of section 2.3: the quotient G/H carries a group structure compatible with the quotient topology exactly when H is normal, and it is Hausdorff exactly when H is closed, so the algebraic and the topological conditions on H line up. Connectedness, developed in section 3.1, enters through the identity component, a closed normal subgroup whose quotient is totally disconnected; connected groups and totally disconnected groups are the two poles between which the general theory is organized. These are the point-set foundations on which the profinite groups of Galois theory and the Lie groups of geometry are later built.

The chapter treats only the topological side of the subject. Haar measure, the convolution algebra $L^1(G)$, and the unitary representations that make locally compact groups the setting for abstract harmonic analysis belong to the theory of unitary representations, taken up in the companion volume devoted to it; here the locally compact groups of section 8.7 are studied as spaces, not yet as measure spaces.

8.1 Definitions and Examples

A topological group is a set carrying two structures at once, a group law and a topology, coupled by a single compatibility demand: the algebraic operations must be continuous. This section fixes that definition, records the small equivalence that lets one check compatibility with a single map rather than two, and stocks the reader with the examples that recur throughout the chapter. The separation behaviour of these groups is subtle and is treated on its own in section 8.4; nothing here presupposes any separation axiom.

The group operations are the multiplication $m\colon G \times G \to G$, $(x, y) \mapsto xy$, and the inversion $\iota\colon G \to G$, $x \mapsto x^{-1}$. Continuity of m is measured against the product topology on $G \times G$ (section 2.2); this is the only reasonable topology to put on the domain, and the choice is what forces the two variables to be controlled jointly rather than one at a time.

Definition 8.1.1: Topological Group

A *topological group* is a group G equipped with a topology such that

1. the multiplication $m\colon G \times G \to G$, $(x, y) \mapsto xy$, is continuous when $G \times G$ carries the product topology, and
2. the inversion $\iota\colon G \to G$, $x \mapsto x^{-1}$, is continuous.

No separation axiom is imposed.

The definition asks continuity of two maps, but a single map records both. The composite $(x, y) \mapsto xy^{-1}$ folds multiplication and inversion together, and its continuity is equivalent to that of the pair. This is the form one most often checks in practice, since a candidate group frequently comes with a natural formula for xy^{-1}.

Proposition 8.1.2: Single-Map Criterion

Let G be a group with a topology. Then G is a topological group if and only if the map

$$d\colon G \times G \to G, \qquad (x, y) \mapsto xy^{-1}$$

is continuous.

***Proof*:**
Suppose first that G is a topological group, so that m and ι are continuous. The map d factors as

$$G \times G \xrightarrow{\mathrm{id}_G \times \iota} G \times G \xrightarrow{m} G$$

where $\mathrm{id}_G \times \iota$ sends (x, y) to (x, y^{-1}). The map $\mathrm{id}_G \times \iota$ is continuous because a map into a product is continuous exactly when its two coordinate functions are (section 2.2), and here the coordinates are the projection $(x, y) \mapsto x$ and the composite $(x, y) \mapsto y \mapsto y^{-1}$, both continuous. Composing with the continuous map m shows that $d = m \circ (\mathrm{id}_G \times \iota)$ is continuous.

Conversely, suppose d is continuous. Setting $x = e$ gives inversion back: the map $y \mapsto (e, y) \mapsto ey^{-1} = y^{-1}$ is the composite of the continuous inclusion $y \mapsto (e, y)$ with d, so ι is continuous. Continuity of the inclusion $y \mapsto (e, y)$ holds because its coordinates are the constant map e

and the identity. Knowing ι is continuous, recover multiplication by feeding an inverted second argument: for $x, y \in G$,

$$xy = x(y^{-1})^{-1} = d(x, y^{-1})$$

so $m = d \circ (\mathrm{id}_G \times \iota)$ is a composite of continuous maps and is therefore continuous. Thus both m and ι are continuous, and G is a topological group, completing the proof.

The criterion is the working tool. To certify that a set with a group law and a topology is a topological group, exhibit a single formula for xy^{-1} and check that formula is continuous; the two-map definition is then automatic. The examples below use it and the raw definition interchangeably, choosing whichever is cleaner for the case at hand.

Two extremes bracket the subject: the discrete topology, where continuity is free and the group structure is unconstrained, and the additive real line, where continuity is the familiar continuity of addition and subtraction from calculus.

Example 8.1.3: Basic Topological Groups
Each of the following is a topological group.

1. *Any group with the discrete topology.* Let G be an arbitrary group and give it the discrete topology, so every subset is open. Then $G \times G$ carries the discrete topology as well, and every map out of a discrete space is continuous. In particular m and ι are continuous, so G is a topological group. A group with the discrete topology is a *discrete group*; the finite groups and the integers $(\mathbb{Z}, +)$ are the standard instances.

2. *The additive reals* $(\mathbb{R}, +)$ *and* $(\mathbb{R}^n, +)$. On $\mathbb{R}$ with its usual topology, the subtraction map $d(x, y) = x - y$ is continuous, being a polynomial in the two coordinates. By proposition 8.1.2, $(\mathbb{R}, +)$ is a topological group. The same argument applies coordinatewise to $(\mathbb{R}^n, +)$: subtraction $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ is $(x, y) \mapsto x - y$, whose components $x_i - y_i$ are each continuous, so the map into the product $\mathbb{R}^n$ is continuous.

3. *The multiplicative reals* $(\mathbb{R}^\times, \cdot)$. Let $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ with the subspace topology (section 2.1) and multiplication as the group law. The map $d(x, y) = xy^{-1} = x/y$ is continuous on $\mathbb{R}^\times \times \mathbb{R}^\times$, since it is a quotient of the continuous coordinate functions with nonvanishing denominator. Thus $(\mathbb{R}^\times, \cdot)$ is a topological group.

4. *The circle* $\mathbb{S}^1$. Realize the circle as the unit complex numbers $\mathbb{S}^1 = \{z \in \mathbb{C} \big| |z| = 1\}$ under multiplication, with the subspace topology from $\mathbb{C}$. Complex multiplication and complex inversion $z \mapsto z^{-1} = \overline{z}$ are continuous on $\mathbb{S}^1$, the latter because inversion agrees with conjugation on the unit circle. Hence $\mathbb{S}^1$ is a topological group. It is the first example whose underlying space is compact and connected, and it recurs as the target of characters throughout the analysis of commutative topological groups.

These four already display the range. Discreteness makes topology inert and leaves group theory untouched; $(\mathbb{R}^n, +)$ is the local model for the groups of geometry and analysis; $(\mathbb{R}^\times, \cdot)$ shows that removing a point to restore invertibility keeps everything continuous; and $\mathbb{S}^1$ is compact, so it behaves very differently from the line under limiting operations. The verifications all reduced to continuity of a rational formula, which is the pattern the single-map criterion is built to exploit.

A structural feature separates topological groups from bare topological spaces: the group can move any point to any other by a continuous map with a continuous inverse. Fixing an element g and translating produces such a self-homeomorphism, and so does inversion. These maps transport local information from one point to another, and in particular from the identity to everywhere.

Proposition 8.1.4: Translations and Inversion Are Homeomorphisms

Let G be a topological group and $g \in G$. Then the left translation $L_g \colon G \to G$, $x \mapsto gx$, the right translation $R_g \colon G \to G$, $x \mapsto xg$, and the inversion $\iota \colon G \to G$, $x \mapsto x^{-1}$, are homeomorphisms of G. Their inverses are $L_{g^{-1}}$, $R_{g^{-1}}$, and ι itself.

***Proof*:**
Consider first the left translation L_g. It is continuous: the map $x \mapsto (g, x)$ from G to $G \times G$ has coordinates the constant g and the identity, hence is continuous, and L_g is its composite with the continuous multiplication m, so $L_g = m \circ (x \mapsto (g, x))$ is continuous. As a group-theoretic identity, $L_g \circ L_{g^{-1}} = L_{gg^{-1}} = L_e = \mathrm{id}_G$ and likewise $L_{g^{-1}} \circ L_g = \mathrm{id}_G$, so L_g is a bijection with inverse $L_{g^{-1}}$. Since $g^{-1} \in G$ is again an element of the group, $L_{g^{-1}}$ is continuous by the same argument applied to g^{-1}. A continuous bijection with continuous inverse is a homeomorphism, so L_g is one.

The right translation R_g is handled identically, replacing $x \mapsto (g, x)$ by $x \mapsto (x, g)$ and using $R_g \circ R_{g^{-1}} = R_{g^{-1}g} = \mathrm{id}_G$; its inverse is $R_{g^{-1}}$.

For inversion, ι is continuous by definition 8.1.1, and $(x^{-1})^{-1} = x$ gives $\iota \circ \iota = \mathrm{id}_G$, so ι is its own inverse. A continuous map equal to its own two-sided inverse is a homeomorphism, so ι is one, as we sought to show.

The content is that G is *homogeneous*: for any two points x, y the left translation $L_{yx^{-1}}$ carries x to y and is a homeomorphism, so no point is topologically distinguished from another. A neighbourhood of one point is, up to translation, a neighbourhood of any other, which is why the topology of a topological group is controlled by the neighbourhoods of the identity alone. That principle is developed in section 8.2; the present proposition supplies the homeomorphisms it runs on. Inversion, meanwhile, is the symmetry that lets one pass between left and right versions of any statement, exchanging L_g with $R_{g^{-1}}$ under conjugation by ι.

The examples so far are commutative. The matrix groups supply the first genuinely noncommutative topological groups and, at the same time, the first whose continuity is not quite a one-line polynomial check, since inversion involves dividing by a determinant. The general linear group is the prototype.

Example 8.1.5: The General Linear Group $\mathrm{GL}_n(\mathbb{R})$

Let $\mathrm{GL}_n(\mathbb{R})$ denote the group of invertible $n \times n$ real matrices under multiplication. Identify the set of all $n \times n$ real matrices with $\mathbb{R}^{n^2}$ by listing the entries, and give $\mathrm{GL}_n(\mathbb{R})$ the subspace topology it inherits as a subset of $\mathbb{R}^{n^2}$.

First, $\mathrm{GL}_n(\mathbb{R})$ is an open subset of $\mathbb{R}^{n^2}$. The determinant $\det \colon \mathbb{R}^{n^2} \to \mathbb{R}$ is a polynomial in the matrix entries, hence continuous, and

$$\mathrm{GL}_n(\mathbb{R}) = \{A \in \mathbb{R}^{n^2} \mid \det A \neq 0\} = \overset{-1}{\det}\big(\mathbb{R} \setminus \{0\}\big)$$

is the preimage of the open set $\mathbb{R} \setminus \{0\}$ under a continuous map, so it is open. It is therefore a topological space in its own right, an open subspace of Euclidean space.

Multiplication is continuous. The (i, j) entry of a product AB is $\sum_{k=1}^{n} A_{ik} B_{kj}$, a polynomial in the $2n^2$ entries of A and B. Each entry of the product map $\mathrm{GL}_n(\mathbb{R}) \times \mathrm{GL}_n(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R})$ is thus continuous, so the map into the product space $\mathbb{R}^{n^2}$ is continuous, and it lands in the open subset $\mathrm{GL}_n(\mathbb{R})$.

Inversion is continuous. Cramer's rule expresses the inverse as

$$(A^{-1})_{ij} = \frac{(-1)^{i+j}\, M_{ji}(A)}{\det A}$$

where $M_{ji}(A)$ is the (j,i) minor of A, itself a polynomial in the entries of A. On $\mathrm{GL}_n(\mathbb{R})$ the denominator $\det A$ never vanishes, so each entry of A^{-1} is a rational function of the entries of A with nonvanishing denominator, hence continuous. Every entry of the inversion map is continuous, so the map into $\mathbb{R}^{n^2}$ is continuous, and again it lands in $\mathrm{GL}_n(\mathbb{R})$. By definition 8.1.1, $\mathrm{GL}_n(\mathbb{R})$ is a topological group, and it is noncommutative for $n \geq 2$.

The determinant does double duty: its nonvanishing carves out $\mathrm{GL}_n(\mathbb{R})$ as an open set, and the same nonvanishing is what makes the Cramer formula for the inverse continuous. This example is the entry point to Lie theory, where $\mathrm{GL}_n(\mathbb{R})$ and its closed subgroups are studied as smooth manifolds and the continuity established here is refined to smoothness; that analytic layer is developed in the series' companion volume on Lie groups.

8.2 Translations and Homogeneity

That each translation L_g is a homeomorphism was recorded when the structure maps were introduced (proposition 8.1.4). This section draws out its consequence: a topological group looks the same at every point, so its topology is stored entirely in the neighbourhoods of the identity. The section extracts a base at e, transports it to every point by translation, and isolates the one closure-theoretic tool (a symmetric neighbourhood that halves under multiplication) that the separation and subgroup arguments of the following sections will use repeatedly.

A space is called *homogeneous* when its self-homeomorphisms act transitively on points: given any two points, some homeomorphism of the space onto itself carries the first to the second. For a group the requisite homeomorphism is already at hand. To move x to y, left-multiply by yx^{-1}; this is the translation $L_{yx^{-1}}$, and translations are homeomorphisms.

Proposition 8.2.1: Homogeneity of a Topological Group

Let G be a topological group. For any two points $x, y \in G$ the left translation $L_{yx^{-1}} : G \to G$ is a homeomorphism carrying x to y. In particular G is a homogeneous space: its self-homeomorphisms act transitively. Consequently, if P is a topological property of a point that is preserved by homeomorphisms and P holds at one point of G, then P holds at every point of G.

***Proof*:**

Fix $x, y \in G$ and set $g = yx^{-1}$. By proposition 8.1.4 the left translation $L_g : G \to G$, $t \mapsto gt$, is a homeomorphism, with continuous inverse $L_{g^{-1}}$. It sends x to $gx = yx^{-1}x = y$, so it carries x to y. Since x and y were arbitrary, the self-homeomorphisms of G act transitively, which is the definition of a homogeneous space.

For the final claim, suppose P is a property of a point invariant under homeomorphisms and that P holds at some point $x_0 \in G$. Let $y \in G$ be arbitrary. The homeomorphism $L_{yx_0^{-1}}$ carries x_0 to y, and P is preserved by homeomorphisms, so P holds at y. As y was arbitrary, P holds at every point of G, as we sought to show.

The force of homogeneity is that a topological group has no distinguished points: any local question (whether the space is locally compact at a point, locally connected there, first countable there, metrizable near there) has the same answer at every point, because a translation moves the answer from one point to another without disturbing it. This is why local hypotheses on a topological group are always phrased at the identity alone. To say "G is locally compact" it suffices to produce one compact neighbourhood of e; translation then furnishes a compact neighbourhood of every point. The identity e is not special as a point of the space, since homogeneity forbids that, but it is distinguished as a *marker*, the one point whose local data are recorded. Everything in the following sections that reads "a neighbourhood of e" is implicitly a statement about every point, converted to a statement about one.

The property in question must genuinely be local and homeomorphism-invariant for the transport to work. Being isolated is such a property, and it already exposes how rigid homogeneity is.

Example 8.2.2: Discreteness Is All-or-Nothing

Let G be a topological group in which the identity e is an isolated point, meaning $\{e\}$ is open. The property "$\{x\}$ is open" is local and preserved by homeomorphisms. It holds at $x = e$, so by proposition 8.2.1 it holds at every $x \in G$: the translate $L_x(\{e\}) = \{x\}$ is the image of an open set under the homeomorphism L_x, hence open. Every singleton is therefore open, and G carries the discrete topology.

Thus a topological group is either discrete or has no isolated points at all; there is no middle case in which some points are isolated and others are not. Concretely, in $(\mathbb{R}, +)$ no point is isolated, so $\mathbb{R}$ is not discrete; in $\mathbb{Z}$ viewed with the subspace topology from $\mathbb{R}$, the point 0 is isolated (the interval $(-1, 1)$ meets $\mathbb{Z}$ only in 0), so $\mathbb{Z}$ is discrete. The dichotomy rules out any topology on $\mathbb{Z}$, compatible with the group operations, that isolates 0 without isolating every integer.

Homogeneity says the local topology is the same everywhere; the next result says how much of it a single point already records. Because every neighbourhood of x is the translate of a neighbourhood of e, and conversely, the neighbourhood filter at e determines the neighbourhood filter at every other point, and hence the whole topology. The following proposition makes this precise in the two forms used in practice: a criterion for openness phrased at e, and the transport of a neighbourhood base at e to a base at each point.

Proposition 8.2.3: The Topology Is Determined at the Identity

Let G be a topological group and let $\mathcal{N}$ denote the filter of neighbourhoods of the identity e.

1. A subset $U \subseteq G$ is open if and only if for every $g \in U$ the set $g^{-1}U$ belongs to $\mathcal{N}$, that is, $g^{-1}U$ is a neighbourhood of e.
2. If $\mathcal{B}$ is a neighbourhood base at e, then for each $g \in G$ the translated family $g\mathcal{B} = \{\, gV \big| V \in \mathcal{B} \,\}$ is a neighbourhood base at g. Consequently $\{\, gV \big| g \in G,\ V \in \mathcal{B} \,\}$ is a base for the topology of G.

***Proof*:**
Throughout, recall that for fixed g the left translation L_g and its inverse $L_{g^{-1}} = L_g^{-1}$ are homeomorphisms (proposition 8.1.4), so L_g and $L_{g^{-1}}$ carry open sets to open sets and neighbourhoods of a point to neighbourhoods of its image. Note that $L_{g^{-1}}(U) = g^{-1}U$ and $L_{g^{-1}}$ sends g to e.

Part (1). Suppose first that U is open and let $g \in U$. Then $g^{-1}U = L_{g^{-1}}(U)$ is open, being the image of the open set U under the homeomorphism $L_{g^{-1}}$; and it contains $L_{g^{-1}}(g) = e$. An open set containing e is a neighbourhood of e, so $g^{-1}U \in \mathcal{N}$.

Conversely, suppose $g^{-1}U \in \mathcal{N}$ for every $g \in U$. Fix $g \in U$. Since $g^{-1}U$ is a neighbourhood of e, its interior $\operatorname{int}(g^{-1}U)$ is an open set containing e. Applying the homeomorphism L_g, the set $L_g\big(\operatorname{int}(g^{-1}U)\big) = g\operatorname{int}(g^{-1}U)$ is open and contains $L_g(e) = g$; and it is contained in $g\,(g^{-1}U) = U$. Thus every point g of U has an open neighbourhood contained in U, so U is open.

Part (2). Fix $g \in G$. Each gV with $V \in \mathcal{B}$ is a neighbourhood of g: writing V as a neighbourhood of e, the homeomorphism L_g carries it to a neighbourhood $L_g(V) = gV$ of $L_g(e) = g$. It remains to check cofinality. Let W be an arbitrary neighbourhood of g. Then $g^{-1}W = L_{g^{-1}}(W)$ is a neighbourhood of $L_{g^{-1}}(g) = e$, so by the base property of $\mathcal{B}$ there is $V \in \mathcal{B}$ with $V \subseteq g^{-1}W$. Applying L_g gives $gV \subseteq g\,(g^{-1}W) = W$. Hence every neighbourhood of g contains a member of $g\mathcal{B}$, and $g\mathcal{B}$ is a neighbourhood base at g.

For the base claim we may assume each $V \in \mathcal{B}$ is open: the interiors $\{\operatorname{int}V \big| V \in \mathcal{B}\}$ again form a neighbourhood base at e (each $\operatorname{int}V$ is an open neighbourhood of e contained in V, and any neighbourhood of e contains some $V \supseteq \operatorname{int}V$), so replacing $\mathcal{B}$ by this base changes nothing in parts (1) and (2). With $\mathcal{B}$ open, every member gV is open, being $L_g(V)$. To see that $\{\, gV \big| g \in G,\ V \in \mathcal{B} \,\}$ is a base, let U be open and $h \in U$. Since $g\mathcal{B}$ with $g = h$ is a neighbourhood base at h and U is a neighbourhood of h, there is $V \in \mathcal{B}$ with $h \in hV \subseteq U$. Thus every open set is a union of members of the family, so it is a base for the topology of G, completing the proof.

The two parts say the same thing from opposite directions. Part (2) is the constructive form: to specify the topology of a group it is enough to hand over a neighbourhood base at e, and translation manufactures a base at every other point automatically. This is how topological groups are most often built: one prescribes the neighbourhoods of the identity and lets the group structure propagate them. Part (1) is the recognition form: to test whether a candidate set is open, one need not inspect it near every point but only ask, at each of its points g, whether sliding that point back to the identity by $L_{g^{-1}}$ lands the set inside a fixed neighbourhood of e. Both forms reduce a global topological question to data at the single point e, which is exactly what homogeneity licensed.

Example 8.2.4: The Neighbourhood Base at the Identity of $(\mathbb{R}, +)$

On the additive group $(\mathbb{R}, +)$ from example 8.1.3, the symmetric intervals $\mathcal{B} = \{ (-\varepsilon, \varepsilon) \,|\, \varepsilon > 0 \}$ form a neighbourhood base at the identity 0: every neighbourhood of 0 contains some $(-\varepsilon, \varepsilon)$, and each $(-\varepsilon, \varepsilon)$ is itself a neighbourhood of 0. Here the group is written additively, so translation by g is the map $x \mapsto g + x$ and the translate of V is $g + V$.

By proposition 8.2.3(2), translating this base by g gives the family $g + \mathcal{B} = \{ (g - \varepsilon, g + \varepsilon) \,|\, \varepsilon > 0 \}$, which is a neighbourhood base at g. These are the open intervals centred at g, and they are precisely the standard basic neighbourhoods of g in $\mathbb{R}$: the group structure reproduces the usual metric topology from the single datum of the intervals around 0. To test openness by proposition 8.2.3(1), a set $U \subseteq \mathbb{R}$ is open exactly when for each $g \in U$ the translate $-g + U = \{ u - g \,|\, u \in U \}$ contains some $(-\varepsilon, \varepsilon)$, that is, when U contains the interval $(g - \varepsilon, g + \varepsilon)$; this recovers the familiar definition of an open subset of $\mathbb{R}$.

Having reduced the topology to the neighbourhoods of e, the remaining task is to control how those neighbourhoods interact with the group operations. In a metric space the basic manoeuvre is to halve a radius: given a ball of radius r, one works inside the ball of radius $r/2$, whose points combine to stay within distance r. A topological group has no metric to halve, but the continuity of multiplication supplies a substitute. The next definition and lemma isolate it.

Two features of a neighbourhood matter for the arguments to come. First, one often needs a neighbourhood together with its set of inverses, and it is convenient when the two coincide. Second, one needs a neighbourhood small enough that the product of any two of its points stays inside a prescribed target. The definition addresses the first feature.

Definition 8.2.5: Symmetric Neighbourhood

Let G be a topological group with identity e. For a subset $V \subseteq G$, write $V^{-1} = \{ v^{-1} \,|\, v \in V \}$ for its image under inversion. A neighbourhood V of e is *symmetric* if $V = V^{-1}$, that is, $v \in V$ if and only if $v^{-1} \in V$.

For any neighbourhood U of e, the sets $U \cap U^{-1}$ and $U \cup U^{-1}$ are symmetric, since inversion is an involution and interchanges the two sets in each pair. Because inversion ι is a homeomorphism fixing e, the set $U^{-1} = \iota(U)$ is again a neighbourhood of e, so $U \cap U^{-1}$ is a symmetric neighbourhood of e contained in U. Symmetric neighbourhoods are therefore plentiful: every neighbourhood of e contains one.

Example 8.2.6: Symmetric Neighbourhoods in $(\mathbb{R}, +)$ and $\mathrm{GL}_n(\mathbb{R})$

In the additive group $(\mathbb{R}, +)$ inversion is $x \mapsto -x$, so $V^{-1} = -V$ and symmetry of a neighbourhood V of 0 means $-V = V$. The interval $(-\varepsilon, \varepsilon)$ is symmetric, whereas $(-1, 2)$ is not: its inverse is $(-2, 1)$, and the intersection $(-1, 2) \cap (-2, 1) = (-1, 1)$ is the symmetric neighbourhood the recipe above extracts from it.

In the matrix group $\mathrm{GL}_n(\mathbb{R})$ from example 8.1.5 inversion is $A \mapsto A^{-1}$, and a neighbourhood V of the identity matrix is symmetric when $A \in V$ forces $A^{-1} \in V$. Fix a submultiplicative matrix norm $\|\cdot\|$ with $\|I\| = 1$. The set $V = \{\, A \in \mathrm{GL}_n(\mathbb{R}) \,\big|\, \|A - I\| < \frac{1}{2} \text{ and } \|A^{-1} - I\| < \frac{1}{2} \,\}$ is a neighbourhood of I, and it is symmetric by construction: the two defining inequalities are interchanged by $A \mapsto A^{-1}$, so $A \in V$ if and only if $A^{-1} \in V$.

The symmetric neighbourhood alone does not yet halve anything; for that one needs the product of two of its points to remain in a target set. Continuity of multiplication at the identity delivers this.

Lemma 8.2.7: Halving a Neighbourhood

Let G be a topological group and let U be a neighbourhood of the identity e. Then there exists a symmetric neighbourhood V of e with

$$VV \subseteq U$$

where $VV = \{\, v_1 v_2 \,\big|\, v_1, v_2 \in V \,\}$. In particular $V \subseteq VV \subseteq U$, and $V^{-1} = V$.

***Proof*:**
Multiplication $m\colon G \times G \to G$ is continuous, and $m(e,e) = e \in U$. Since U is a neighbourhood of e, it contains an open set U' with $e \in U'$. By continuity of m at the point (e,e), the preimage $m^{-1}(U')$ is an open subset of $G \times G$ containing (e,e). By the definition of the product topology (section 2.2), a basic open neighbourhood of (e,e) has the form $W_1 \times W_2$ with W_1, W_2 open in G; hence there exist open sets W_1, W_2 with $e \in W_1$, $e \in W_2$, and

$$W_1 \times W_2 \subseteq m^{-1}(U')$$

Applying m, this says $w_1 w_2 \in U'$ for all $w_1 \in W_1$ and $w_2 \in W_2$; that is, $W_1 W_2 \subseteq U' \subseteq U$.

Set $W = W_1 \cap W_2$. Then W is an open neighbourhood of e with $WW \subseteq W_1 W_2 \subseteq U$, since $W \subseteq W_1$ and $W \subseteq W_2$. To make the neighbourhood symmetric, put

$$V = W \cap W^{-1}$$

As noted after definition 8.2.5, $W^{-1} = \iota(W)$ is an open neighbourhood of e because inversion ι is a homeomorphism fixing e; hence V is an open neighbourhood of e, and $V = V^{-1}$ because $V^{-1} = (W \cap W^{-1})^{-1} = W^{-1} \cap W = V$. Since $V \subseteq W$, we have $VV \subseteq WW \subseteq U$. Finally $e \in V$ gives $V = Ve \subseteq VV$, so $V \subseteq VV \subseteq U$, completing the proof.

This lemma is the group-theoretic replacement for halving a radius, and the analogy is exact. The target U plays the role of the ball of radius r; the symmetric V plays the role of the ball of radius $r/2$; and the inclusion $VV \subseteq U$ is the triangle inequality $\frac{r}{2} + \frac{r}{2} = r$, rewritten multiplicatively. Symmetry, $V = V^{-1}$, is the counterpart of a ball being centred: it lets one pass freely between v and v^{-1} without leaving V, just as a centred ball treats x and $-x$ alike. The two properties combine into the estimate the following sections lean on: if a and b each lie within V of a point, then $a^{-1}b \in VV \subseteq U$, so a and b are within U of each other. This is precisely the manipulation a metric makes routine, now available in any topological group with no metric in sight.

The halving can be iterated, which is what turns a single application into a genuine substitute for a sequence of radii $r, r/2, r/4, \ldots$ tending to 0.

Example 8.2.8: Iterated Halving and Cubes of a Neighbourhood

Let U be a neighbourhood of e in a topological group G. Applying lemma 8.2.7 to U produces a symmetric V with $VV \subseteq U$. Applying it again to V produces a symmetric V' with $V'V' \subseteq V$; then

$$V'V'V'V' \subseteq VV \subseteq U$$

so V' is a symmetric neighbourhood whose fourfold product lands in U. Repeating n times yields, for any n, a symmetric neighbourhood $V^{(n)}$ of e with 2^n factors of $V^{(n)}$ contained in U. This is the exact analogue of subdividing a radius r into 2^n steps of size $r/2^n$.

A single application already gives control of triple products, which is the form the closure arguments use. Given U, take the symmetric V with $VV \subseteq U$, then a symmetric V_1 with $V_1V_1 \subseteq V$. Then

$$V_1V_1V_1 \subseteq V_1V_1V_1V_1 \subseteq VV \subseteq U$$

using $e \in V_1$ to insert a fourth factor. Thus every neighbourhood U of e contains a symmetric V_1 with $V_1V_1V_1 \subseteq U$; concretely in $(\mathbb{R}, +)$, starting from $U = (-1, 1)$ one may take $V = (-\frac{1}{2}, \frac{1}{2})$ and $V_1 = (-\frac{1}{4}, \frac{1}{4})$, and indeed $V_1 + V_1 + V_1 = (-\frac{3}{4}, \frac{3}{4}) \subseteq (-1, 1)$.

8.3 Subgroups and Closures

The subsets that matter most in a topological group are the ones that respect both structures at once: the subgroups. This section fixes how a subgroup inherits a topology, and then asks how the two structures interact when a subgroup is enlarged to its closure or is already open. Three facts emerge, each a consequence of the continuity of the operations rather than of any special feature of the group. A subgroup is again a topological group; the closure of a subgroup is a subgroup, and stays normal if the subgroup was; and an open subgroup is automatically closed. The last of these, together with a symmetric neighbourhood argument, pins down when a discrete subgroup sits inside its ambient group as a closed set.

The starting point is that a subgroup carries a topology for free, from the ambient space, and that this topology is compatible with its group law.

Proposition 8.3.1: Subgroups Are Topological Groups

Let G be a topological group and $H \leq G$ a subgroup. Equipped with the subspace topology from G (section 2.1), H is a topological group.

***Proof*:**

The subspace topology makes the inclusion $j \colon H \to G$ continuous, and it is the coarsest topology with this property. Two maps must be shown continuous: the multiplication $m_H \colon H \times H \to H$ and the inversion $\iota_H \colon H \to H$ of the subgroup.

Multiplication. Since H is closed under the group law, m_H is the map $(h, h') \mapsto hh'$ obtained by restricting the ambient multiplication $m \colon G \times G \to G$ to $H \times H$ and observing that its values lie in H. Concretely, $j \circ m_H = m \circ (j \times j)$ as maps $H \times H \to G$. The product map $j \times j \colon H \times H \to G \times G$ is continuous because each factor j is, and m is continuous by hypothesis, so the composite $m \circ (j \times j)$ is continuous into G. Its image lies in H, and by the universal property of the subspace topology a map into H is continuous exactly when its composite with j is continuous into G. Hence m_H is continuous.

Inversion. The same argument applies to $\iota_H \colon H \to H$, $h \mapsto h^{-1}$. Here $j \circ \iota_H = \iota \circ j$, the composite of the continuous inclusion with the continuous ambient inversion ι, so $j \circ \iota_H$ is continuous into G with image in H; therefore ι_H is continuous. Both operations are continuous, so H is a topological group, as we sought to show.

The proposition says nothing surprising and everything necessary: it is the sanity check that the definition of a topological group is stable under passing to subgroups, so that every subgroup encountered from now on is silently a topological group in its own right. The mechanism is entirely formal, resting only on the two defining properties of the subspace topology, that the inclusion is continuous and that continuity of a map into H is tested by composing with the inclusion. Nothing about the group is used beyond closure of H under the operations.

Example 8.3.2: The Circle Inside the Nonzero Complexes

The unit circle $\mathbb{S}^1 = \{z \in \mathbb{C} \big| |z| = 1\}$ is a subgroup of the multiplicative group $\mathbb{C}^\times = (\mathbb{C} \setminus \{0\}, \cdot)$, since $|zw| = |z|\,|w|$ and $|z^{-1}| = |z|^{-1}$. Give $\mathbb{C}^\times$ its topology as a subspace of $\mathbb{C} \cong \mathbb{R}^2$. By proposition 8.3.1, $\mathbb{S}^1$ is a topological group in the subspace topology, and that subspace topology is exactly the usual topology of the circle: a basic open set is the intersection of $\mathbb{S}^1$ with an open disc in $\mathbb{C}^\times$, which is an open arc. Multiplication of unit complex numbers adds arguments, and this operation is continuous in the arc topology, matching the general statement. The same reasoning realizes the group μ_n of n-th roots of unity as a (finite, hence discrete) subgroup of $\mathbb{S}^1$.

Enlarging a subgroup to its closure produces a larger set, and the question is whether that larger set remains a subgroup. It does, and the reason is that the group operations are continuous: continuity forces a map to send the closure of a set into the closure of its image, which is enough to carry the subgroup axioms across to the closure. The one inclusion that does the work is $\overline{A}\,\overline{B} \subseteq \overline{AB}$, valid for any two subsets.

Proposition 8.3.3: The Closure of a Subgroup Is a Subgroup

Let G be a topological group and $H \leq G$ a subgroup. Then the closure $\overline{H}$ is a subgroup of G. If moreover $H \trianglelefteq G$, then $\overline{H} \trianglelefteq G$.

***Proof*:**
The argument rests on one inclusion, which is worth isolating. For any subsets $A, B \subseteq G$, write $AB = \{ab \big| a \in A,\ b \in B\}$ for their product. Then

$$\overline{A}\,\overline{B} \subseteq \overline{AB} \tag{8.1}$$

To see (8.1), recall that a continuous map f satisfies $f(\overline{S}) \subseteq \overline{f(S)}$ for every set S, since $f^{-1}(\overline{f(S)})$ is a closed set containing S and hence containing $\overline{S}$. Apply this to the continuous multiplication $m \colon G \times G \to G$ with $S = A \times B$. In the product topology $\overline{A \times B} = \overline{A} \times \overline{B}$ (section 2.2), and $m(A \times B) = AB$, so

$$\overline{A}\,\overline{B} = m\left(\overline{A} \times \overline{B}\right) = m\left(\overline{A \times B}\right) \subseteq \overline{m(A \times B)} = \overline{AB}$$

which is (8.1).

Closure under multiplication. Since H is a subgroup, $HH = H$. Taking $A = B = H$ in (8.1) gives $\overline{H}\,\overline{H} \subseteq \overline{HH} = \overline{H}$, so the product of any two elements of $\overline{H}$ again lies in $\overline{H}$.

Closure under inversion. The inversion $\iota\colon G \to G$ is a homeomorphism, being continuous with continuous inverse ι itself. A homeomorphism commutes with closure, so $\iota(\overline{H}) = \overline{\iota(H)}$. Now $\iota(H) = H^{-1} = H$ because H is a subgroup, whence $\overline{H}^{-1} = \iota(\overline{H}) = \overline{H}$: the closure is closed under inversion. Finally $e \in H \subseteq \overline{H}$, so $\overline{H}$ contains the identity. Thus $\overline{H}$ is a subgroup.

Normality. Suppose $H \trianglelefteq G$, and fix $g \in G$. Conjugation $c_g\colon x \mapsto gxg^{-1}$ is the composite $L_g \circ R_{g^{-1}}$ of two translations, each a homeomorphism, so c_g is a homeomorphism and again commutes with closure: $c_g(\overline{H}) = \overline{c_g(H)}$. Normality of H gives $c_g(H) = gHg^{-1} = H$, so $g\overline{H}g^{-1} = c_g(\overline{H}) = \overline{H}$. As $g \in G$ was arbitrary, $\overline{H}$ is normal, completing the proof.

The proposition lets one pass freely between a subgroup and its closure while staying inside the category of subgroups, a move used constantly: to build closed subgroups, one takes closures without fear of leaving the class, and normality survives the passage. The inclusion (8.1) is the entire content, and it is one instance of the recurring principle that continuity of the structure maps transports algebraic closure properties to topological closures. The same inclusion, applied to a single point $A = \{e\}$, will identify $\overline{\{e\}}$ as a normal subgroup and drive the separation theory of the next section.

Example 8.3.4: The Rationals in the Reals

Take $G = (\mathbb{R}, +)$ and the subgroup $H = \mathbb{Q}$ of rationals, written additively. The rationals are dense in $\mathbb{R}$, so $\overline{\mathbb{Q}} = \mathbb{R}$. Proposition 8.3.3 predicts that $\overline{\mathbb{Q}}$ is a subgroup, and indeed $\mathbb{R}$ is a subgroup of itself, the whole group. This is the extreme case, where the closure swallows all of G. A less degenerate instance is the cyclic subgroup $\mathbb{Z}\alpha = \{n\alpha \big| n \in \mathbb{Z}\}$ generated by an irrational α inside the circle $\mathbb{R}/\mathbb{Z}$: its closure is the entire circle, again a subgroup, whereas for rational $\alpha = p/q$ the subgroup $\mathbb{Z}\alpha$ is already the finite (hence closed) group of q-th roots of the identity. In each case the closure is a subgroup, exactly as the proposition asserts.

There is a second way a subgroup can interact with the topology, complementary to being closed: it can be open. Openness is a strong hypothesis for a subgroup, because a subgroup is a union of cosets and each coset is a translate of the subgroup. Translating an open set by a homeomorphism keeps it open, so if the subgroup is open then so is every one of its cosets, and the complement, being a union of the other cosets, is open as well.

Proposition 8.3.5: Open Subgroups Are Closed

Let G be a topological group and $H \leq G$ an open subgroup. Then H is closed.

***Proof*:**

The group G is the disjoint union of the left cosets of H: distinct cosets are disjoint, and every $g \in G$ lies in gH. The complement of H is therefore the union of all cosets other than H itself,

$$G \setminus H = \bigcup_{gH \neq H} gH$$

where the union runs over the nontrivial cosets. Each coset gH is the image $L_g(H)$ of the open set H under the left translation L_g, which is a homeomorphism, so gH is open. A union of open sets is open, so $G \setminus H$ is open, and hence H is closed, as we sought to show.

The proposition exposes a rigidity peculiar to groups: for a subgroup there is no room between open and closed, since being open forces being closed. The reason is homogeneity. A single open subgroup is copied by translation onto each of its cosets, tiling the whole group into open pieces of equal size, and H is one tile among a partition of the space into open tiles. The complement is a union of the remaining tiles, so it too is open. Read through the quotient, the statement says that an open subgroup has *discrete* quotient: the cosets are the points of G/H, each is open, so every point of G/H is open and the quotient topology is discrete. An open subgroup thus carves off a discrete quotient, and the coarser the group is near the identity, the fewer open subgroups it admits.

Example 8.3.6: Open Subgroups of the Reals and of a Discrete Group

In $(\mathbb{R}, +)$ the only open subgroup is $\mathbb{R}$ itself. Any subgroup containing an open interval around 0 contains an interval $(-\varepsilon, \varepsilon)$, hence all integer multiples of every point in it, hence all of $\mathbb{R}$ by the Archimedean property; and $\{0\}$ is not open. So proposition 8.3.5 says only that $\mathbb{R}$ is closed in $\mathbb{R}$, which is vacuous, and this reflects that $\mathbb{R}$ is connected and thus has no proper open-and-closed subgroup. The content of the proposition appears where a group has many open subgroups. In a discrete group every subgroup is open, and the proposition recovers the obvious fact that every subgroup is then also closed. The useful setting is intermediate, such as the additive group $\mathbb{Z}_p$ of p-adic integers, where the subgroups $p^n\mathbb{Z}_p$ are open, and proposition 8.3.5 certifies each of them closed with discrete finite quotient $\mathbb{Z}_p/p^n\mathbb{Z}_p \cong \mathbb{Z}/p^n\mathbb{Z}$; these are the profinite groups of Galois theory.

The interplay between open and closed subgroups feeds directly into a question about discrete subgroups. A discrete subgroup need not be open, since the ambient topology can be far finer than the discrete one, and so the previous proposition does not apply to it. Nevertheless a discrete subgroup of a Hausdorff group is closed, and the proof isolates each of its points by a small symmetric neighbourhood, using the Hausdorff hypothesis to rule out any limit point outside the subgroup. The Hausdorff hypothesis is stated explicitly because the argument leans on it directly: it separates a putative limit point from the unique subgroup element sitting near it, and that separation is exactly what Hausdorffness provides. Without separation the conclusion fails, as the trivial subgroup $\{e\}$ shows in the indiscrete topology on any group with more than one element, where $\overline{\{e\}}$ is the whole group even though $\{e\}$ is discrete.

Corollary 8.3.7: Discrete Subgroups of Hausdorff Groups Are Closed

Let G be a Hausdorff topological group and $H \leq G$ a discrete subgroup, meaning that H carries the discrete topology as a subspace of G. Then H is closed in G.

***Proof*:**

Discreteness of H means that the singleton $\{e\}$ is open in the subspace topology, so there is an open set U of G with

$$U \cap H = \{e\} \tag{8.2}$$

By lemma 8.2.7 choose a symmetric neighbourhood V of e, so $V = V^{-1}$, with $VV \subseteq U$. The claim is that $\overline{H} = H$, and since $H \subseteq \overline{H}$ always, it suffices to prove $\overline{H} \subseteq H$. Fix $x \in \overline{H}$.

Step 1: the neighbourhood xV of x meets H in at most one point. Suppose $h_1, h_2 \in xV \cap H$, say $h_1 = xv_1$ and $h_2 = xv_2$ with $v_1, v_2 \in V$. Then

$$h_2^{-1}h_1 = v_2^{-1}x^{-1}xv_1 = v_2^{-1}v_1 \in V^{-1}V = VV \subseteq U$$

using $V = V^{-1}$ for the equality $V^{-1}V = VV$. On the other hand $h_2^{-1}h_1 \in H$ because H is a subgroup, so $h_2^{-1}h_1 \in U \cap H = \{e\}$ by (8.2). Therefore $h_2^{-1}h_1 = e$, that is $h_1 = h_2$. So $xV \cap H$ has at most one element.

Step 2: the neighbourhood xV of x meets H. The translate $xV = L_x(V)$ is a neighbourhood of x, being the image of a neighbourhood of e under the homeomorphism L_x. Since $x \in \overline{H}$, every neighbourhood of x contains a point of H, so $xV \cap H \neq \emptyset$. Combined with Step 1, this gives $xV \cap H = \{h\}$ for a single element $h \in H$.

Step 3: the Hausdorff hypothesis forces $x = h$. Suppose to the contrary that $x \neq h$. Because G is Hausdorff, there is an open set N with $x \in N$ and $h \notin N$. Then $N \cap xV$ is again a neighbourhood of x, and it excludes h, so

$$(N \cap xV) \cap H \subseteq (xV \cap H) \setminus \{h\} = \emptyset$$

Thus the neighbourhood $N \cap xV$ of x contains no point of H, contradicting $x \in \overline{H}$. Therefore $x = h$, and in particular $x \in H$.

Every $x \in \overline{H}$ lies in H, so $\overline{H} = H$ and H is closed, completing the proof.

The corollary is the point-set foundation for treating discrete subgroups as closed subobjects, so that quotients by them, such as the tori $\mathbb{R}^n/\mathbb{Z}^n$ or the modular quotients built from lattices in a Lie group, inherit good separation properties from the ambient group. The proof is a template worth remembering: a discrete subgroup comes with a neighbourhood of the identity that isolates it, one shrinks that neighbourhood symmetrically so that translates overlap at most trivially, and Hausdorffness then prevents any stray limit point from hiding between an element of the subgroup and its would-be accumulation. The three ingredients, a symmetric shrinking neighbourhood, the subgroup structure, and the separation axiom, recur throughout the analysis of discrete subgroups.

Example 8.3.8: Lattices in Euclidean Space

The integer lattice $\mathbb{Z}^n$ is a discrete subgroup of the Hausdorff group $(\mathbb{R}^n, +)$: the open ball U of radius $\frac{1}{2}$ about the origin meets $\mathbb{Z}^n$ only in 0, so $\{0\}$ is open in the subspace topology and $\mathbb{Z}^n$ is discrete. By corollary 8.3.7, $\mathbb{Z}^n$ is closed in $\mathbb{R}^n$, which one also sees directly since its complement is a union of open unit cubes with the lattice points removed. Closedness is what makes the quotient $\mathbb{R}^n/\mathbb{Z}^n$, the n-torus, a Hausdorff topological group rather than a pathological quotient. Contrast this with the dense subgroup $\mathbb{Z} + \mathbb{Z}\sqrt{2}$ of $\mathbb{R}$: it is a subgroup, but it is *not* discrete, since every neighbourhood of 0 contains infinitely many of its points, and correspondingly it is not closed, its closure being all of $\mathbb{R}$. The corollary applies precisely to the discrete case, and its hypothesis of discreteness is exactly what the dense example fails.

8.4 Separation

The separation axioms of chapter 4 form a strict hierarchy for a general space, and telling one level from the next can be delicate. On a topological group homogeneity flattens the hierarchy, dragging the weakest axioms up to the strongest. The task of this section is to make that collapse precise, tracing it back to a single object, the closure $\overline{\{e\}}$ of the identity.

The identity of a group is the natural place to test how well points can be told apart, because translation moves the test to e without loss. If two distinct points x and y cannot be separated, then e and $x^{-1}y$ cannot be separated either, so the failure of separation is entirely encoded in which points cluster around e. The set of such points is exactly $\overline{\{e\}}$, and its algebraic character is the first thing to establish.

Proposition 8.4.1: Closure of the Identity

Let G be a topological group. The closure $\overline{\{e\}}$ is a closed normal subgroup of G, and it is contained in every neighbourhood of e.

***Proof*:**
The singleton $\{e\}$ is a subgroup of G, and it is normal, since $geg^{-1} = e$ for every $g \in G$. By proposition 8.3.3, the closure of a subgroup is a subgroup and the closure of a normal subgroup is normal; applying this to $\{e\}$ shows that $\overline{\{e\}}$ is a normal subgroup of G. It is closed because a closure is always closed.

It remains to show that $\overline{\{e\}}$ is contained in every neighbourhood of e. Let U be a neighbourhood of e and let $x \in \overline{\{e\}}$. Membership in the closure means every neighbourhood of x meets $\{e\}$, that is, every neighbourhood of x contains e. The interior $\mathrm{int}(U)$ is an open neighbourhood of e, and since inversion ι is a homeomorphism, the set $V = \mathrm{int}(U)^{-1}$ is again an open neighbourhood of e. Applying the homeomorphism L_x to V yields the open neighbourhood xV of x. Because $x \in \overline{\{e\}}$, this neighbourhood contains e, so $e = xv$ for some $v \in V$. Then $x = v^{-1} \in V^{-1} = \mathrm{int}(U) \subseteq U$. Every point of $\overline{\{e\}}$ therefore lies in U, completing the proof.

The containment in the statement makes $\overline{\{e\}}$ a subset of every neighbourhood of e, hence of their intersection; the equivalence theorem below sharpens this into an identity of the two sets when G is Hausdorff. This subgroup measures the failure of the group to separate points near the identity. When it is trivial the group will turn out to be Hausdorff, and when it is all of G the topology is indiscrete. Because it is normal, one can always quotient it away, a construction the next section makes into the standard route to a Hausdorff group.

Example 8.4.2: The Trivial Topology on a Group
Let G be any group carrying the indiscrete topology, whose only open sets are $\emptyset$ and G. Both m and ι are continuous, since the preimage of an open set is open, so G is a topological group. The only closed sets are $\emptyset$ and G, so the smallest closed set containing e is G itself, giving $\overline{\{e\}} = G$. Consistently, the only neighbourhood of e is G, so the intersection of the neighbourhoods of e is G. At the opposite extreme, take G discrete: then $\{e\}$ is already closed, $\overline{\{e\}} = \{e\}$, and the intersection of the neighbourhoods of e is $\{e\}$ as well.

With $\overline{\{e\}}$ identified as the obstruction to separation, the collapse of the hierarchy can be stated as an equivalence. The list below runs from the weakest classical axiom, T_0, up through Hausdorffness, and then to two group-theoretic reformulations that hold only because the space is homogeneous. That a mere T_0 hypothesis on a topological group already forces the full Hausdorff conclusion is what distinguishes these spaces from arbitrary ones.

Theorem 8.4.3: Separation Criteria for Topological Groups

Let G be a topological group. The following are equivalent.

1. G is T_0.
2. G is T_1.
3. G is Hausdorff.
4. The singleton $\{e\}$ is closed.
5. The intersection of all neighbourhoods of e is $\{e\}$.

The proof runs as a cycle, (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) $\Rightarrow$ (1), so that each condition implies the next and the last closes back on the first. Only one link, (3) $\Rightarrow$ (4), holds for a general space: in any space a Hausdorff topology has closed singletons. The other four, (1) $\Rightarrow$ (2), (2) $\Rightarrow$ (3), (4) $\Rightarrow$ (5), and (5) $\Rightarrow$ (1), use the group structure in an essential way, through translation or the continuity of $(a, b) \mapsto a^{-1}b$; it is exactly these that fail for an arbitrary space, where T_1 does not force Hausdorff.

***Proof*:**
(1) $\Rightarrow$ (2). Assume G is T_0. It suffices to show that $\{e\}$ is closed, since translation then carries this to every point: for any g, the singleton $\{g\} = L_g(\{e\})$ is the image of a closed set under the homeomorphism L_g, hence closed, and a space in which every singleton is closed is T_1. To prove $\{e\}$ closed, fix $x \neq e$ and produce a neighbourhood of x excluding e, which shows $x \notin \overline{\{e\}}$. By T_0, some open set contains exactly one of e and x. If an open set W contains x but not e, then W is a neighbourhood of x avoiding e. If instead an open set U contains e but not x, then U^{-1} is open, since inversion ι is a homeomorphism, and xU^{-1} is an open neighbourhood of x; here $e \in xU^{-1}$ would give $x^{-1} \in U^{-1}$, hence $x \in U$, a contradiction, so xU^{-1} avoids e. In both cases x has a neighbourhood excluding e, so $x \notin \overline{\{e\}}$. As $x \neq e$ was arbitrary, $\overline{\{e\}} = \{e\}$, so $\{e\}$ is closed and G is T_1.

(2) $\Rightarrow$ (3). Assume G is T_1, so every singleton is closed; in particular $\{e\}$ is closed. Let $x \neq y$ in G. Then $x^{-1}y \neq e$, and since $\{e\}$ is closed, its complement $G \setminus \{e\}$ is open. The map $d\colon G \times G \to G$ sending $(a, b) \mapsto a^{-1}b$ is continuous, being the composite of ι on the first coordinate with m. Since $d(x, y) = x^{-1}y$ lies in the open set $G \setminus \{e\}$, continuity of d provides open neighbourhoods A of x and B of y with $d(A \times B) \subseteq G \setminus \{e\}$, that is, $a^{-1}b \neq e$ for all $a \in A$, $b \in B$. Equivalently $a \neq b$ whenever $a \in A$ and $b \in B$, so $A \cap B = \emptyset$. Thus x and y have disjoint open neighbourhoods, and G is Hausdorff (section 1.5).

(3) $\Rightarrow$ (4). Assume G is Hausdorff. In a Hausdorff space every singleton is closed: given $x \neq e$, disjoint open neighbourhoods U of x and V of e exist, so U is a neighbourhood of x missing e, whence $x \notin \overline{\{e\}}$. As x was arbitrary, $\overline{\{e\}} = \{e\}$, so $\{e\}$ is closed.

(4) $\Rightarrow$ (5). Assume $\{e\}$ is closed. Write I for the intersection of all neighbourhoods of e. Since $\{e\}$ is closed, $\overline{\{e\}} = \{e\}$, and by proposition 8.4.1 the set $\overline{\{e\}}$ is contained in every neighbourhood of e, so $\{e\} \subseteq I$. For the reverse inclusion, let $x \neq e$. Applying the homeomorphism L_x to the closed set $\{e\}$ shows $\{x\}$ is closed, so $G \setminus \{x\}$ is an open neighbourhood of e that excludes x;

hence $x \notin I$. Every point of I therefore equals e, giving $I \subseteq \{e\}$ and $I = \{e\}$.

(5) $\Rightarrow$ (1). Assume the intersection of all neighbourhoods of e is $\{e\}$. Let $x \neq y$. Then $x^{-1}y \neq e$, so by hypothesis $x^{-1}y$ fails to lie in some neighbourhood of e; choose an open neighbourhood U of e with $x^{-1}y \notin U$. The set xU is an open neighbourhood of x, and $y \notin xU$, since $y \in xU$ would give $x^{-1}y \in U$. Thus x has an open neighbourhood excluding y, which is the T_0 condition for the pair x, y. As the pair was arbitrary, G is T_0, completing the cycle and with it the proof.

The equivalence collapses the bottom of the separation tower onto a single condition, and it does so for a structural reason: homogeneity means a separation property need only be verified at e, and each of the five statements is a way of asserting that e can be separated from every other point. Condition (5) is the most useful in practice, because it is checked against a neighbourhood base at e rather than against all pairs of points. In the profinite groups of Galois theory, for instance, the open subgroups form such a base, and the group is Hausdorff exactly when their intersection is trivial, which is the separatedness built into the inverse-limit topology. From here on "Hausdorff topological group" may be used interchangeably with T_1, or with the demand that $\{e\}$ be closed.

Example 8.4.4: The Circle and the Line with a Collapsed Coordinate

The circle group $\mathbb{S}^1 \subseteq \mathbb{C}^\times$, with its subspace topology, is Hausdorff, so by the theorem $\{1\}$ is closed, every singleton is closed, and the intersection of the neighbourhoods of 1 is $\{1\}$: the arcs about 1 shrink to the point. For a group where every condition of the theorem *fails*, take $G = \mathbb{R}^2$ under coordinatewise addition, but topologized by pulling the standard topology of $\mathbb{R}$ back along the projection $p\colon (s,t) \mapsto s$: the open sets of G are exactly the sets $U \times \mathbb{R}$ with $U \subseteq \mathbb{R}$ open. Addition and negation are continuous, since each acts on the first coordinate through the continuous operations of $\mathbb{R}$ and ignores the second, so G is a topological group. The closed sets are the sets $C \times \mathbb{R}$ with $C \subseteq \mathbb{R}$ closed, so the smallest closed set containing the identity $(0,0)$ is $\{0\} \times \mathbb{R}$, giving $\overline{\{(0,0)\}} = \{0\} \times \mathbb{R}$. This is a nontrivial closed normal subgroup, so $\{(0,0)\}$ is not closed and G is not T_1; the points $(0,0)$ and $(0,1)$ lie in exactly the same open sets and cannot be told apart even by the T_0 axiom. Every neighbourhood of $(0,0)$ contains $\{0\} \times \mathbb{R}$, and their intersection is precisely $\{0\} \times \mathbb{R}$, consistent with the containment proposition 8.4.1 guarantees. Collapsing the second coordinate by the quotient of the next section returns the Hausdorff group $\mathbb{R}$.

A Hausdorff group satisfies more than the axioms of the theorem: it separates a point from a closed set, not merely a point from a point. This is the regularity axiom T_3, and for a general space it is a genuine strengthening of Hausdorffness that must be imposed by hand. For a topological group it comes for free, and the reason is again translation together with the symmetric neighbourhoods of lemma 8.2.7: one shrinks a neighbourhood of e so that even its closure stays small.

Proposition 8.4.5: Hausdorff Groups are Regular

Every Hausdorff topological group is regular; that is, given a point x and a closed set A with $x \notin A$, there exist disjoint open sets $O_1 \ni x$ and $O_2 \supseteq A$.

***Proof*:**
Let $A \subseteq G$ be closed and $x \notin A$. Translating by $L_{x^{-1}}$, which is a homeomorphism, replaces A by the closed set $x^{-1}A$ and x by e, and $e \notin x^{-1}A$; a separation of e from $x^{-1}A$ translates back

by L_x to a separation of x from A. So it suffices to separate e from a closed set $B = x^{-1}A$ not containing it.

Since B is closed and $e \notin B$, the complement $G \setminus B$ is an open neighbourhood of e. By lemma 8.2.7, there is a symmetric neighbourhood V of e, so $V = V^{-1}$, with

$$VV \subseteq G \setminus B$$

which says $VV \cap B = \emptyset$. Set $O_1 = \operatorname{int}(V)$, an open neighbourhood of e, and $O_2 = G \setminus \overline{V}$. The two steps below show that O_2 is an open set containing B and that it is disjoint from O_1, using that a symmetric V controls its own closure.

Step 1: the closure of V misses B. Suppose $b \in \overline{V} \cap B$. Since $b \in \overline{V}$, every neighbourhood of b meets V; in particular bV is a neighbourhood of b, being the image of the neighbourhood V of e under the homeomorphism L_b, so $bV \cap V \neq \emptyset$. Thus $bv_1 = v_2$ for some $v_1, v_2 \in V$, giving $b = v_2 v_1^{-1}$. Because V is symmetric, $v_1^{-1} \in V$, so $b \in VV \subseteq G \setminus B$, contradicting $b \in B$. Hence $\overline{V} \cap B = \emptyset$, so $B \subseteq G \setminus \overline{V} = O_2$.

Step 2: the two sets are disjoint. The set $O_2 = G \setminus \overline{V}$ is open, since $\overline{V}$ is closed, and it contains B by Step 1. Its complement is $\overline{V}$, so $O_2 \cap \overline{V} = \emptyset$, and $O_1 = \operatorname{int}(V) \subseteq V \subseteq \overline{V}$ gives $O_1 \cap O_2 = \emptyset$. Thus O_1 and O_2 are disjoint open sets with $e \in O_1$ and $B \subseteq O_2$. Translating back to x and A separates x from A, as we sought to show.

Because a Hausdorff group is T_1 by the equivalence above and regular by this proposition, it is T_3 in the strong sense that includes point-closedness, and so every Hausdorff topological group is automatically a T_3 space. The passage from "open neighbourhood U of e" to "open neighbourhood V with $\overline{V} \subseteq U$" is exactly the shrinking that regularity demands, and lemma 8.2.7 supplies it uniformly at e, after which homogeneity spreads it to every point. Iterating the construction yields neighbourhoods whose closures nest arbitrarily deep, which is the input to a stronger fact left to chapter 4: a Hausdorff topological group is completely regular, so that a continuous real-valued function separates any point from any disjoint closed set (a Urysohn function).

8.5 Quotient Groups

Passing to a quotient is the one construction where the group structure and the topology have to negotiate. Set-theoretically the coset space G/H is fixed the moment H is chosen; the question is which topology it should carry, and whether that topology is compatible with the group law inherited when H is normal. The quotient topology (section 2.3) is the forced choice, and this section checks that it does everything one wants: the projection is not merely continuous but open, the quotient of a topological group by a normal subgroup is again a topological group, and the separation of G/H is controlled by a single condition on H. The homogeneity that governs G itself reappears at the end, now attached to G/H even when H is not normal, as the notion of a homogeneous space.

Throughout, $H \leq G$ is a subgroup, $G/H = \{ gH : g \in G \}$ is the set of left cosets, and $\pi \colon G \to G/H$ is the map $g \mapsto gH$.

Definition 8.5.1: Quotient Topology on a Coset Space

Let $H \leq G$ be a subgroup of a topological group. The *coset space* G/H is the set of left cosets gH, and $\pi\colon G \to G/H$, $\pi(g) = gH$, is the *quotient map*. The *quotient topology* on G/H is the topology induced by π (section 2.3): a set $W \subseteq G/H$ is open exactly when $\pi^{-1}(W)$ is open in G. When $H \trianglelefteq G$ is normal, G/H carries its usual group structure, and it is then called the *quotient group*.

The definition asks nothing new: it simply installs the quotient topology from section 2.3 on the coset space, so that π becomes the universal continuous surjection out of G that is constant on cosets. A set downstairs is open precisely when its full preimage upstairs is open, which is the only topology making π continuous while identifying as many open sets as possible. What must be verified is not that this topology exists but that it interacts well with translation and, when H is normal, with the group operations, and that verification is the content of the results below. The preimage $\pi^{-1}(gH)$ of a single coset is the coset gH itself, viewed as a subset of G; this identification of points of G/H with subsets of G is used constantly.

Example 8.5.2: The Circle as a Quotient

Take $G = (\mathbb{R}, +)$ and $H = \mathbb{Z}$. The coset space $\mathbb{R}/\mathbb{Z}$ consists of the cosets $x + \mathbb{Z}$, and $\pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{Z}$ sends x to its class modulo 1. A set $W \subseteq \mathbb{R}/\mathbb{Z}$ is open when $\pi^{-1}(W)$ is a union of the translates by integers of some open subset of $\mathbb{R}$, that is, a $\mathbb{Z}$-periodic open set. Wrapping the line around the circle, the map

$$x + \mathbb{Z} \mapsto (\cos 2\pi x, \sin 2\pi x)$$

is a bijection of $\mathbb{R}/\mathbb{Z}$ onto $\mathbb{S}^1 \subseteq \mathbb{R}^2$, and one checks directly that a $\mathbb{Z}$-periodic set is open in $\mathbb{R}$ if and only if its image is open in $\mathbb{S}^1$. So the quotient topology on $\mathbb{R}/\mathbb{Z}$ is exactly the circle's topology, and $\mathbb{R}/\mathbb{Z}$ recovers the compact group $\mathbb{S}^1$ as a coset space of the line.

The circle example already exhibits the phenomenon that makes quotients of topological groups better behaved than quotients of arbitrary spaces: the projection $\mathbb{R} \to \mathbb{R}/\mathbb{Z}$ carries open intervals to open arcs. This is not an accident of the circle but a structural feature, and it comes from the fact that translating an open set by the elements of H produces an open set. The next result isolates this.

Proposition 8.5.3: The Quotient Map is Open

Let $H \leq G$ and let $\pi\colon G \to G/H$ be the quotient map. Then π is an open map: for every open $U \subseteq G$, the image $\pi(U)$ is open in G/H. Moreover

$$\pi^{-1}(\pi(U)) = UH = \bigcup_{h \in H} Uh \tag{8.3}$$

***Proof*:**

By the definition of the quotient topology, $\pi(U)$ is open in G/H if and only if $\pi^{-1}(\pi(U))$ is open in G, so it suffices to prove the identity (8.3) and then observe that its right-hand side is open.

First we identify $\pi^{-1}(\pi(U))$. An element $g \in G$ lies in $\pi^{-1}(\pi(U))$ exactly when $gH = uH$ for some $u \in U$, which holds exactly when $g \in uH$ for some $u \in U$, that is, when $g \in UH$. Writing the saturation coset by coset,

$$UH = \{\, uh : u \in U,\ h \in H \,\} = \bigcup_{h \in H} Uh$$

which proves (8.3).

It remains to see that UH is open. For each fixed $h \in H$ the right translation $R_h \colon x \mapsto xh$ is a homeomorphism of G (proposition 8.1.4), so $Uh = R_h(U)$ is open. A union of open sets is open, so $UH = \bigcup_{h \in H} Uh$ is open in G. Therefore $\pi^{-1}(\pi(U))$ is open, and $\pi(U)$ is open in G/H, completing the proof.

Openness of π, beyond its mere continuity, is what the constructions below rest on. The set UH appearing in (8.3) is the *saturation* of U: the smallest H-invariant set containing U, obtained by sliding U along all of H. Because each slide Uh is a translate of an open set, and translations are homeomorphisms, the saturation is again open, which is precisely where the group structure feeds the topology. The payoff is that π preserves open sets in both directions, so images and preimages of open sets are open, which is exactly the regularity one needs to transport the continuity of the multiplication upstairs to a continuity statement downstairs. Openness of π also fails for general quotient maps of topological spaces; it is special to the coset-space situation, and every structural result that follows leans on it.

With openness in hand, the compatibility of the group law with the topology on G/H follows almost formally, provided H is normal so that a group law exists at all.

Theorem 8.5.4: Quotients by Normal Subgroups

Let $H \trianglelefteq G$ be a normal subgroup of a topological group G, and give G/H the quotient topology. Then G/H is a topological group, and the quotient map $\pi \colon G \to G/H$ is a continuous, open, surjective homomorphism.

***Proof*:**
Since H is normal, the left cosets form a group under $(gH)(g'H) = (gg')H$, with identity eH and inversion $(gH)^{-1} = g^{-1}H$, and π is a group homomorphism by construction. Continuity of π is immediate from the definition of the quotient topology, and openness is proposition 8.5.3; surjectivity is clear. It remains to prove that the multiplication

$$\bar{m} \colon G/H \times G/H \to G/H, \qquad (gH,\, g'H) \mapsto (gg')H$$

and the inversion $\bar{\imath} \colon G/H \to G/H$, $gH \mapsto g^{-1}H$, are continuous.

Multiplication. Let $m \colon G \times G \to G$ be the multiplication of G. The two projections fit into the square

$$\bar{m} \circ (\pi \times \pi) = \pi \circ m$$

as maps $G \times G \to G/H$, since both send (g, g') to $(gg')H$. Now $\pi \times \pi \colon G \times G \to G/H \times G/H$ is a surjective open map, because a product of open surjections is an open surjection: for open $U, U' \subseteq G$ the image $(\pi \times \pi)(U \times U') = \pi(U) \times \pi(U')$ is a basic open set of the product topology (section 2.2) by proposition 8.5.3, and every open subset of $G \times G$ is a union of such rectangles, so its image is open. A surjective open map is a quotient map, hence a set $W \subseteq G/H$ has

$\bar{m}^{-1}(W)$ open if and only if $(\pi \times \pi)^{-1}(\bar{m}^{-1}(W))$ is open in $G \times G$. But

$$(\pi \times \pi)^{-1}(\bar{m}^{-1}(W)) = (\bar{m} \circ (\pi \times \pi))^{-1}(W) = (\pi \circ m)^{-1}(W) = m^{-1}(\pi^{-1}(W))$$

which is open because $\pi^{-1}(W)$ is open (continuity of π) and m is continuous. Therefore $\bar{m}^{-1}(W)$ is open whenever W is, so $\bar{m}$ is continuous.

Inversion. The same argument, with the inversion $\iota\colon G \to G$ in place of m and the open surjection π in place of $\pi \times \pi$, applies. From $\bar{\iota} \circ \pi = \pi \circ \iota$ and the fact that π is a quotient map, a set $W \subseteq G/H$ has $\bar{\iota}^{-1}(W)$ open if and only if $\pi^{-1}(\bar{\iota}^{-1}(W)) = \iota^{-1}(\pi^{-1}(W))$ is open, and the latter is open because $\pi^{-1}(W)$ is open and ι is continuous. Hence $\bar{\iota}$ is continuous.

Both structure maps are continuous, so G/H is a topological group, and π is a continuous open surjective homomorphism, as we sought to show.

The proof is worth reading as a template rather than a computation, because it shows how openness of π converts π into a quotient map in the topological sense and thereby lets one check continuity of a map *out of* G/H by lifting it to a map out of G. Continuity downstairs becomes continuity of the composite upstairs, which is already known. This is the general principle: a continuous map $f\colon G \to X$ that is constant on the cosets of H factors uniquely as $f = \bar{f} \circ \pi$ through a continuous map $\bar{f}\colon G/H \to X$, and $\bar{f}$ is continuous precisely because π is a quotient map. That factorization is the *universal property* of the quotient: G/H is the initial topological group receiving a continuous homomorphism from G that kills H, and every continuous homomorphism out of G trivial on H descends to it. When f is a continuous homomorphism with kernel exactly H, the descended map $\bar{f}$ is a continuous bijective homomorphism onto its image, the topological form of the first isomorphism theorem; it need not be a homeomorphism unless f is itself open onto its image.

Example 8.5.5: The Quotient $\mathrm{GL}_n(\mathbb{R})$ by Its Center

Let $G = \mathrm{GL}_n(\mathbb{R})$ and let H be its center, the subgroup of nonzero scalar matrices $\{\, \lambda I : \lambda \in \mathbb{R}^\times \,\}$, which is normal. By theorem 8.5.4 the projective linear group $G/H = \mathrm{PGL}_n(\mathbb{R})$ is a topological group, and $\pi\colon \mathrm{GL}_n(\mathbb{R}) \to \mathrm{PGL}_n(\mathbb{R})$ is a continuous open homomorphism. Concretely, a set of scalar-classes of matrices is open in $\mathrm{PGL}_n(\mathbb{R})$ exactly when the set of all matrices representing those classes is open in $\mathrm{GL}_n(\mathbb{R})$; and the determinant, which is not invariant under scaling, does *not* descend, whereas the vanishing locus of any homogeneous polynomial in the entries does, illustrating the universal property in action.

Continuity and openness settle the algebraic-topological compatibility, but they say nothing about separation: the quotient of a Hausdorff group can fail to be Hausdorff, and one wants to know exactly when it does not. The separation results of section 8.4 reduce every such question to a condition on the single point eH, and that reduction pins the answer to a property of H.

Proposition 8.5.6: Separation of the Quotient

Let $H \leq G$ be a subgroup of a topological group G, and give G/H the quotient topology.

1. G/H is Hausdorff if and only if H is closed in G.
2. G/H is discrete if and only if H is open in G.

***Proof*:**
Part (1). The point eH of G/H has $\pi^{-1}(\{eH\}) = H$, so $\{eH\}$ is closed in G/H if and only if its preimage H is closed in G, by the definition of the quotient topology. When H is normal, G/H is a topological group by theorem 8.5.4, and theorem 8.4.3 says a topological group is Hausdorff if and only if its identity is closed; combining these, G/H is Hausdorff if and only if $\{eH\}$ is closed if and only if H is closed.

For a general (not necessarily normal) subgroup H we argue directly. If G/H is Hausdorff, then points are closed, so $\{eH\}$ is closed and hence $H = \pi^{-1}(\{eH\})$ is closed. Conversely, suppose H is closed. Let $gH \neq g'H$ in G/H and set $a = g^{-1}g'$, so $a \notin H$. Since H is closed, $G \setminus H$ is an open neighbourhood of a. The map $\mu\colon G \times G \to G$, $(x, y) \mapsto xay$, is continuous and sends (e, e) to $a \in G \setminus H$, so $\mu^{-1}(G \setminus H)$ is an open neighbourhood of (e, e) and contains a product neighbourhood $V' \times V''$. Setting $V = V' \cap (V')^{-1} \cap V'' \cap (V'')^{-1}$ gives a symmetric neighbourhood of e (lemma 8.2.7) with

$$VaV \subseteq G \setminus H, \qquad \text{that is,} \quad VaV \cap H = \varnothing$$

Then $\pi(gV)$ and $\pi(g'V)$ are open neighbourhoods of gH and $g'H$ in G/H by proposition 8.5.3, and they are disjoint. Indeed, if they met, some gv and $g'v'$ with $v, v' \in V$ would satisfy $gvH = g'v'H$, hence $(gv)^{-1}(g'v') \in H$, that is,

$$v^{-1}g^{-1}g'v' = v^{-1}av' \in H$$

But $v^{-1} \in V$ by symmetry and $v' \in V$, so $v^{-1}av' \in VaV$, contradicting $VaV \cap H = \varnothing$. Hence $\pi(gV)$ and $\pi(g'V)$ separate gH from $g'H$, and G/H is Hausdorff.

Part (2). The singleton $\{eH\}$ is open in G/H if and only if $\pi^{-1}(\{eH\}) = H$ is open in G. Suppose H is open. For any gH the coset gH is open in G, being the translate $L_g(H)$ of the open set H, so its image $\{gH\} = \pi(gH)$ is open in G/H by proposition 8.5.3. Thus every singleton is open and G/H is discrete. Conversely, if G/H is discrete then $\{eH\}$ is open, so $H = \pi^{-1}(\{eH\})$ is open, completing the proof.

The first statement is the practical criterion one uses constantly: to build a Hausdorff quotient, quotient by a closed subgroup, and nothing more is required. It also explains why the non-Hausdorff quotients are exactly the ones obtained by collapsing a subgroup that is not already topologically closed, since the coset eH then cannot be separated from its limit points, which are the other cosets accumulating on it. The reduction is entirely local: separation of the whole quotient collapses to closedness of a single fiber, because homogeneity spreads the separation at eH to every pair of points. The second statement is the discrete counterpart: an open subgroup produces a discrete quotient, matching the fact from section 8.3 that open subgroups are also closed, so a discrete quotient is automatically Hausdorff. Between these extremes lie the interesting cases, closed but not open subgroups, whose quotients are Hausdorff but not discrete, as the circle $\mathbb{R}/\mathbb{Z}$ already shows.

Example 8.5.7: A Non-Hausdorff Quotient
Take $G = (\mathbb{R}, +)$ and $H = \mathbb{Q}$, the subgroup of rationals. Then $\mathbb{Q}$ is a dense subgroup of $\mathbb{R}$, so it is not closed: $\overline{\mathbb{Q}} = \mathbb{R}$. By proposition 8.5.6 the quotient $\mathbb{R}/\mathbb{Q}$ is not Hausdorff. In fact its topology is indiscrete. If $W \subseteq \mathbb{R}/\mathbb{Q}$ is nonempty and open, then $\pi^{-1}(W)$ is a nonempty open $\mathbb{Q}$-invariant subset of $\mathbb{R}$; being $\mathbb{Q}$-invariant it is invariant under translation by a dense set, and a nonempty open set invariant under a dense group of translations is all of $\mathbb{R}$, so $\pi^{-1}(W) = \mathbb{R}$ and $W = \mathbb{R}/\mathbb{Q}$.

The only nonempty open set is the whole space, which is as far from Hausdorff as a topology can be, and the failure traces back exactly to the non-closedness of $\mathbb{Q}$.

Normality of H was used only to give G/H a group law; the topology and its separation properties never needed it. Dropping normality, the coset space G/H is no longer a group, but G still acts on it by left translation, and this action inherits all the good properties of the quotient. The resulting object, a space on which G acts transitively and continuously, is the natural home for the many geometric spaces that arise as G/H, and it deserves a name.

Definition 8.5.8: Homogeneous Space

Let G be a topological group and $H \leq G$ a closed subgroup, not necessarily normal. The coset space G/H, with the quotient topology, is called a *homogeneous space* for G. The group G acts on G/H by left translation,

$$a \cdot (gH) = (ag)H \qquad (a \in G,\ gH \in G/H)$$

and this action $G \times G/H \to G/H$ is continuous and transitive. A *continuous action* of G on a space X is a continuous map $G \times X \to X$, $(a, x) \mapsto a \cdot x$, with $e \cdot x = x$ and $a \cdot (b \cdot x) = (ab) \cdot x$; it is *transitive* when for every pair $x, y \in X$ there is $a \in G$ with $a \cdot x = y$.

The two claims folded into the definition, that the left-translation action is continuous and transitive, are quick to verify and worth spelling out. Transitivity is immediate: given cosets gH and $g'H$, the element $a = g'g^{-1}$ satisfies $a \cdot (gH) = (g'g^{-1}g)H = g'H$, so G carries any coset to any other. Continuity is the same quotient-map argument used throughout the section. The action map factors as the outer route in

$$G \times G/H \xrightarrow{\alpha} G/H, \qquad \alpha \circ (\mathrm{id}_G \times \pi) = \pi \circ m'$$

where $m' \colon G \times G \to G$ is the multiplication and $\mathrm{id}_G \times \pi \colon G \times G \to G \times G/H$ is an open surjection (a product of an open surjection with the identity), hence a quotient map; since $\pi \circ m'$ is continuous, α is continuous. A closed H is required only so that G/H is Hausdorff (proposition 8.5.6), the setting in which homogeneous spaces are studied. The homogeneity of section 8.2 was the special case $H = \{e\}$, where $G/H = G$ acts on itself; the general definition says that every quotient looks the same from every point, because G moves any point to any other.

Example 8.5.9: Spheres as Homogeneous Spaces

The orthogonal group $\mathrm{O}(n+1)$ acts on the unit sphere $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$ by rotation, and this action is transitive: any unit vector can be rotated to any other. The stabilizer of the north pole $e_{n+1} = (0, \ldots, 0, 1)$ is the subgroup of orthogonal matrices fixing that vector, a copy of $\mathrm{O}(n)$ acting on the orthogonal complement, and it is closed in $\mathrm{O}(n+1)$ as the preimage of the closed point e_{n+1} under the continuous evaluation $A \mapsto Ae_{n+1}$. Sending a coset $A\,\mathrm{O}(n)$ to the point Ae_{n+1} is then a continuous bijection $\mathrm{O}(n+1)/\mathrm{O}(n) \to \mathbb{S}^n$ intertwining the two actions, exhibiting the sphere as the homogeneous space $\mathbb{S}^n \cong \mathrm{O}(n+1)/\mathrm{O}(n)$. The single algebraic identity $\mathbb{S}^n = \mathrm{O}(n+1)/\mathrm{O}(n)$ encodes both the topology of the sphere and the symmetry group acting on it, which is precisely what the homogeneous-space viewpoint is for.

8.6 Connectedness and Total Disconnectedness

Homogeneity forces the connectedness structure of a topological group to be uniform across the space, and this section extracts the algebraic object that records it. The connected component of the identity is a subgroup, not merely a subset, so a group splits along a canonical exact sequence: a connected piece sitting inside, and a purely disconnected piece as the quotient. The disconnected quotients that arise this way are exactly the objects that carry the arithmetic of the later chapters, so pinning down their defining property closes the point-set theory of this chapter.

The starting point is the connected component of e. Connectedness of a subset was developed in section 3.1; the only new input is that in a group one component is distinguished, the one through the identity, and its translates cover the space.

Definition 8.6.1: Identity Component

Let G be a topological group with identity e. The *identity component* of G, written G°, is the connected component of e, that is, the union of all connected subsets of G that contain e.

By the general theory of components (section 3.1), G° is the largest connected subset of G containing e, and the components of G partition it into disjoint connected pieces. Because left translation L_g is a homeomorphism, it carries the component of e onto the component of $g \cdot e = g$; so every component of G is a translate gG° of the identity component, and the components are precisely the left cosets of G°. The topology of G near any point is thus a translated copy of the topology near e, and the identity component is the one representative from which every other component is recovered by sliding.

Example 8.6.2: The Identity Component of $\mathrm{GL}_n(\mathbb{R})$

The determinant $\det\colon \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times$ is continuous and surjective, and $\mathbb{R}^\times$ is disconnected into the two rays $(0, \infty)$ and $(-\infty, 0)$. Hence $\mathrm{GL}_n(\mathbb{R})$ is disconnected: the two open sets

$$\mathrm{GL}_n^+(\mathbb{R}) = \{A \,\big|\, \det A > 0\}, \qquad \mathrm{GL}_n^-(\mathbb{R}) = \{A \,\big|\, \det A < 0\}$$

separate it. The identity I has $\det I = 1 > 0$, so $\mathrm{GL}_n(\mathbb{R})^\circ \subseteq \mathrm{GL}_n^+(\mathbb{R})$. In fact equality holds: $\mathrm{GL}_n^+(\mathbb{R})$ is path-connected, since any A with $\det A > 0$ can be joined to I by a continuous path of invertible matrices (row-reduce A toward I through elementary operations that never let the determinant vanish, keeping it positive throughout), and a path-connected set is connected. Therefore

$$\mathrm{GL}_n(\mathbb{R})^\circ = \mathrm{GL}_n^+(\mathbb{R})$$

the matrices of positive determinant. The other component is the coset $\mathrm{GL}_n^-(\mathbb{R}) = D \cdot \mathrm{GL}_n^+(\mathbb{R})$, where D is any fixed matrix of negative determinant, in accordance with the coset description above.

The example already exhibits the phenomenon this section formalizes: the identity component is not just a connected subset but a subgroup ($\mathrm{GL}_n^+(\mathbb{R})$ is closed under multiplication and inversion), it is normal (det is a homomorphism, so its sign class is conjugation-invariant), and the quotient $\mathrm{GL}_n(\mathbb{R})/\mathrm{GL}_n^+(\mathbb{R}) \cong \{\pm 1\}$ is a discrete two-point group, as disconnected as a group can be. The next result shows this is general.

Proposition 8.6.3: The Identity Component is a Closed Normal Subgroup

Let G be a topological group. Then G° is a closed normal subgroup of G.

***Proof*:**
Three properties must be checked: G° is a subgroup, it is normal, and it is closed.

Step 1: G° is a subgroup. It suffices to show that G° is closed under the map $(x, y) \mapsto xy^{-1}$ and contains e; the latter is immediate. Fix $g \in G^\circ$. Inversion $\iota\colon G \to G$ is a homeomorphism fixing e, so $\iota(G^\circ)$ is a connected set containing $\iota(e) = e$; by maximality of G° among connected sets through e, we have $\iota(G^\circ) \subseteq G^\circ$, that is, $g^{-1} \in G^\circ$. Next, left translation L_g is a homeomorphism, so $L_g(G^\circ) = gG^\circ$ is connected, and it contains $g \cdot e = g \in G^\circ$. Two connected sets that share the point g have connected union, so $G^\circ \cup gG^\circ$ is connected and contains e; by maximality again, $gG^\circ \subseteq G^\circ$. Thus for every $g, h \in G^\circ$ we have $gh \in gG^\circ \subseteq G^\circ$, and combined with $h^{-1} \in G^\circ$ this gives $gh^{-1} \in G^\circ$. Hence G° is a subgroup.

Step 2: G° is normal. Fix $g \in G$ and consider the conjugation map $c_g\colon x \mapsto gxg^{-1}$. It is the composite $L_g \circ R_{g^{-1}}$ of two homeomorphisms, hence a homeomorphism, and it fixes e since $geg^{-1} = e$. Therefore $c_g(G^\circ)$ is a connected set containing e, and by maximality $c_g(G^\circ) \subseteq G^\circ$, that is, $gG^\circ g^{-1} \subseteq G^\circ$ for every g. This holds for g and for g^{-1}, so in fact $gG^\circ g^{-1} = G^\circ$, and $G^\circ \trianglelefteq G$.

Step 3: G° is closed. The connected components of any topological space are closed, since the closure of a connected set is connected: if A is connected then $\overline{A}$ is connected, so $\overline{G^\circ}$ is a connected set containing e, and by maximality $\overline{G^\circ} \subseteq G^\circ$, forcing $\overline{G^\circ} = G^\circ$. Hence G° is closed, completing the proof.

The proof rests on a single mechanism used three times: a homeomorphism of G that fixes e must map the identity component into itself, because it sends connected-sets-through-e to connected-sets-through-e and G° is the maximal such set. Inversion, left translation followed by absorption, and conjugation are all instances. This is the algebraic dividend of homogeneity: the purely topological notion of "the component of e" is automatically stable under the group operations, so it hands back a subgroup for free. The normality in Step 2 is what makes the quotient G/G° a group, and Step 3 (closedness) is what makes that quotient Hausdorff.

Since G° is a closed normal subgroup, the quotient G/G° is a topological group by theorem 8.5.4, and it is Hausdorff because G° is closed. The next result identifies its connectedness structure: it has none. This is the sense in which G° captures *all* of the connectedness of G, leaving a quotient in which the only connected sets are points.

Proposition 8.6.4: The Component Quotient is Totally Disconnected

Let G be a topological group and let $\pi\colon G \to G/G^\circ$ be the quotient map. Then the only connected subsets of G/G° are the singletons and the empty set. In particular the identity component of G/G° is the trivial subgroup $\{\overline{e}\}$, where $\overline{e} = \pi(e)$.

***Proof*:**
Write $Q = G/G^\circ$. Since Q is homogeneous (translation by any element is a homeomorphism carrying the component of $\bar{e}$ onto the component of that element), it suffices to prove that the component of $\bar{e}$ in Q is $\{\bar{e}\}$: then every component of Q is a single point, and a set with two points, meeting two distinct components, cannot be connected.

Let C be the component of $\bar{e}$ in Q, and let $S = \pi^{-1}(C) \subseteq G$. Since $\pi(e) = \bar{e} \in C$, the set S contains G°. The claim is $S = G^\circ$; granting this, $C = \pi(S) = \pi(G^\circ) = \{\bar{e}\}$, as required, so the whole proof reduces to showing $S = G^\circ$.

Step 1: S is connected. The preimage of a connected set need not be connected in general, so this step uses the specific structure of π. The map π is open (proposition 8.5.3), and $S = \pi^{-1}(C)$ is saturated: it is a union of the cosets gG°, which are exactly the components of G. Suppose, for contradiction, that $S = A \sqcup B$ with A, B nonempty, disjoint, and open in the subspace S. Each of A, B is a union of full cosets gG°: a coset is connected, so it cannot meet both of the disjoint relatively open sets A and B, and it therefore lies entirely in one of them. Hence A and B are saturated, $A = \pi^{-1}(\pi(A))$ and $B = \pi^{-1}(\pi(B))$.

The images $\pi(A)$ and $\pi(B)$ separate C. First, $\pi(A)$ is open in C: writing $A = S \cap W$ with W open in G, the equality $\pi(A) = C \cap \pi(W)$ holds. The inclusion $\pi(A) \subseteq C \cap \pi(W)$ is immediate; for the reverse, if $q \in C \cap \pi(W)$ then $\pi^{-1}(q) \subseteq S$ (as $q \in C$), and some $x \in W$ has $\pi(x) = q$, so $x \in \pi^{-1}(q) \cap W \subseteq S \cap W = A$, giving $q \in \pi(A)$. Since π is open, $\pi(W)$ is open in Q, so $\pi(A) = C \cap \pi(W)$ is open in C; symmetrically $\pi(B)$ is open in C. They are nonempty (as A, B are), they cover C (as $A \cup B = S = \pi^{-1}(C)$), and they are disjoint (as A, B are saturated and disjoint, so no point of C has preimages in both). This separates C, contradicting its connectedness. Therefore S is connected.

Step 2: $S = G^\circ$. By Step 1, S is a connected subset of G containing e (since $G^\circ \subseteq S$). By the maximality of the identity component among connected sets through e, $S \subseteq G^\circ$. Combined with the inclusion $G^\circ \subseteq S$ already noted, $S = G^\circ$.

Thus $C = \pi(S) = \pi(G^\circ) = \{\bar{e}\}$, the component of $\bar{e}$ in Q is trivial, and by homogeneity every component of Q is a singleton, completing the proof.

Every topological group is therefore an extension of a totally disconnected group by a connected one: the closed normal subgroup G° is connected, and the quotient G/G° has no connectedness at all. Symbolically, the short exact sequence

$$1 \longrightarrow G^\circ \longrightarrow G \longrightarrow G/G^\circ \longrightarrow 1$$

separates the two regimes cleanly, and the study of topological groups splits along it into two nearly disjoint theories. The connected part G° is the province of Lie theory and the exponential map, where the group is locally coordinatized by a vector space; the quotient part G/G° is the province of arithmetic, built from finite and profinite data. For $\mathrm{GL}_n(\mathbb{R})$ the sequence reads $1 \to \mathrm{GL}_n^+(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R}) \to \{\pm 1\} \to 1$, the sign of the determinant. The name for the quotient regime is the following.

Definition 8.6.5: Totally Disconnected Group

A topological group G is *totally disconnected* if its only connected subsets are the empty set and the singletons, equivalently if $G^\circ = \{e\}$.

The two formulations agree by homogeneity: if $G^\circ = \{e\}$ then every component, being a translate of

G°, is a single point, so no subset with two points can be connected; conversely if the only connected subsets are singletons then in particular the component of e is $\{e\}$. Proposition 8.6.4 says precisely that G/G° is totally disconnected for every G, so this class of groups is exactly what remains after the connected part is quotiented away. A totally disconnected group can still be far from discrete: total disconnectedness forbids nontrivial connected subsets, but it permits limit points, so the group may be perfect (no isolated points) while every point is its own component.

Example 8.6.6: The p-adic Integers as a Totally Disconnected Group
Fix a prime p and equip $\mathbb{Z}$ with the p-adic metric $d(x, y) = p^{-v}$, where p^v is the exact power of p dividing $x - y$ (and $d(x, x) = 0$). The completion of $(\mathbb{Z}, d)$ is a topological group $\mathbb{Z}_p$ under addition, in which a neighbourhood base at 0 is given by the subgroups $p^n\mathbb{Z}_p$ for $n \geq 0$. Each $p^n\mathbb{Z}_p$ equals the ball $\{x \in \mathbb{Z}_p : d(x, 0) \leq p^{-n}\}$, which is open because balls are open in the ultrametric d; it is also closed, being the complement of the union of the finitely many other cosets $a + p^n\mathbb{Z}_p$ with $0 \leq a < p^n$, each of which is open. A set that is both open and closed and contains a given point x but omits a point $y \neq x$ shows that no connected subset can contain both x and y: choose n large enough that $d(x, y) > p^{-n}$, so that the open-closed coset $x + p^n\mathbb{Z}_p$ contains x but not y, and it separates any prospective connected set through x and y into two nonempty relatively open pieces. Hence the only connected subsets of $\mathbb{Z}_p$ are singletons, and $\mathbb{Z}_p$ is totally disconnected. It is not discrete: 0 is a limit of the sequence $p, p^2, p^3, \ldots$, so no point is isolated. The group $\mathbb{Z}_p$ is thus a totally disconnected group that is compact (it is the limit of the finite quotients $\mathbb{Z}/p^n\mathbb{Z}$) yet has no isolated points, the antithesis of a connected Lie group.

The compact totally disconnected groups are the profinite groups of Galois theory: a profinite group is by construction a limit of finite discrete groups, and such a limit is compact, Hausdorff, and totally disconnected, with a neighbourhood base at the identity consisting of open normal subgroups of finite index. The p-adic integers $\mathbb{Z}_p$ of the example are the profinite completion of $\mathbb{Z}$ along the powers of p, and the absolute Galois group of a field, with its Krull topology, is the archetype: its open subgroups are the stabilizers of finite subextensions, and its total disconnectedness is the topological shadow of the fact that a field has arbitrarily small finite extensions. The connectedness dichotomy of this section is therefore the point-set gateway to that arithmetic theory, developed in full in the companion volumes on Galois cohomology and class field theory.

8.7 Locally Compact Groups

The separation and quotient theory of the preceding sections placed no demand on the local size of the space. Adding one such demand, local compactness, singles out the class of groups on which the analytic theory of the subject rests, and it is a demand the neighbourhood structure at e controls completely. This section isolates that class, records the constructions that preserve it, and marks the door through which measure and representation theory enter.

A locally compact group is one that, as a space, has a compact neighbourhood of each point. Because translations are homeomorphisms and the space is homogeneous, it is enough to test this at the identity: a compact neighbourhood of e is carried by L_g to a compact neighbourhood of any g. So the condition reduces, as every local condition on a topological group does, to a single statement about the neighbourhoods of e.

Definition 8.7.1: Locally Compact Group

A topological group G is *locally compact* if its underlying space is Hausdorff and each point has a neighbourhood whose closure is compact. Equivalently, by homogeneity, if the identity e has a neighbourhood with compact closure.

The Hausdorff clause is a convention, adopted here once and kept throughout. Some authors define "locally compact" for arbitrary spaces and impose Hausdorffness separately; in the presence of a group law the two are almost always wanted together, since it is the Hausdorff locally compact groups that carry the invariant measure discussed at the end of this section, and a group that fails to be Hausdorff can be replaced by its largest Hausdorff quotient (section 1.5) without losing the analysis. Under this convention every locally compact group is in particular Hausdorff, so the separation results of section 1.5 and chapter 4 apply without further comment.

The two clauses of the definition agree because homogeneity moves a single compact neighbourhood everywhere. If C is a neighbourhood of e with $\overline{C}$ compact, then for any g the set $gC = L_g(C)$ is a neighbourhood of g, and $\overline{gC} = L_g(\overline{C})$ is compact as the continuous image of a compact set under the homeomorphism L_g; so a compact neighbourhood at e propagates to a compact neighbourhood at every point. The identity alone governs the whole group.

The immediate examples cover the groups already in hand and one that foreshadows the totally disconnected theory of section 3.1.

Example 8.7.2: Basic Locally Compact Groups

Each of the following is a locally compact group.

1. The additive group $\mathbb{R}^n$. The closed unit ball is a compact neighbourhood of the origin, by the Heine-Borel theorem, so $\mathbb{R}^n$ is locally compact; every closed bounded set is compact.
2. Any discrete group G. The singleton $\{e\}$ is open, hence an open neighbourhood of e, and it is its own closure and is compact (a one-point space is compact). Discreteness is the extreme case in which the compact neighbourhood is a point.
3. Any compact group G. The whole space G is a neighbourhood of e with compact closure $\overline{G} = G$, so compactness is local compactness in its strongest form. In particular the circle $\mathbb{S}$ and any finite group qualify.
4. The general linear group $\mathrm{GL}_n(\mathbb{R})$. It is an open subset of the space of $n \times n$ real matrices $\mathbb{R}^{n^2}$, being the preimage of $\mathbb{R} \setminus \{0\}$ under the continuous determinant map. A point $A \in \mathrm{GL}_n(\mathbb{R})$ therefore has an open ball B around it that lies inside $\mathrm{GL}_n(\mathbb{R})$, and a slightly smaller closed ball $B' \subseteq B$ is a compact neighbourhood of A; so $\mathrm{GL}_n(\mathbb{R})$ is locally compact.

These four span the range the definition is meant to capture: the continuous model $\mathbb{R}^n$, the discrete groups at the opposite pole, the compact groups where local compactness is global, and the matrix groups that are locally Euclidean without being either compact or discrete. A fifth kind of example sits outside the Euclidean picture entirely. The p-adic integers $\mathbb{Z}_p$, the limit of the finite quotients $\mathbb{Z}/p^n\mathbb{Z}$, form a group whose topology is at once compact and totally disconnected: it has no connected piece larger than a point, yet every point has arbitrarily small compact-open neighbourhoods, the cosets of the subgroups $p^n\mathbb{Z}_p$. Such totally disconnected locally compact groups are the topological shape of the profinite groups that organize Galois theory, and they are the reason the theory below is not merely a theory of manifolds; the connectedness apparatus of section 3.1 distinguishes them from the Lie-type examples above.

Local compactness is stable under the two constructions the previous sections built: passing to a closed subgroup, and passing to a Hausdorff quotient. Both preservations reduce to a fact about the underlying spaces, married to the subgroup and quotient theory already established.

Proposition 8.7.3: Closed Subgroups and Hausdorff Quotients

Let G be a locally compact group.

1. If $H \subseteq G$ is a closed subgroup, then H, with the subspace topology, is a locally compact group.
2. If $H \trianglelefteq G$ is a closed normal subgroup, then the quotient G/H is a locally compact group.

***Proof*:**

1. By proposition 8.3.1, H with the subspace topology is a topological group, and as a subspace of the Hausdorff space G it is Hausdorff. It remains to produce a compact neighbourhood of the identity in H. Since G is locally compact, e has a neighbourhood C in G with $\overline{C}$ compact; choose an open set U of G with $e \in U \subseteq C$. Then $U \cap H$ is an open neighbourhood of e in H, and its closure in H is $\overline{U \cap H} \cap H$, which is contained in $\overline{C} \cap H$. Now $\overline{C}$ is compact and H is closed in G, so $\overline{C} \cap H$ is a closed subset of the compact set $\overline{C}$ and is therefore compact. A closed subset of the compact set $\overline{C} \cap H$ is again compact, so the closure of $U \cap H$ in H is compact. Hence $U \cap H$ is a neighbourhood of the identity in H with compact closure, and H is locally compact.
2. Since $H \trianglelefteq G$ is a closed normal subgroup, theorem 8.5.4 makes G/H a topological group, and proposition 8.5.6 makes it Hausdorff, H being closed. Let $\pi\colon G \to G/H$ be the quotient map. By proposition 8.5.3, π is open, so it carries open neighbourhoods of e to open neighbourhoods of $\pi(e)$, the identity coset. Take a neighbourhood C of e in G with $\overline{C}$ compact, and an open set U with $e \in U \subseteq C$. Then $\pi(U)$ is an open neighbourhood of the identity of G/H, and $\pi(U) \subseteq \pi(\overline{C})$. The image $\pi(\overline{C})$ is the continuous image of the compact set $\overline{C}$ under π, hence compact, and a compact subset of the Hausdorff space G/H is closed. So $\pi(\overline{C})$ is a compact set containing the open neighbourhood $\pi(U)$ of the identity, whence the closure of $\pi(U)$ is a closed subset of the compact set $\pi(\overline{C})$ and is compact. Thus the identity of G/H has a neighbourhood with compact closure, and G/H is locally compact.

The two halves rest on the same mechanism read in opposite directions. A closed subgroup inherits compact closures because intersecting a compact set with a closed set leaves it compact; a Hausdorff quotient inherits them because the open quotient map pushes a compact closure forward to a compact set, and Hausdorffness makes that pushed-forward set closed. The closedness hypotheses are not decoration: a subgroup that is not closed need not be locally compact in the subspace topology (the rationals inside $\mathbb{R}$ are the standard failure, being neither closed nor locally compact), and G/H is not even Hausdorff, let alone locally compact, when H is not closed (proposition 8.5.6). With them, local compactness travels intact through the constructions of section 8.3 and section 8.5, so the category of locally compact groups is closed under exactly the operations one uses to build new groups from old.

This closure property is what makes local compactness the right standing hypothesis for the analytic continuation of the subject, taken up in the companion theory rather than here. The forward pointer below marks the transition, and it is the one place in this chapter where the group is studied as more than a space.

8.7.4 Toward Harmonic Analysis A locally compact Hausdorff group carries a nonzero measure that is invariant under left translation, the *Haar measure*, and this measure is unique up to a positive scalar. Its existence is the point at which the topology of G begins to do analytic work. Integration against Haar measure turns functions on G into a Banach space $L^1(G)$, whose multiplication is convolution; completing the picture with the adjoint structure gives the group C^*-algebra, and the representations of G by unitary operators on Hilbert space organize into the subject of abstract harmonic analysis. That subject (Haar measure, $L^1(G)$ and convolution, the unitary dual, and the Fourier theory of G) is developed in the companion volume on unitary representation theory, where local compactness is the running hypothesis throughout. In the present chapter such groups enter only as spaces: the structure recorded above is exactly the topological input those analytic constructions require, and no measure is built here.